Year 11 Unit 2 Mathematics Solutions

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Year 11 Unit 2 Mathematics

Basic Arithmetic & Algebra

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Exercise 1

Rational Numbers & Surds

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1)

a) $\frac{3}{4}+\frac{5}{6}=\frac{9}{12}+\frac{10}{12}=\frac{19}{12}=1\frac{7}{12}$

b) $\frac{2}{11}+\frac{1}{13}=\frac{26}{143}+\frac{11}{143}=\frac{37}{143}$

c) $1\frac{1}{7}+3\frac{3}{5}=1+3+\frac{5}{35}+\frac{21}{35}=4\frac{26}{35}$

d) $\frac{4}{11}-\frac{4}{22}=\frac{4}{11}-\frac{2}{11}=\frac{2}{11}$

e) $2\frac{2}{3}-3\frac{1}{4}=\frac{8}{3}-\frac{13}{4}=\frac{32}{12}-\frac{39}{12}=-\frac{7}{12}$

f) $\frac{5}{9}-\frac{35}{63}=\frac{35}{63}-\frac{35}{63}=0$

2)

a) $\frac{3}{4}\times\frac{8}{9}=\frac{24}{36}=\frac{2}{3}$

b) $\frac{1}{11}\times\frac{33}{12}=\frac{33}{132}=\frac{1}{4}$

c) $\frac{7}{2}\times\frac{1}{15}=\frac{7}{30}$

d) $\frac{3}{2}\div\frac{9}{10}=\frac{3}{2}\times\frac{10}{9}=\frac{30}{18}=\frac{5}{3}=1\frac{2}{3}$

e) $\frac{12}{39}\div\frac{36}{26}=\frac{12}{39}\times\frac{26}{36}=\frac{1}{3}\times\frac{2}{3}=\frac{2}{9}$

f) $\frac{1}{2}\div\frac{2}{3}=\frac{1}{2}\times\frac{3}{2}=\frac{3}{4}$

3) $\frac{2}{5}\div\frac{3}{10}=\frac{2}{5}\times\frac{10}{3}=\frac{20}{15}=1\frac{1}{3}$

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4)

a) $\frac{1}{4} = 0.25$

b) $\frac{2}{3} = 0.\dot{6}$

c) $\frac{1}{8} = 0.125$

d) $\frac{1}{12} = 0.08\dot{3}$

5)

a) $\frac{1}{5} = 20\%$

b) $\frac{3}{4} = 75\%$

c) $\frac{3}{8} = 37.5\%$

d) $\frac{1}{3} = 33.\dot{3}\%$

e) $\frac{7}{1000} = 0.7\%$

6)

a) $30\% = \frac{3}{10}$

b) $12.5\% = \frac{1}{12}$

c) $0.4\% = \frac{0.4}{100} = \frac{4}{1000} = \frac{1}{250}$

d) $2.5\% = \frac{1}{40}$

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7)

a) $0.01=\frac{1}{100}$

b) $0.4=\frac{4}{10}$

c) $0.625=\frac{5}{8}$

d) Let $x = 0.\dot{1}\dot{5}$, then $100x = 15.\dot{1}\dot{5} = 15 + x$

Therefore $99x$=15, so $x = \frac{15}{99} = \frac{5}{33}$

e) Let $x = 4.0\dot{1}\dot{5}$, then $10x = 40.\dot{1}\dot{5} = 40 + \frac{5}{33}$

Therefore $x = 4 + \frac{5}{330} = 4\frac{1}{66}$

8)

a) $\left(\frac{3}{5}\times 498\right) + \left(\frac{3}{5}\times 2\right) = \frac{3}{5}\times(498+2) = \frac{3}{5}\times 500 = 300$

b) $\left(\frac{2}{3}\times 2\right) - \left(\frac{2}{3}\times\frac{1}{2}\right) = \frac{2}{3}\left(2-\frac{1}{2}\right) = \frac{2}{3}\times\frac{3}{2} = 1$

c) $(a+1)(a^2) + (a+1)(2a) + a + 1 = (a+1)(a^2+2a+1) = (a+1)^3$

d) $\frac{(a-1)2x+3(a-1)}{a(2x+3)-2x-3} = \frac{(a-1)(2x+3)}{a(2x+3)-(2x+3)} = \frac{(a-1)(2x+3)}{(a-1)(2x+3)} = 1$

9)

a) $42731 = 4.27\times 10^4$

b) $0.91326 = 9.13\times 10^{-1}$

c) $6139900 = 6.14\times 10^6$

d) $0.034 = 3.40\times 10^{-2}$

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10)

		4 d.p.	4 s.f
a)	0.043176	0.0432	0.04318
b)	0.2565443	0.2565	0.2565
c)	0.00012739	0.0001	0.0001274
d)	1.128755	1.1288	1.129

11)

a) $6\sqrt{2} + 2\sqrt{2} = 8\sqrt{2}$

b) $4\sqrt{8} + 2\sqrt{2} = 4\sqrt{4 \times 2} + 2\sqrt{2} = 8\sqrt{2} + 2\sqrt{2} = 10\sqrt{2}$

c) $\sqrt{27} + 2\sqrt{3} = \sqrt{9 \times 3} + 2\sqrt{3} = 3\sqrt{3} + 2\sqrt{3} = 5\sqrt{3}$

d) $2\sqrt{45} + 3\sqrt{20} = 2\sqrt{9 \times 5} + 3\sqrt{4 \times 5} = 6\sqrt{5} + 6\sqrt{5} = 12\sqrt{5}$

12)

a) $\sqrt{108} - \sqrt{48} = \sqrt{36 \times 3} - \sqrt{16 \times 3} = 6\sqrt{3} - 4\sqrt{3} = 2\sqrt{3}$

b) $\sqrt{32} - \sqrt{18} = \sqrt{16 \times 2} - \sqrt{9 \times 2} = 4\sqrt{2} - 3\sqrt{2} = \sqrt{2}$

c) $3x\sqrt{18} - x\sqrt{128} = x\left(3\sqrt{18} - \sqrt{128}\right) = x(3\sqrt{9 \times 2} - \sqrt{64 \times 2} =$
$x\left(9\sqrt{2} - 8\sqrt{2}\right)$
$= x\sqrt{2}$

d) $2\sqrt{20} - 4\sqrt{5} = 2\sqrt{4 \times 5} - 4\sqrt{5} = 4\sqrt{5} - 4\sqrt{5} = 0$

13)

a) $\sqrt{12} \times \sqrt{3} = \sqrt{36} = 6$

b) $\sqrt{3} \times \sqrt{27} = \sqrt{81} = 9$

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c) $\sqrt{8} \times \sqrt{50} = \sqrt{400} = 20$

d) $\sqrt{18} \times \sqrt{8} = \sqrt{144} = 12$

e) $\sqrt{1.6} \times \sqrt{5} = \sqrt{8} = 2\sqrt{2}$

f) $\sqrt{\frac{3}{4}} \times \sqrt{\frac{12}{25}} = \sqrt{\frac{36}{100}} = \frac{6}{10} = \frac{3}{5}$

g) $\sqrt{\frac{2}{3}} \times \sqrt{\frac{4.5}{27}} = \sqrt{\frac{9}{81}} = \sqrt{\frac{1}{9}} = \frac{1}{3}$

14)

a) $\frac{3}{1-\sqrt{2}} = \frac{3}{1-\sqrt{2}} \times \frac{1+\sqrt{2}}{1+\sqrt{2}} = \frac{3+3\sqrt{2}}{1-2} = -3(1+\sqrt{2})$

b) $\frac{\sqrt{6}}{4-\sqrt{6}} = \frac{\sqrt{6}}{4-\sqrt{6}} \times \frac{4+\sqrt{6}}{4+\sqrt{6}} = \frac{4\sqrt{6}+6}{16-6} = \frac{2(2\sqrt{6}+3)}{10} = \frac{2\sqrt{6}+3}{5}$

c) $\frac{32}{\sqrt{2}+1} = \frac{32}{\sqrt{2}+1} \times \frac{\sqrt{2}-1}{\sqrt{2}-1} = \frac{32(\sqrt{2}-1)}{2-1} = 32(\sqrt{2}-1)$

d) $\frac{4a}{\sqrt{3}-1} + \frac{2b}{1-\sqrt{3}} = \left(\frac{4a}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1}\right) + \left(\frac{2b}{1-\sqrt{3}} \times \frac{1+\sqrt{3}}{1+\sqrt{3}}\right)$

$$= \left(\frac{4a(\sqrt{3}+1)}{2}\right) + \left(\frac{2b(1+\sqrt{3})}{-2}\right) = \left(2a(\sqrt{3}+1)\right) + \left(-b(\sqrt{3}+1)\right)$$

$$= (2a-b)(\sqrt{3}+1)$$

15)

$$\frac{a}{2+\sqrt{5}} - \frac{b}{9-4\sqrt{5}} = \left(\frac{a}{2+\sqrt{5}} \times \frac{2-\sqrt{5}}{2-\sqrt{5}}\right) - \left(\frac{b}{9-4\sqrt{5}} \times \frac{9+4\sqrt{5}}{9+4\sqrt{5}}\right)$$

$$= \left(\frac{2a-a\sqrt{5}}{4-5}\right) - \left(\frac{9b+4b\sqrt{5}}{81-80}\right) = \left(-2a+a\sqrt{5}\right) - \left(9b+4b\sqrt{5}\right)$$

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$= (-2a - 9b) + (a - 4b)(\sqrt{5})$. To be rational, cannot be a surd. Therefore

$a - 4b = 0, a = 4b; (a, b\ rational)$. NOTE if $a - 4b = \sqrt{5}$ to rationalise the surd, the first part of the expression becomes irrational

16)

a) $\sqrt{1.69} = 1.3$

b) $\sqrt{5\frac{4}{9}} = \sqrt{\frac{49}{9}} = \frac{7}{3}$

c) $\left(4\frac{1}{3}\right)^2 = 16\frac{1}{9}$

d) $\sqrt[3]{0.027} = 0.3$

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Exercise 2

Inequalities & Absolute Values

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1)

a) $x < 5 - 2,$
$x < 3$

b) $x > 4 = 3,$
$x > 7$

c) $t \geq \frac{42}{7},$
$t \geq 6$

d) $y < \frac{-30}{5}$
$y < -6$

e) $x > 9 \times 3,$
$x > 27$

f) $y \leq 10 \times \frac{3}{2},$
$y \leq 15$

g) $x > -6$

h) $y \geq -\frac{3}{2}$

2)

a) $2a \geq 2,$
$a \geq 1$

b) $3p \leq 15,$
$p \leq 5$

c) $-3x > 9,$
$x < -3$

d) $-4g < 6,$
$g > -\frac{3}{2}$

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e) $\frac{x}{3} \leq -2, \ \ x \leq -6$

f) $\frac{-x}{2} > 8, \ \ x < -16$

g) $\frac{x}{3} - x < 1, \ -\frac{2}{3}x < 1, \ \ x > -\frac{3}{2}$

h) $\frac{-x}{2} > x - 2, \ -\frac{x}{2} - x > -2, \ -$

3)

a) $2t + 3 \leq t - 6$

b) $3y - 5 > -y - 4$

c) $2(x - 1) - x < \frac{x}{2}$

d) $3(4 - y) + 2y \geq \frac{1}{2}y$

e) $\frac{x}{2} + \frac{x}{3} \leq 4$

$$\frac{5x}{6} \leq 4$$

$$x \leq \frac{24}{5}$$

f) $\frac{-x}{3} - \frac{2x}{4} > 6$

$$\frac{-4x - 6x}{12} > 6$$

$$-10x > 72$$

$$x < -\frac{72}{10}$$

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g) $\frac{-2p}{3} + \frac{p}{-2} \leq -4$

$\frac{4p + 3p}{-6} \leq -4$

$7p \geq 24$

$p \geq \frac{24}{7}$

h) $\frac{-t}{3} - \frac{t}{2} > 7$

$\frac{-5t}{6} > 7$

$t < -\frac{42}{5}$

4)

a) $x = 3 \; or \; x = -3$

b) $x = 5 \; or \; x = -5$

c) $|x| = 8$

$x = 8 \; or \; x = -8$

d) No solution as $|x|$ is always positive

5)

a) $|x + 2| = 7$

$x + 2 = 7 \; or \; -(x + 2) = 7$

$x = 5 \; or \; x = -9$

b) $|x - 3| = 4$

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$x - 3 = 4\ or\ -(x - 3) = 4$
$x = 7\ or\ x = -1$

c) $-|x + 4| = -10$

$-(x + 4) = -10\ or\ (x + 4) = -10$

$-x = -6\ or\ x = -14$

$x = 6\ or\ x = -14$

d) $-|x - 2| = \frac{3}{2}$

No solution

6)

a) $|2x - 3| - 2 = 6$

$|2x - 3| = 8$

$2x - 3 = 8\ or\ -(2x - 3) = 8$

$2x = 11\ or\ -2x = 5$

$x = \frac{11}{2}\ or\ x = -\frac{5}{2}$

b) $|3x + 4| - 5 = \frac{1}{2}$

$|3x + 4| = 5\frac{1}{2}$

$3x + 4 = \frac{11}{2}\ or\ -(3x + 4) = \frac{11}{2}$

$3x = \frac{3}{2}\ or\ -3x = \frac{19}{2}$

$x = \frac{1}{2}\ or\ x = -\frac{19}{6}$

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c) $|2x+1|-4=3$

$2x+1=7\ or\ -(2x+1)=7$

$x=3\ or\ x=-4$

d) $|x-2|+\frac{1}{2}=\frac{2}{3}$

$x-2=\frac{1}{6}\ or\ -(x-2)=\frac{1}{6}$

$x=2\frac{1}{6}\ or\ x=\frac{11}{6}$

7)

a) $|x^2+6x-3|=4$

Case 1

$x^2+6x-3=4$

$x^2+6x-7=0$

$(x+7)(x-1)=0$

$x=-7\ or\ x=1$

Case 2

$-(x^2+6x-3)=4$

$-x^2-6x+3=4$

$-x^2-6x-1=0$

$x^2+6x+1=0$

Applying the quadratic formula (or using calculator) gives

$x=-3+2\sqrt{2}\ or\ x=-3-2\sqrt{2}$

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b) $|x^2 + 2x - 1| = 2$

Case 1

$x^2 + 2x - 1 = 2$

$x^2 + 2x - 3 = 0$

$(x + 3)(x - 1) = 0$

$x = -3 \; or \; x = 1$

Case 2

$-(x^2 + 2x - 1) = 2$

$-x^2 - 2x + 1 = 2$

$-x^2 - 2x - 1 = 0$

$x^2 + 2x + 1 = 0$

$(x + 1)^2 = 0$

$x = -1$

c) $|x^2 + 5x - 2| = -6$

No solution

d) $|x^2 - 13| = 4$

Case 1

$x^2 - 13 = 4$

$x^2 - 17 = 0$

$x = \pm\sqrt{17}$

Case 2

$-(x^2 - 13) = 4$

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$-x^2 + 13 = 4$

$-x^2 + 9 = 0$

$x^2 - 9 = 0$

$x = \pm 3$

8) Note that answers should be substituted back into equation for validity and not all solutions to equations of the type $|a + b| = |p + q| + c$ will be valid

a) $|x + 1| = |x - 2|$

Case 1

$x + 1 = -(x - 2)$

$x = 1 = -x + 2$

$x = \frac{1}{2}$

Case 2

$-(x + 1) = x - 2$

$-x - 1 = x - 2$

$-2x = -1$

$x = \frac{1}{2}$

b) $\left|x - \frac{1}{2}\right| = |x + 3|$

Case 1

$x - \frac{1}{2} = -(x + 3)$

$x - \frac{1}{2} = -x - 3$

$2x = -\frac{5}{2}$

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$$x = -\frac{5}{4}$$

Case 2

$$-\left(x - \frac{1}{2}\right) = x + 3$$

$$-x + \frac{1}{2} = x + 3$$

$$-2x = \frac{5}{2}$$

$$x = -\frac{5}{4}$$

c) $|2x - 3| = |x + 1|$

Case 1

$$2x - 3 = -(x + 1)$$

$$2x - 3 = -x - 1$$

$$3x = 2$$

$$x = \frac{2}{3}$$

Case 2

$$-(2x - 3) = x + 1$$

$$-2x + 3 = x + 1$$

$$-3x = -2$$

$$x = \frac{2}{3}$$

Case 3

$$2x - 3 = x + 1$$

$$x = 4$$

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Case 4

$-(2x-3) = -(x+1)$

$-2x+3 = -x-1$

$-x = -4$

$x = 4$

d) $|3x+1| = |x+3|$

Case 1

$3x+1 = x+3$

$2x = 2$

$x = 1$

Case 2

$3x+1 = -(x+3)$

$3x+1 = -x-3$

$4x = -4$

$x = -1$

Case 3

$-(3x+1) = x+3$

$-3x-1 = x+3$

$-4x = 4$

$x = -1$

Case 4

$-(3x+1) = -(x+3)$

$-3x-1 = -x-3$

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$-2x = -2$

$x = 1$

e) $\left|\frac{1}{2}x + 1\right| = |2x - 3| + 1$

Case 1

$\frac{1}{2}x + 1 = 2x - 3 + 1$

$\frac{1}{2}x + 1 = 2x - 2$

$-\frac{3}{2}x = -3$

$x = 2$; on substitution solution is valid

Case 2

$\frac{1}{2}x + 1 = -(2x - 3) + 1$

$\frac{1}{2}x + 1 = -2x + 3 + 1$

$\frac{5}{2}x = 3$

$x = \frac{6}{5}$; on substitution solution is valid

Case 3

$-\left(\frac{1}{2}x + 1\right) = 2x - 3 + 1$

$-\frac{1}{2}x - 1 = 2x - 2$

$-\frac{5}{2}x = -1$

$x = \frac{2}{5}$; on substitution solution is NOT valid

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Case 4

$$-\left(\frac{1}{2}x+1\right)=-(2x-3)+1$$

$$-\frac{1}{2}x-1=-2x+3+1$$

$$\frac{3}{2}x=5$$

$x=\frac{10}{3}$; on substitution solution is not valid

f) $|2x-5|=|x-3|+6$

Case 1

$$2x-5=x-3+6$$

$$x=8$$

Case 2

$$-(2x-5)=-(x-3)+6$$

$$-2x+5=-x+3+6$$

$$-x=4$$

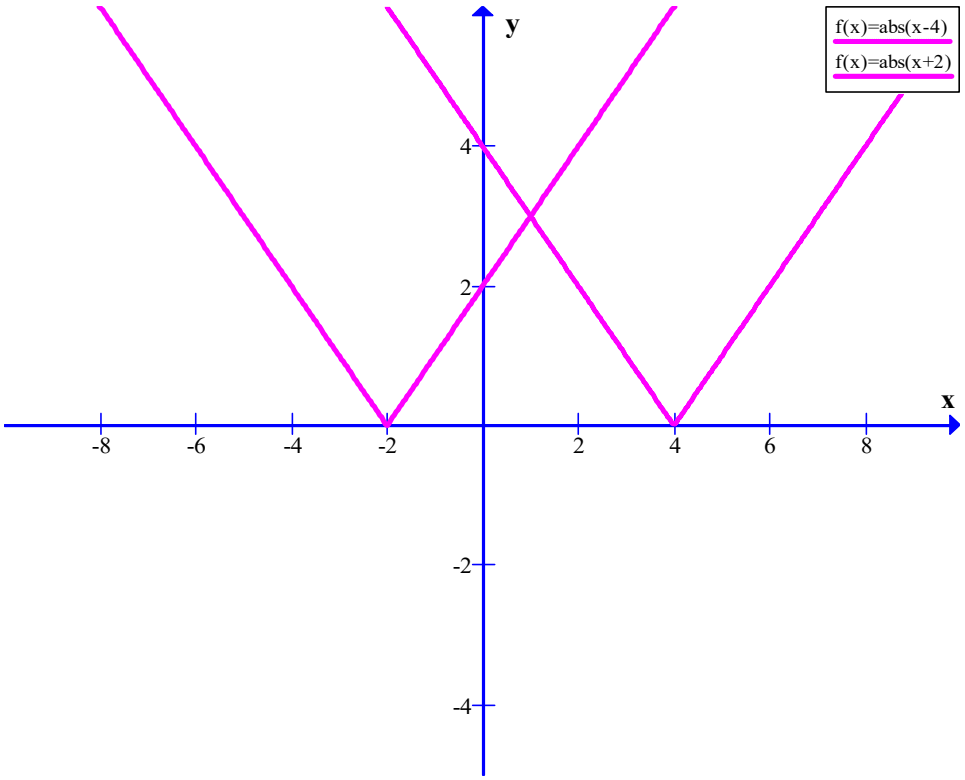
$$x=-4$$

Note for the other two cases invalid solutions are found

9)

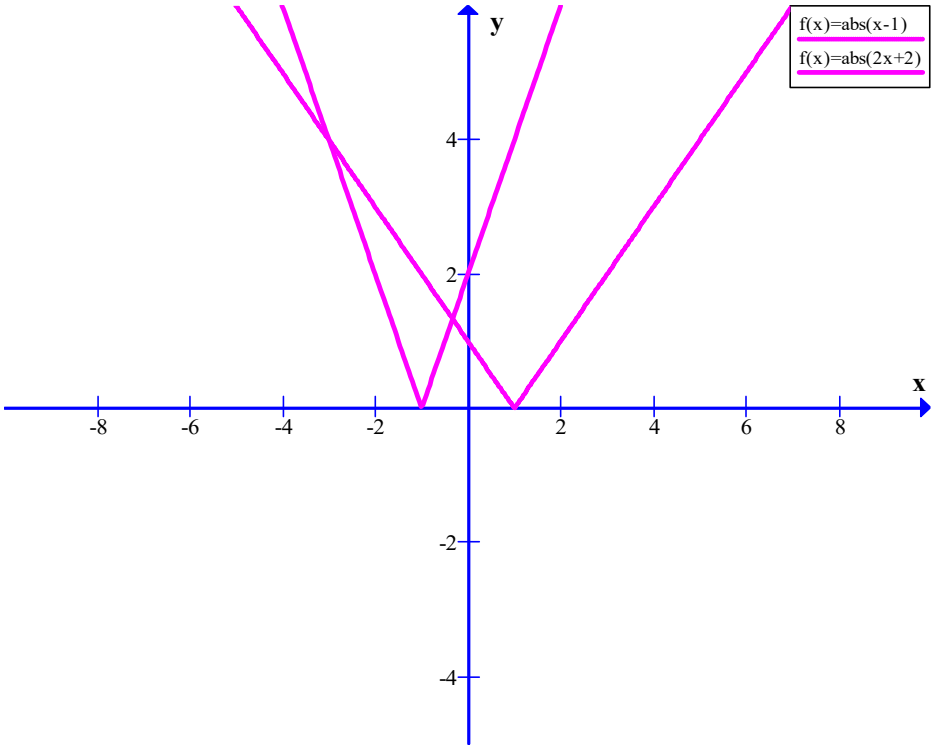
a) $|x-4|=|x+2|$

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Solution is $x = 1$

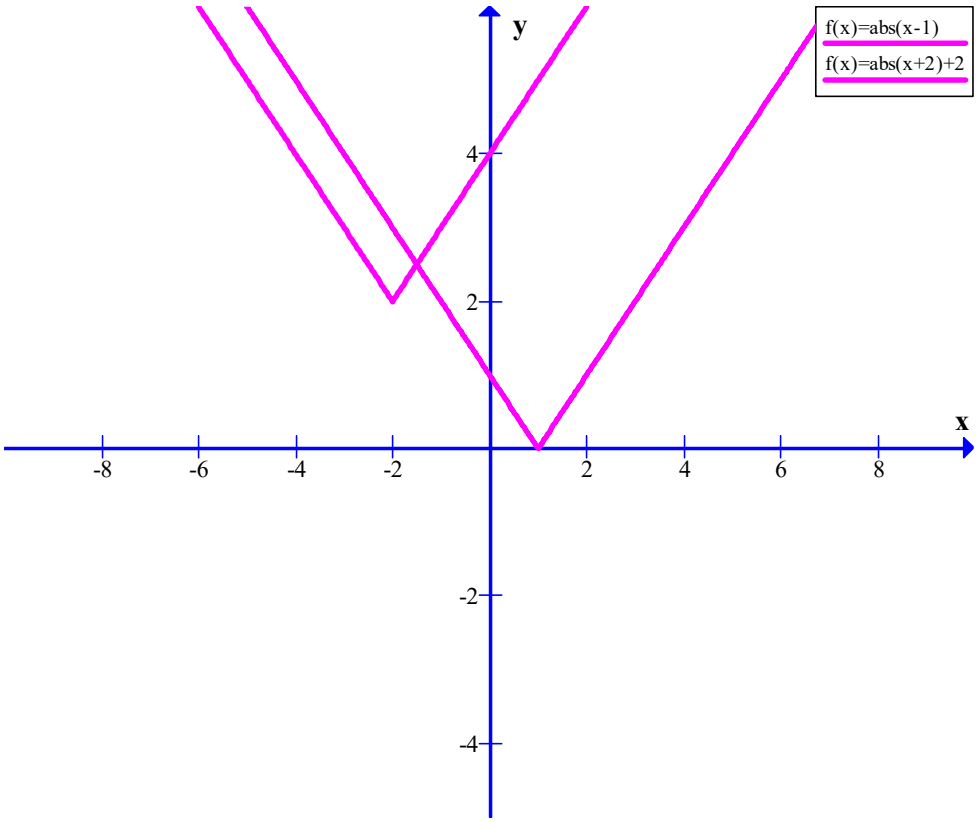
b) $|2x + 2| = |x - 1|$



Solutions are $x = -\frac{1}{3,}\ , x = -3$

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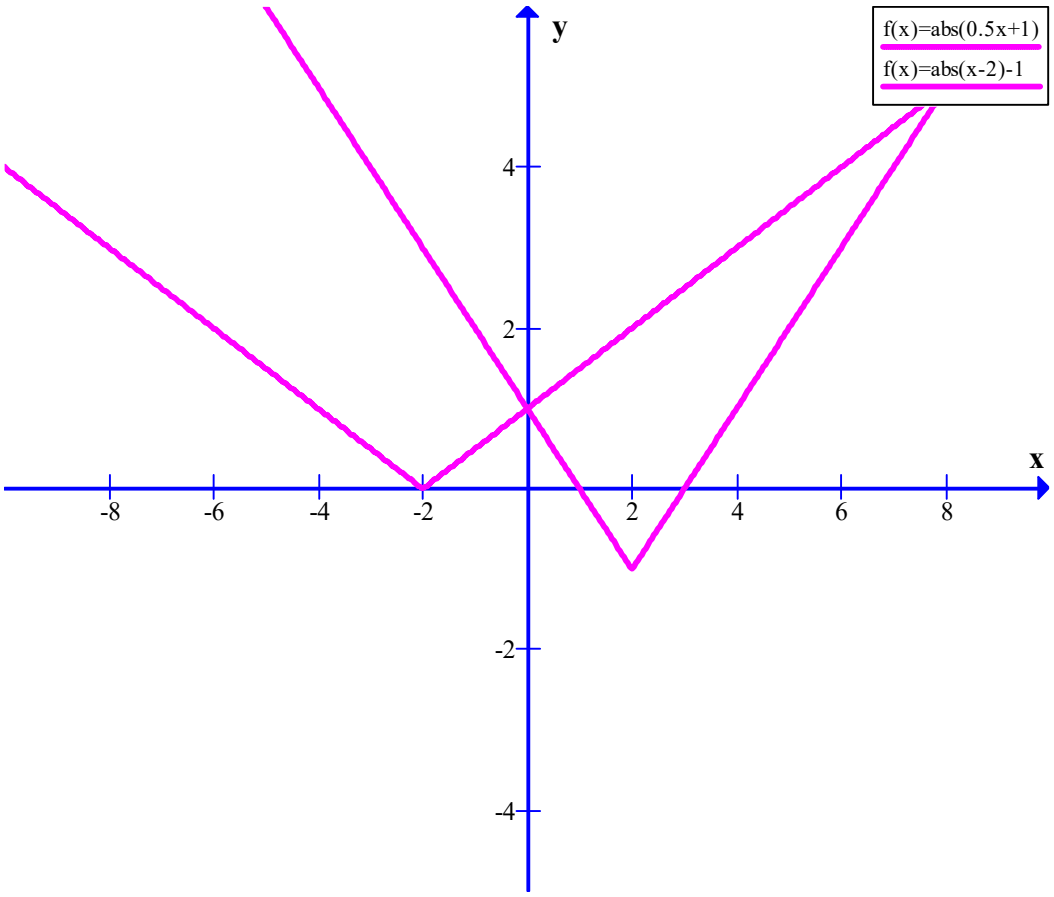
c) $|x-1| = |x+2| + 2$



Solution is $x = -\frac{3}{2}$

d) $\left|\frac{1}{2}x + 1\right| = |x-2| - 1$

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Solution is $x = 0$, $x = 8$

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Exercise 3

Algebraic Expressions

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1)

a) $3x + 4(x - 2) = 3x + 4x - 8 = 7x - 8$

b) $2(3 - y) + 3(2y + 2) = 6 - 2y + 6y + 4 = 10 + 4y$

c) $-(5 - 2t) - (-3 + 4t) = -5 + 2t + 3 - 4t = -2 - 2t$

d) $\frac{1}{2}(4y + 2) - \frac{1}{3}(3 - 6y) = 2y + 1 - 1 + 2y = 4y$

e) $2 + 5(2y - 4) - 4(4 - y) = 2 + 10y - 20 - 16 + 4y = 14y - 34$

f) $-2 - (-3 - x) + (2x - 3) = -2 + 3 + x + 2x - 3 = 3x - 2$

2)

a) $(x^2 - 4x + 3) + (2x^2 + 2x - 6) = 3x^2 - 2x - 3$

b) $x(2 - x^2) + x^2(2x + 4) = 2x - x^3 + 2x^3 + 4x^2 = x^3 + 4x^2 + 2x$

c) $4(9x - x^2) - 3x(4 - x) = 36x - 4x^2 - 12x + 3x^2 = 24x - x^2$

d) $-2(2x - 2) - 3(-3x - 3) = -4x + 4 + 9x + 9 = 5x + 5$

3)

a) $(2 + 5y)(2y - 4) = 4y + 10y^2 - 8 - 20y = 10y^2 - 16y - 8$

b) $(3x - 2) + (2x + 3) = 3x - 2 + 2x + 3 = 5x + 1$

c) $\frac{1}{2}x + 4 - (x - 3) = \frac{1}{2}x + 4 - x + 3 = -\frac{1}{2}x + 7$

d) $\frac{1}{2}(x + 4) - (x - 3) = \frac{1}{2}x + 2 - x + 3 = -\frac{1}{2}x + 5$

e) $2x(3x + 1)(x - 2) = 2x(3x^2 - 5x - 2) = 6x^3 - 10x^2 - 4x = 6x^3 - 4x^2 - 4x$

f) $-x\big((x + 2) + (x^2 + x)\big) = -x(x^2 + 2x + 2) = -x^3 - 2x^2 - 2x$

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4) If $x = \sqrt{2}$

a) $x^2 = 2$

b) $x^8 = 16$

c) $x^4 - x^2 = 4 - 2 = 2$

d) $\frac{2}{x^2} = \frac{2}{2} = 1$

e) $\left(\frac{2}{x}\right)^2 = \left(\frac{2}{\sqrt{2}}\right)^2 = \frac{4}{2} = 2$

5) $x = \frac{a^2 b}{c^2}$

a) $a = 1, b = 2, c = 3; x = \frac{1 \times 2}{9} = \frac{1}{3}$

b) $a = (\frac{1}{5})^2, b = \left(\frac{2}{5}\right)^{\frac{1}{2}}, c = \sqrt[4]{2}; x = \frac{(\frac{1}{5})^4 \times \left(\frac{2}{5}\right)^{\frac{1}{2}}}{2^{\frac{1}{2}}} = \frac{1}{(5)^{\frac{9}{2}}} \cong 1397.54$

c) $a = \frac{1}{2b}, c = 2\sqrt{b}; x = \frac{1xb}{4b^2 \times 4b} = \frac{1}{16b^2}$

d) $a = \frac{1}{2b}, c = \frac{1}{2\sqrt{b}}; x = \frac{4b \times b}{4b^2} = 1$

6) $12 = \pi r^2$

$$r = \sqrt{\frac{12}{\pi}} \cong \sqrt{3.82} \cong 1.95cm$$

7) Note: Convert mass to kilograms and velocity to meters per second before calculating

a) $KE = \frac{1}{2} \times 2 \times 4^2 = 16\ (Joules)$

b) $KE = \frac{1}{2} \times \frac{1}{2} \times 10^2 = 25\ (Joules)$

c) $KE = \frac{1}{2} \times 10 \times 10{,}000^2 = 5 \times 10^8\ (Joules)$

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d) $KE = \frac{1}{2} \times 0.25 \times 4^2 = 2\ (Joules)$

8) $V = \frac{1}{3}\pi r^2 h$

$$r = \sqrt{\frac{3V}{\pi h}} = \sqrt{\frac{3 \times 1200}{\pi \times 100}} = \sqrt{\frac{36}{\pi}} \cong 3.39cm$$

9)

a) $\frac{1}{R} = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2}$

$$R = \frac{2}{3}$$

b) $\frac{1}{R} = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{13}{12}$

$$R = \frac{12}{13}$$

c) $\frac{1}{R} = \frac{1}{\frac{1}{2}} + \frac{1}{2} + \frac{1}{\frac{1}{4}} = 2 + \frac{1}{2} + 4 = \frac{13}{2}$

$$R = \frac{2}{13}$$

d) $\frac{1}{R} = \frac{1}{\frac{1}{2}R_2} + \frac{1}{\frac{1}{2}R_3} + R_3 = \frac{1}{\frac{1}{4}R_3} + \frac{1}{\frac{1}{2}R_3} + R_3 = \frac{4}{R_3} + \frac{2}{R_3} + R_3 = \frac{6+R_3^2}{R_3}$

$$R = \frac{R_3}{6 + R_3^2}$$

10)

a) $4p^2 - 2p = 2p(2p - 1)$

b) $3y^3 + 2y = y(3y^2 + 2)$

c) $6x^2 + 4x + 2 = 2(3x^2 + 2x + 1)$

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d) $10x^3 - 4x^2 + 8x = 2x(5x^2 - 2x + 4)$

e) $-4x - 3x^2 = -x(4 + 3x)$

f) $\frac{1}{x} + \frac{1}{x^2} = \frac{1}{x^2}(x + 1)$

11)

a) $x^2 - 4 = (x - 2)(x + 2)$

b) $4y^2 - 9 = (2y - 3)(2y + 3)$

c) $25p^2 - 25 = 25(p^2 - 1) = 25(p - 1)(p + 1)$

d) $\frac{x^2}{4} - \frac{1}{16} = \left(\frac{x}{2} - \frac{1}{4}\right)\left(\frac{x}{2} + \frac{1}{4}\right)$

e) $y^4 - 100y^2 = (y^2 - 10y)(y^2 + 10y)$

f) $x^2 - 2 = \left(x - \sqrt{2}\right)(x + \sqrt{2})$

12)

a) $x^2 - 6x + 9 = (x - 3)^2$

b) $x^2 + 4x - 5 = (x + 5)(x - 1)$

c) $x^2 - 8x + 12 = (x - 6)(x - 2)$

d) $2x^2 + 9x + 10 = (2x + 5)(x + 2)$

e) $3x^2 + 5x - 12 = (3x - 4)(x + 3)$

f) $6x^2 - 14x + 8 = (6x - 8)(x - 1)$

13)

a) $bx + by - 3x - 3y = (b - 3)(x + y)$

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b) $x^2 + xy - \frac{1}{4} - \frac{1}{2}y = \left(x - \frac{1}{2}\right)\left(x + \frac{1}{2}\right) + y\left(x - \frac{1}{2}\right) = \left(x + y + \frac{1}{2}\right)(x - \frac{1}{2})$

c) $4x^2 - y^2 + 8x - 4y = (2x - y)(2x + y) + 4(2x - y) = (2x + y + 4)(2x - y)$

d) $x^3 + 1 = (x + 1)(x^2 - x + 1)$

e) $x^3 - 27 = (x - 3)(x^2 + 3x + 9)$

f) $x^3 + 125 = (x + 5)(x^2 - 5x + 25)$

14)

a) $\frac{2p+3t}{8p+12t} = \frac{2p+3t}{4(2p+3t)} = \frac{1}{4}$

b) $\frac{3x-y}{4y-12x} = \frac{3x-y}{-4(3x-y)} = -\frac{1}{4}$

c) $\frac{1-x}{1-x^2} = \frac{1-x}{(1-x)(1+x)} = \frac{1}{1+x}$

d) $\frac{x^2+5x+6}{x+3} = \frac{(x+3)(x+2)}{x+3} = x + 2$

e) $\frac{y^2-16}{y-4} = \frac{(y-4)(y+4)}{y-4} = y + 4$

f) $\frac{x^2+9}{x-3}$ cannot be simplified

15)

a) $\frac{x}{x-2} \times \frac{x^2-4}{xy+x} = \frac{x}{x-2} \times \frac{(x-2)(x+2)}{x(y+1)} = \frac{x+2}{y+1}$

b) $\frac{2}{x+1} \times \frac{(x+1)^2}{4} = \frac{x+1}{2}$

c) $\frac{x^2-8x+7}{t^2-6t+9} \times \frac{t-3}{x-7} = \frac{(x-7)(x-1)}{(t-3)(t-3)} \times \frac{t-3}{x-7} = \frac{x-1}{t-3}$

d) $\frac{2}{p-2} \div \frac{4p}{p^2-4} = \frac{2}{p-2} \times \frac{p^2-4}{4p} = \frac{1}{p-2} \times \frac{(p-2)(p+2)}{2p} = \frac{p+2}{2p} = \frac{p}{2} + \frac{2}{p}$

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e) $\frac{3x}{2x-1} \div \frac{6x}{4x-2} = \frac{3x}{2x-1} \times \frac{4x-2}{6x} = \frac{1}{2x-1} \times \frac{2(2x-1)}{2} = 1$

f) $\frac{xy}{x+1} \div \frac{x^2y^2}{4x+4} = \frac{xy}{x+1} \times \frac{4(x+1)}{(xy)^2} = \frac{4}{xy}$

16)

a) $\frac{a+b}{3} + \frac{a-b}{9} = \frac{3(a+b)+(a-b)}{9} = \frac{4a+2b}{9}$

b) $\frac{2x+3y}{2} - \frac{2y-3x}{3} = \frac{3(2x+3y)-2(2y-3x)}{6} = \frac{6x+9y-4y+6x}{6} = \frac{12x+5y}{6}$

c) $\frac{2-p}{p} + \frac{p+2}{2} = \frac{2(2-p)+p(p+2)}{2p} = \frac{4-2p+p^2+2p}{2p} = \frac{p^2+4}{2p} = \frac{p}{2} + \frac{2}{p}$

d) $\frac{4}{y} - \frac{6}{y(y-1)} = \frac{4(y-1)-6}{y(y-1)} = \frac{4y-10}{y(y-1)}$

e) $\frac{1}{x^2+5x+4} + \frac{2}{x+1} = \frac{1}{(x+1)(x+4)} + \frac{2}{x+1} = \frac{1+2(x+4)}{(x+1)(x+4)} = \frac{2x+5}{(x+1)(x+4)}$

f) $\frac{1}{x^2-4} + \frac{1}{x^2+4x+4} = \frac{1}{(x-2)(x+2)} + \frac{1}{(x+2)(x+2)} = \frac{x+2+(x-2)}{(x-2)(x+2)(x+2)}$

$= \frac{2x}{(x-2)(x+2)^2}$

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Exercise 4

Linear & Quadratic Expressions

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1)

a) $2x + 4 = 10$

$2x = 6$

$x = 3$

b) $3x + 7 = 4$

$3x = -3$

$x = -1$

c) $\frac{1}{2}x - 4 = 5$

$\frac{1}{2}x = 9$

$x = 18$

d) $\frac{2}{3}x + 6 = 8$

$\frac{2}{3}x = 2$

$x = 3$

e) $2 - \frac{4}{5}x = -6$

$-\frac{4}{5}x = -8$

$x = 10$

f) $11 - \frac{1}{2}x = 11$

$-\frac{1}{2}x = 0$

$x = 0$

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2)

a) $\frac{4x+1}{x} = 3$

$4x + 1 = 3x$

$x = -1$

b) $\frac{2x-6}{x} = -4$

$2x - 6 = -4x$

$6x = 6$

$x = 1$

c) $\frac{4x-2}{x-2} = 8$

$4x - 2 = 8x - 16$

$-4x = -14$

$x = \frac{7}{2}$

d) $\frac{3x+6}{x-4} = 10$

$3x + 6 = 10x - 40$

$-7x = -46$

$x = \frac{46}{7}$

e) $\frac{\frac{1}{2}x-4}{x+1} = 6$

$\frac{1}{2}x - 4 = 6x + 6$

$-\frac{11}{2}x = 10$

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$$x = -\frac{20}{11}$$

f) $\frac{2-\frac{1}{3}x}{x-3} = 3$

$$2 - \frac{1}{3}x = 3x - 9$$

$$-\frac{10}{3}x = -11$$

$$x = \frac{33}{10}$$

3) Solve the following linear equations

a) $\frac{x-4}{x+2} = \frac{x-3}{x+1}$

$$x^2 - 3x - 4 = x^2 - x - 6$$

$$-2x = -2$$

$$x = -1$$

b) $\frac{x+1}{x-1} = \frac{x+2}{x-6}$

$$x^2 - 5x - 6 = x^2 + x - 2$$

$$-6x = 4$$

$$x = -\frac{2}{3}$$

c) $\frac{2x-3}{3x-2} = \frac{2x+1}{3x-1}$

$$6x^2 - 11x + 3 = 6x^2 - x - 2$$

$$-10x = -5$$

$$x = \frac{1}{2}$$

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d) $\frac{\sqrt{x}+2}{\sqrt{x}-1}=\frac{\sqrt{x}-4}{\sqrt{x}+1}$

$x+3\sqrt{x}+2=x-5\sqrt{x}+4$

$8\sqrt{x}=2$

$\sqrt{x}=\frac{1}{4}$

$x=\pm\frac{1}{2}$

4)

a) $2x+2>6$

$2x>4$

$x>2$

b) $4x-3\leq 9$

$4x\leq 12$

$x\leq 3$

c) $2-6x\geq -10$

$-6x\geq -12$

$x\leq 2$

d) $1-\frac{1}{2}x<3$

$-\frac{1}{2}x<2$

$x>-4$

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e) $|x-2|<5$

$x-2<5\ and\ -x+2<5$

$x<7\ and\ -x<3$

$x<7\ and\ x>-3$

f) $|x+1|\geq 3$

$x+1\geq 3\ and\ -x-1\geq 3$

$x\geq 2\ and\ -x\geq 4$

$x\geq 2\ and\ x\leq -4$

g) $|x+1|+1<2$

$|x+1|<1$

$x+1<1\ and\ -x-1<1$

$x<0\ and\ -x<2$

$x<0\ and\ x>-2$

5)

a) $x^2+5x-6=0$

$(x+6)(x-1)=0$

$x=-6\ or\ x=1$

b) $x^2-5x+6=0$

$(x-3)(x-2)=0$

$x=3\ or\ x=2$

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c) $x^2 + 2x + 1 = 0$

$(x + 1)^2 = 0$

$x = -1$

d) $2x^2 + 7x - 9 = 0$

$(2x + 9)(x - 1) = 0$

$x = -\frac{9}{2}$ or $x = 1$

e) $6x^2 - 14x + 8 = 0$

$(3x - 4)(2x - 2) = 0$

$x = \frac{4}{3}$ or $x = 1$

f) $10x^2 + 6x - 4 = 0$

$5x^2 + 3x - 2 = 0$

$(5x - 2)(x + 1) = 0$

$x = \frac{2}{5}$ or $x = -1$

g) $10x^2 - 6x - 4 = 0$

$5x^2 - 3x - 2 = 0$

$(5x + 2)(x - 1) = 0$

$x = -\frac{2}{5}$ or $x = 1$

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6) Solve the following equations using the most appropriate method

a) $6 - x^2 = x$

$x^2 + x - 6 = 0$

$(x + 3)(x - 2) = 0$

$x = -3 \; or \; x = 2$

b) $8x^2 + 2x - 1 = 0$

$(4x - 1)(2x + 1) = 0$

$x = \frac{1}{4} \; or \; x = -\frac{1}{2}$

c) $x^2 = 8x$

$x^2 - 8x = 0$

$x(x - 8) = 0$

$x = 0 \; or \; x = 8$

Note if we cancel x from both sides of the initial equation we lose the solution $x = 0$

d) $(x - 4)^2 = 9$

$x - 4 = 3 \; or \; x - 4 = -3$

$x = 7 \; or \; x = 1$

e) $2x^2 + 4x + 4 = 0$

Cannot factorize

Use quadratic formula

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$$x = \frac{-4 \pm \sqrt{4^2 - 4 \times 2 \times 4}}{4}$$

$Since\ discriminant > 0, there\ are\ no\ real\ solutions$

f) $x^2 = 4x - 2$

$x^2 - 4x + 2 = 0$

Cannot factorize

Use quadratic formula

$$x = \frac{4 \pm \sqrt{16 - (4 \times 1 \times 2)}}{2}$$

$$x = \frac{4 \pm \sqrt{8}}{2}$$

$x = 2 \pm \sqrt{2}$

7)

a) $2x + 3y = 5$ and $x + y = 2$

Multiply equation 2 by 2

$2x + 3y = 5\ and\ 2x + 2y = 4$

Subtract equation 2 from equation 1

$y = 1$

Substituting gives $x = 1$

Solution is (1, 1)

b) $x - 4y = -10$ and $x - y = -1$

Subtract equation 2 from equation 1

$-3y = -9$

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$y = 3$

Substituting gives $x = 2$

Solution is (2, 3)

c) $x + \frac{3}{2}y = -\frac{5}{2}$ and $2x - y = 3$

Multiply equation 1 by 2

$2x + 3y = -5 \; and \; 2x - y = 3$

Subtracting equation 2 from equation 1

$4y = -8$

$y = -2$

Substituting gives $x = \frac{1}{2}$

Solution is $\left(\frac{1}{2}, -2\right)$

d) $4x - 2y = 3$ and $x + y = 0$

Multiply equation 2 by 4

$4x - 2y = 3 \; and \; 4x + 4y = 0$

Subtract equation 2 from equation 1

$-6y = 3$

$y = -\frac{1}{2}$

Substituting gives $x = \frac{1}{2}$

Solution is $\left(\frac{1}{2}, -\frac{1}{2}\right)$

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e) $x - y = 4$ and $-x + y = 8$

Add equations

$4 = 8$

Therefore no solution

f) $x - y = 2$ and $-x + y = -2$

Add equations

$0 = 0$

Therefore there is a solution, but must use another method

$y = x - 2 \; and \; y = -2 + x$

These equations are the same; therefore any pair satisfying the condition is valid

Solution is (a, b) such that $b = a - 2$

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Year 11 Unit 2 Mathematics

Real Functions

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Exercise 1

Range, Domain & Variables

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1)

a) $f(x) = x^2 - 1$. Domain is all real x; range is all $y \geq -1$

b) $f(x) = \sqrt{x}$. Domain is all $x \geq 0$; range is all $y \geq 0$

c) $f(x) = \frac{1}{x}$. Domain is all $x \neq 0$; range is all $y \neq 0$

d) $f(x) = \frac{1}{\sqrt{x}}$. Domain is all $x > 0$; range is all $y > 0$

e) $f(x) = x + 1$. Domain and range is set of all real numbers

f) $f(x) = \sqrt{-x}$. Domain is all $x \leq 0$; range is all $y \geq 0$

2)

a) $f(x) = \sqrt{x} + 2$. Domain is all $x \geq 0$; range is all $y \geq 2$

b) $f(x) = \sqrt{x} + 1$. Domain is all $x \geq 0$; range is all $y \geq 1$

c) $f(x) = \sqrt{x} - 2$. Domain is all $x \geq 0$; range is all $y \geq -2$

d) $f(x) = \sqrt{x} + c$. Domain is all $x \geq 0$; range is all $y \geq c$

3)

a) $f(x) = x^2$. Domain is all x; range is all $y \geq 0$

b) $f(x) = x^2 + 1$. Domain is all x; range is all $y \geq 1$

c) $f(x) = x^2 - 2$. Domain is all x; range is all $y \geq -2$

d) $f(x) = x^2 + c$. Domain is all x; range is all $y \geq c$

4)

a) $f(x) = (x + 1)^2$. Domain is all x; range is all $y \geq 0$

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b) $f(x) = (x-2)^2$. Domain is all x; range is all $y \geq 0$

c) $f(x) = (x+4)^2$. Domain is all x; range is all $y \geq 0$

d) $f(x) = (x+c)^2$. Domain is all x; range is all $y \geq 0$

5) Which of the following are not functions; give reasons for those considered non-functions

a) $f(x) = \frac{1}{\sqrt{x}}$

b) $f(x) = 2$

c) $f(x)^2 - 2x = 3$. There is more than one x value for each value of y

d) $f(x^2) - 2x = 3$

e) $x = -2$. Every co-ordinate is of the form $(-2, y)$

f) $x^2 + y^2 = 4$. This forms a circle which is not a function

6)

a) $y = \frac{1}{x}$. Domain is all $x \neq 0$; range is all y

b) $y = \frac{1}{x} + 1$. Domain is all $x \neq 0$; range is all y

c) $y = \frac{1}{x} - 1$. Domain is all $x \neq 0$; range is all y

d) $y = \frac{1}{x} + c$. Domain is all $x \neq 0$; range is all y

7)

a) $y = \frac{1}{x^2}$. Domain is all $x \neq 0$; range is all $y > 0$

b) $y = \frac{1}{x^2+1}$. Domain is all x; range is $0 < y \leq 1$

c) $y = \frac{1}{x^2-1}$. Domain is all x, except $x = \pm 1$; range is all $y \neq 0$

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d) $y = \frac{1}{x^2+c}$. Domain is all x, except $x = \pm c$; range is all $y \neq 0$ for c negative, $0 < y \leq \frac{1}{c}$ for c positive

8)

a) $y = |x|$. Domain is all x; range is all $y \geq 0$

b) $y = |x + 1|$. Domain is all x; range is all $y \geq 0$

c) $y = |x - 2|$. Domain is all x; range is all $y \geq 0$

d) $y = |x + c|$. Domain is all x; range is all $y \geq 0$

e) $y = |x| + 1$. Domain is all x; range is all $y \geq 1$

f) $y = |x| - 2$. Domain is all x; range is all $y \geq -2$

g) $y = |x| + c$. Domain is all x; range is all $y \geq c$

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Exercise 2

Properties of Graphs of Real Functions

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For each question below, sketch the graph of the function, and determine the following properties.

- x intercept
- y intercept
- Where the function is increasing
- Where the function is decreasing
- Where the function is positive, negative, and zero
- Any horizontal or vertical asymptotes
- The maximum and minimum values of the function
- If there are any discontinuities

Use the last equation in each question to generalize the above properties of functions of that type

1) Linear functions

Function	x int	y int	inc	dec	+ve	-ve	asymp	Max/min	disc
$y = 2x$	$x = 0$	$y = 0$	always	never	For x>0	For x<0	none	$\infty/-\infty$	none
$y = 3x + 1$	$x = -\frac{1}{3}$	$y = 1$	always	never	$x > -\frac{1}{3}$	$x < -\frac{1}{3}$	none	$\infty/-\infty$	none
$y = 4x - 2$	$x = \frac{1}{2}$	$y = -2$	always	never	$x > \frac{1}{2}$	$x < \frac{1}{2}$	none	$\infty/-\infty$	none
$y = ax + b$	$x = -\frac{b}{a}$	$y = b$	always	never	$x > -\frac{b}{a}$	$x < -\frac{b}{a}$	none	$\infty/-\infty$	none

2) Quadratic functions

Function	x int	y int	inc	dec	+ve	-ve	asymp	Max/min	disc
$y = x^2$	$x = 0$	$y = 0$	$x > 0$	$x < 0$	$x \neq 0$	never	none	$\infty/0$	none
$y = x^2 + 1$	none	$y = 1$	$x > 0$	$x < 0$	all x	never	none	$\infty/1$	none
$y = x^2 - 2$	$x = \pm\sqrt{2}$	$y = -2$	$x > 0$	$x < 0$	$\lvert x\rvert > \sqrt{2}$	$\lvert x\rvert < \sqrt{2}$	none	$\infty/-2$	none
$y = x^2 + c$	For $c > 0$, none For $c \leq 0$, $x = \pm\sqrt{-c}$	$y = c$	$x > 0$	$x < 0$	For $c > 0$, always For $c < 0$ $\lvert x\rvert < \sqrt{-c}$	For $c > 0$, never For $c < 0$, $\lvert x\rvert < \sqrt{-c}$	none	∞/c	none

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3)

Function	x int	y int	inc	dec	+ve	-ve	asymp	Max/ min	disc
$y=\frac{1}{x}$	none	none	never	As $x \to 0$ from -∞ and $as\ x \to \infty$	$x>0$	$x<0$	$x=0,$ $y=0$	$\infty, -\infty$	$x=0$
$y=\frac{1}{x+1}$	none	$y=1$	never	As $x \to -1$ from $-\infty$ and as $x \to \infty$	$x>-1$	$x<-1$	$x=-1$ $y=0$	$\infty, -\infty$	$x=-1$
$y=\frac{1}{x-2}$	none	none	never	As $x \to 2$ from $-\infty$ and as $x \to \infty$	$x>2$	$x<2$	$x=2$ $y=0$	$\infty, -\infty$	$x=2$
$y=\frac{1}{x}+1$	none	$x=-1$	never	As $x \to 0$ from $-\infty$ and as $x \to \infty$	$x<-1$ $x>0$	$x>-1$ $x<0$	$x=0$ $y=1$	$\infty, -\infty$	$x=0$
$y=\frac{1}{x}-2$	none	$x=\frac{1}{2}$	never	As $x \to 0$ from $-\infty$ and as $x \to \infty$	$x<\frac{1}{2}$ $x>0$	$x>\frac{1}{2}$ $x<0$	$x=0$ $y=-2$	$\infty, -\infty$	$x=0$
$y=\frac{1}{x+a}+b$	none	$x=-\frac{1}{b}$	never	As $x \to a$ from $-\infty$ and as $x \to \infty$	$x<-\frac{1}{b}$ $x>-a$	$x>-\frac{1}{b}$ $x<-a$	$x=-a$ $y=b$	$\infty, -\infty$	$x=-b$

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4)

Function	x int	y int	inc	dec	+ve	-ve	asymp	Max/ min	disc
$y=\sqrt{x}$	$x=0$	$y=0$	$x>0$	never	$x>0$	never	none	$0,\infty$	none
$y=\sqrt{x+1}$	$x=-1$	$y=1$	$x>-1$	never	$x>-1$	never	none	$0,\infty$	none
$y=\sqrt{x-2}$	$x=2$	none	$x>2$	never	$x>2$	never	none	$0,\infty$	none
$y=\sqrt{x}+1$	none	$y=1$	$x>0$	never	$x>0$	never	none	$1,\infty$	none
$y=\sqrt{x}-2$	$x=4$	$y=-2$	$x>0$	never	$x>4$	$0\le x\le 4$	none	$-2,\infty$	none
$y=\sqrt{x+a}$	$x=-a$	$y=\sqrt{a}$, $a>0$	$x>-a$	never	$x>-a$	never	none	$0,\infty$	none
$y=\sqrt{x}+b$	$x=b^2$ $b<0$	$y=b^2$	$x>0$	never	$x>0$ for b>0 $x=b^2$ for b<0	$0\le x\le b^2$ for b<0	none	b,∞	none

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5)

Function	x int	y int	inc	dec	+ve	-ve	asymp	Max/ min	disc
$y = \|x\|$	$x = 0$	$y = 0$	All except at $x = 0$	never	$x > 0$	never	none	$0, \infty$	$x = 0$
$y = \|x + 1\|$	$x = -1$	$y = 1$	All except at $x = -1$	never	$x > -1$	never	none	$0, \infty$	$x = -1$
$y = \|x - 2\|$	$x = 2$	$y = 2$	All except at $x = 2$	never	$x > 2$	never	none	$0, \infty$	$x = 2$
$y = \|x\| + 1$	none	$y = 1$	All except at $x = 0$	never	$x > 0$	never	none	$1, \infty$	$x = 0$
$y = \|x\| - 2$	$x = \pm 2$	$y = -2$	All except at $x = 0$	never	$x > \pm 2$	$\|x\| < 2$	none	$-2, \infty$	$x = 0$
$y = \|x + a\|$	$x = -a$	$y = a$	All except at $x = -a$	never	$x > -a$	never	none	$0, \infty$	$x = -a$
$y = \|x\| + b$	$x = \pm b$ for b<0	$y = b$	All except at $x = 0$	never	$x = \pm b$ for b<0 $x > 0$ for b>0	never	none	b, ∞	$x = 0$

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6) Miscellaneous functions

Function	x int	y int	inc	dec	+ve	-ve	asymp	Max/ min	disc
$y = x \; for \; k \leq x \leq k+1$	$0 \leq x < 1$	$y = 0$	never	never	$x > 1$	$x < 0$	none	$-\infty, \infty$	$x = k+1$
$y = \frac{1}{x^2}$	never	never	always	never	$x > 0$	never	$x = 0,$ $y = 0$	$-\infty, \infty$	$x = 0$
$y = \frac{1}{\sqrt{x}}$	never	never	never	always	$x > 0$	never	$x = 0$ $y = 0$	$\infty, 0$	none
$y = \frac{1}{\lvert x \rvert}$	never	never	always	never	$x > 0$	never	$x = 0,$ $y = 0$	$-\infty, \infty$	$x = 0$
$y = x^3 + c$	$x = \sqrt[3]{c}$	$y = c$	All except $x = 0$	never	$x > \sqrt[3]{c}$	$x < \sqrt[3]{c}$	none	$-\infty, \infty$	none

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Exercise 3

Geometric Representation

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1)

a) $x^2 + y^2 = 1$

b) $x^2 + y^2 = 4$

c) $x^2 + (y-1)^2 = 4$

d) $(x-1)^2 + (y+1)^2 = 9$

e) $(x-2)^2 + (y-3)^2 = 16$

f) $\left(x - \frac{1}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 = 2.25$

2)

a) $x^2 + y^2 = 9$

Centre at origin, radius 3

b) $x^2 + y^2 - 4y + 2 = 0$

$x^2 + (y-2)^2 - 2 = 0$

$x^2 + (y-2)^2 = 2$

Centre at $(0, 2)$, radius $\sqrt{2}$

c) $x^2 + y^2 - 2x - 2y - \frac{3}{2} = 0$

$$(x-1)^2 + (y-1)^2 - \frac{1}{2} = 0$$

$$(x-1)^2 + (y-1)^2 = \frac{1}{2}$$

Centre at $(1, 1)$, radius $\frac{1}{\sqrt{2}}$

d) $x^2 + 4x + y^2 - 4y = 2$

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$x^2 + 4x + y^2 - 4y = 2$

$(x + 2)^2 + (y - 2)^2 = 2$

Centre at$(-2, 2)$, radius $\sqrt{2}$

e) $x^2 + 6x + y^2 + 2y + 9 = 0$

$(x + 3)^2 + (y + 1)^2 - 1 = 0$

$(x + 3)^2 + (y + 1)^2 = 1$

Centre at$(-3, -1)$, radius 1

f) $x^2 + x + y^2 + y = 0$

$\left(x + \frac{1}{2}\right)^2 + \left(y + \frac{1}{2}\right)^2 - \frac{1}{2} = 0$

$\left(x + \frac{1}{2}\right)^2 + \left(y + \frac{1}{2}\right)^2 = \frac{1}{2}$

Centre at$\left(-\frac{1}{2}, -\frac{1}{2}\right)$, radius $\frac{1}{\sqrt{2}}$

3) (Put equations into standard parabolic form)

a) $y = \frac{x^2}{4}$

$(x - 0)^2 = 4(y - 0)^2$

Vertex is (0, 0)

Focus is (0, 1)

b) $y = 2x^2 - 4x + 4$

$\frac{1}{2}y = x^2 - 2x + 2$

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$\frac{1}{2}y = (x-1)^2 + 1$

$(x-1)^2 = \frac{1}{2}y - 1$

$(x-1)^2 = \frac{1}{2}(y-2)$

Vertex at (1, 2)

Focus at $(1, \frac{17}{8})$

c) $6y = x^2 + 4x + 16$

$6y = (x+2)^2 + 12$

$(x+2)^2 = 6(y-2)$

Vertex at (-2, 2)

Focus at $(-2, \frac{7}{2})$

d) $16y = x^2 + 6x + 73$

$16y = (x+3)^2 + 64$

$(x+3)^2 = 16(y-4)$

Vertex at $(-3, 4)$

Focus at $(-3, 8)$

e) $8x = -y^2 + 4y - 12$

$(y-2)^2 + 8 = -8x$

$(y-2)^2 = -8(x+1)$

Vertex at $(2, -1)$

Focus at $(0, -1)$

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f) $26y = -x^2 - x + \frac{155}{4}$

$x^2 + x - \frac{155}{4} = -26y$

$\left(x + \frac{1}{2}\right)^2 - \frac{156}{4} = -26y$

$\left(x + \frac{1}{2}\right)^2 = -26y + 39$

$\left(x + \frac{1}{2}\right)^2 = -26\left(y - \frac{3}{2}\right)$

Vertex at $\left(-\frac{1}{2}, \frac{3}{2}\right)$

Focus at $\left(-\frac{1}{2}, -5\right)$

4)

a) Vertex at $(-1, 3)$, focus at (-1, -3)

$x_0 = -1, y_0 = 3, A = -6$

$(x + 1)^2 = -24(y - 3)$

$x^2 + 2x + 1 = -24y + 72$

$24y = -(x^2 + 2x - 71)$

b) Vertex at $\left(0, -\frac{1}{2}\right)$, focus at (0, 4)

$x_0 = 0, y_0 = -\frac{1}{2}, A = \frac{9}{8}$

$x^2 = \frac{36}{8}\left(y + \frac{1}{2}\right)$

$x^2 = \frac{36}{8}y + \frac{9}{4}$

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$18y = 4x^2 - 9$

c) Vertex at (3, -1), focus at (3, 5)

$x_0 = 3, y_0 = -1, A = 6$

$(x-3)^2 = 24(y+1)$

$x^2 - 6x + 9 = 24y + 24$

$24y = x^2 - 6x - 15$

d) Vertex at $\left(-\frac{3}{4}, \frac{1}{4}\right)$, focus at $\left(-\frac{3}{4}, 0\right)$

$x_0 = -\frac{3}{4}, y_0 = \frac{1}{4}, A = -1$

$\left(y - \frac{1}{4}\right)^2 = -\left(x + \frac{3}{4}\right)$

$-x - \frac{3}{4} = y^2 - \frac{1}{2}y + \frac{1}{16}$

$x = -y^2 + \frac{1}{2}y - \frac{13}{16}$

$16x = -16y^2 + 8y - 13$

e) Vertex at (0, 0), focus at (0, 1.5)

$x_0 = 0, y_0 = 0, A = 1.5$

$x^2 = 6y$

$y = \frac{x^2}{6}$

f) Vertex at (0, -1), focus at (2, -1)

$x_0 = 0, y_0 = -1, A = 2$

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$$8(x) = (y+1)^2$$

$$8x = y^2 + 2y + 1$$

5) Find the equation of the parabola that has

a) Vertex at (0, 0), directrix $y = 2$

$$x_0 = 0, y_0 = 0, A = -2$$

$$x^2 = -8y$$

$$y = -\frac{x^2}{8}$$

b) Vertex at (-1, 2), directrix $y = -3$

$$x_0 = -1, y_0 = 2, A = 3$$

$$(x+1)^2 = 12(y-2)$$

$$x^2 + 2x + 1 = 12y - 24$$

$$12y = x^2 + 2x + 25$$

c) Vertex at $\left(\frac{1}{2}, 1\right)$, directrix $y = 4$

$$x_0 = \frac{1}{2}, y_0 = 1, A = -4$$

$$\left(x - \frac{1}{2}\right)^2 = -16(y-1)$$

$$x^2 - x + \frac{1}{4} = -16y + 16$$

$$16y = -x^2 + x + \frac{63}{4}$$

d) Vertex at (1, 1) directrix $x = 2$

$$x_0 = 1, y_0 = 1, A = -2$$

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$-8(x-1)=(y-1)^2$

$-8x+8=y^2-2y+1$

$8x=-y^2+2y+7$

e) Vertex at $\left(\frac{3}{4},-\frac{1}{2}\right)$, directrix $y=3$

$x_0=\frac{3}{4},y_o=-\frac{1}{2},A=-3$

$\left(x-\frac{3}{4}\right)^2=-12\left(y+\frac{1}{2}\right)$

$x^2-\frac{3}{2}x+\frac{9}{16}=-12y-6$

$12y=-x^2+\frac{3}{2}x-\frac{105}{16}$

f) Vertex at (3, 2), directrix $x=0$

$x_0=3,y_0=2,A=-3$

$-12(x-3)=(y-2)^2$

$-12x+36=y^2-4y+4$

$12x=-y^2+4y+32$

6) Find the equation of the parabola that has

a) Focus at (0, 0), directrix $y=-2$

Vertex $=(-1,0)$

$x_0=-1,y_0=0,A=2$

$(x+1)^2=8y$

$8y=x^2+2x+1$

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Note to check answer, substitute vertex not focus back into equation

b) Focus at (2, -2), directrix $y = \frac{3}{2}$

Vertex $= \left(2, -\frac{1}{4}\right)$

$x_0 = 2, y_0 = -\frac{1}{4}, A = -\frac{3}{2}$

$(x-2)^2 = -6\left(y + \frac{1}{4}\right)$

$x^2 - 4x + 4 = -6y - \frac{3}{2}$

$6y = -x^2 + 4x - \frac{11}{2}$

c) Focus at $\left(\frac{1}{2}, -\frac{1}{2}\right)$, directrix $y = \frac{7}{2}$

Vertex $= \left(\frac{1}{2}, \frac{3}{2}\right)$

$x_0 = \frac{1}{2}, y_0 = \frac{3}{2}, A = -\frac{7}{2}$

$\left(x - \frac{1}{2}\right)^2 = -14\left(y - \frac{3}{2}\right)$

$x^2 - x + \frac{1}{4} = -14y + 21$

$14y = -x^2 + x + \frac{83}{4}$

d) Focus at (1, 1), directrix $x = 3$

Vertex $= (2,1)$

$x_0 = 2, y_0 = 1, A = -3$

$-12(x-2) = (y-1)^2$

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$-12x + 24 = y^2 - 2y = 1$

$12x = -y^2 + 2y + 23$

e) Focus at $\left(2, \frac{3}{4}\right)$, directrix $y = -\frac{7}{4}$

Vertex $= \left(2, -\frac{1}{2}\right)$

$x_0 = 2, y_0 = -\frac{1}{2}, A = \frac{7}{4}$

$(x-2)^2 = \frac{7}{4}\left(y + \frac{1}{2}\right)$

$x^2 - 4x + 4 = \frac{7}{4}y + \frac{7}{8}$

$4x^2 - 16x + 16 = 7y + \frac{7}{2}$

$7y = 4x^2 - 16x + \frac{25}{2}$

f) Focus at (-2, 3), directrix $x = -5$

Vertex $= \left(-\frac{7}{2}, 3\right)$

$x_0 = -\frac{7}{2}, y_0 = 3, A = 5$

$20\left(x + \frac{7}{2}\right) = (y-3)^2$

$20x + 70 = y^2 - 6y + 9$

$20x = y^2 - 6y - 61$

7) Sketch the following curves, showing centre and radius for circles; and focus, directrix and vertex for parabolas

a) $x^2 + y^2 = 16$

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Centre= $(0,0)$, radius $= 4$

b) $y = 4x^2 - 2x + 8$

$\frac{1}{4}y = x^2 - \frac{1}{2}x + 2$

$\frac{1}{4}y = \left(x - \frac{1}{4}\right)^2 + \frac{31}{16}$

$\frac{1}{4}\left(y - \frac{31}{4}\right) = \left(x - \frac{1}{4}\right)^2$

$x_0 = \frac{1}{4}, y_0 = \frac{31}{4}, A = \frac{1}{16}$

Vertex= $\left(\frac{1}{4}, \frac{31}{4}\right)$, Focus= $\left(\frac{5}{16}, \frac{31}{4}\right)$, Directrix is $x = \frac{3}{16}$

c) $x^2 + y^2 + 4x - 6y = 10$

$(x+2)^2 + (y-3)^2 = 15$

Centre= $(-2,3)$, radius $= \sqrt{15}$

d) $y = 3x^2 - 6x + 3$

$\frac{y}{3} = x^2 - 2x + 1$

$\frac{y}{3} = (x-1)^2$

$x_0 = 1, y_0 = 0, A = \frac{4}{3}$

Vertex= $(1,0)$, Focus= $\left(1, \frac{4}{3}\right)$, Directrix is $y = -\frac{4}{3}$

e) $y^2 + 2 = 2x - 4 + 6y$

$y^2 - 6y + 6 = 2x$

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$(y-3)^2-3=2x$

$2\left(x+\frac{3}{2}\right)=(y-3)^2$

$x_0=-\frac{3}{2}, y_0=3, A=\frac{1}{2}$

Vertex$=\left(-\frac{3}{2},3\right)$, Focus$=(1,3)$, Directrix is $x=-2$

f) $y^2+8y-4x=-x^2-2$

$y^2+8y+x^2-4x+2=0$

$(y+4)^2+(x-2)^2-10=0$

Centre$(2,-4)$, radius $\sqrt{10}$

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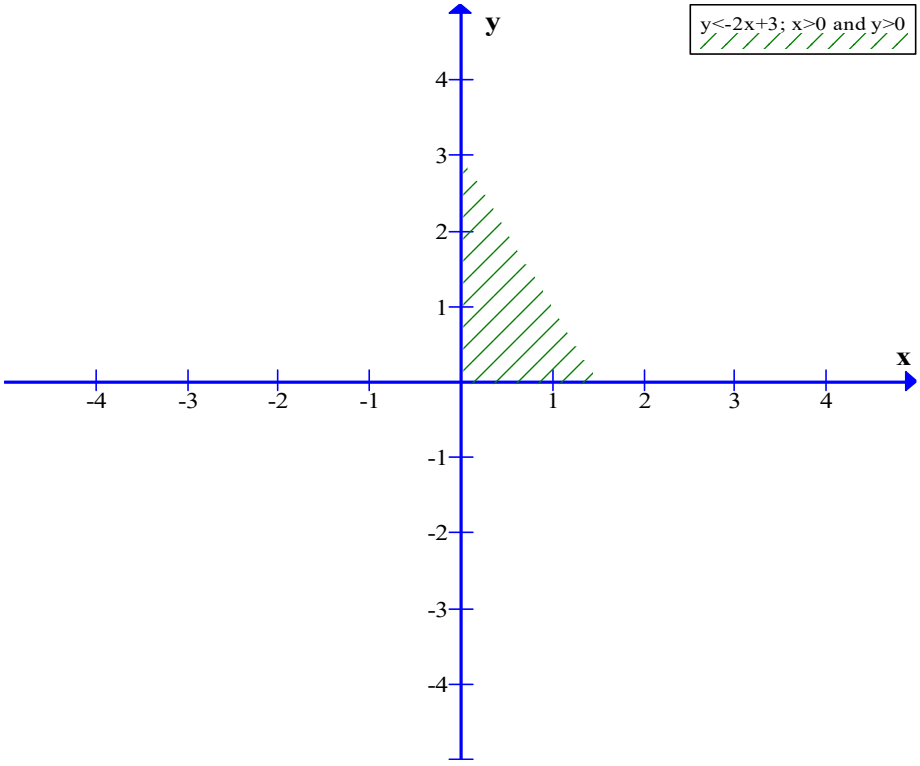
Exercise 4

Graphing Inequalities

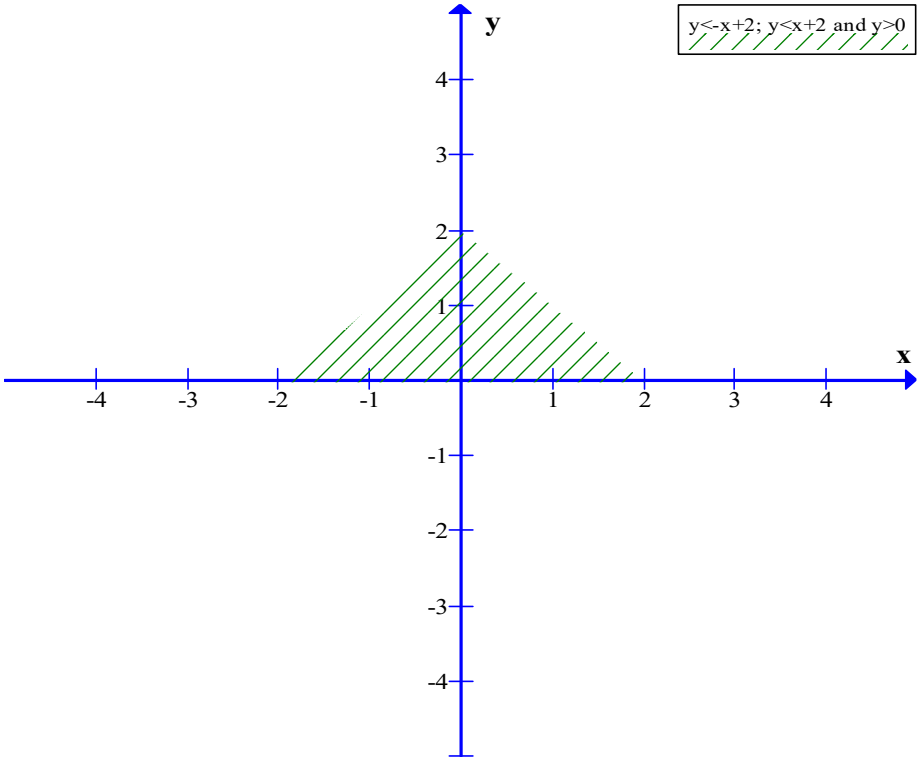
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1)

a) The x axis, the y axis, and the inequality $y < -2x + 3$

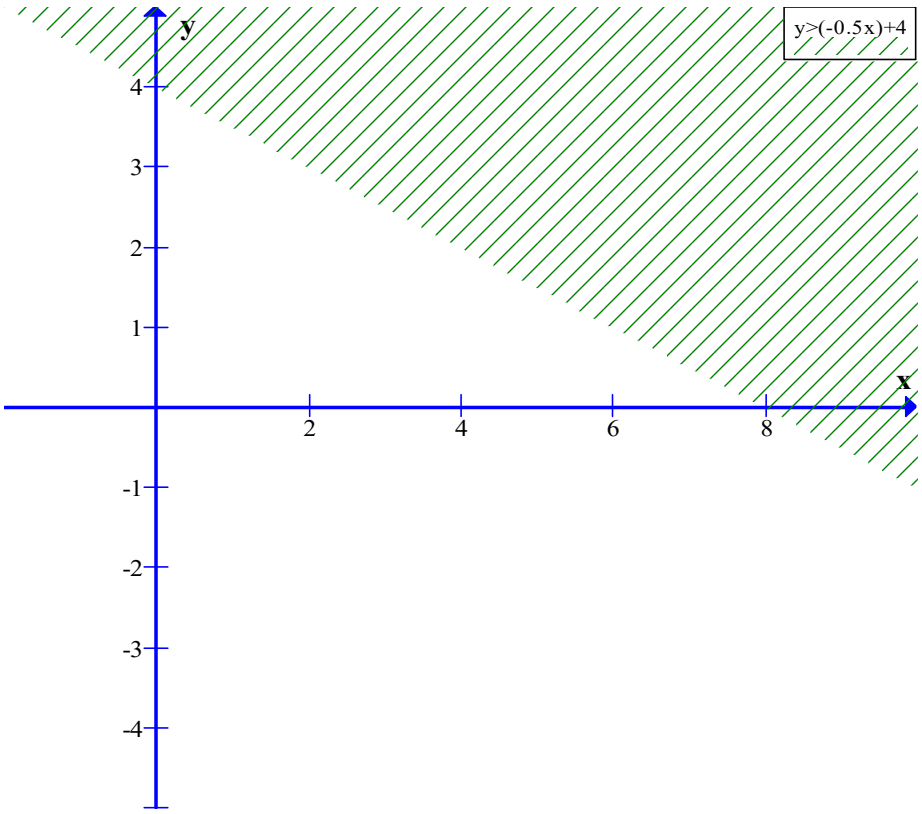


b) The x axis and the inequalities $y < -x + 2$ and $y < x + 2$

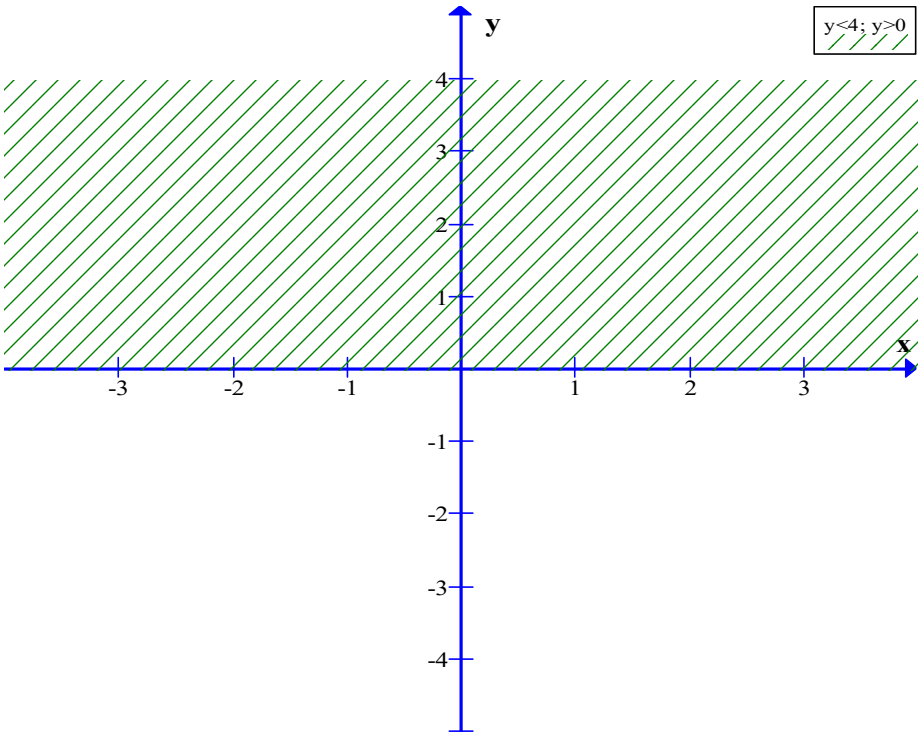


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c) The inequality $y > -\frac{1}{2}x + 4$

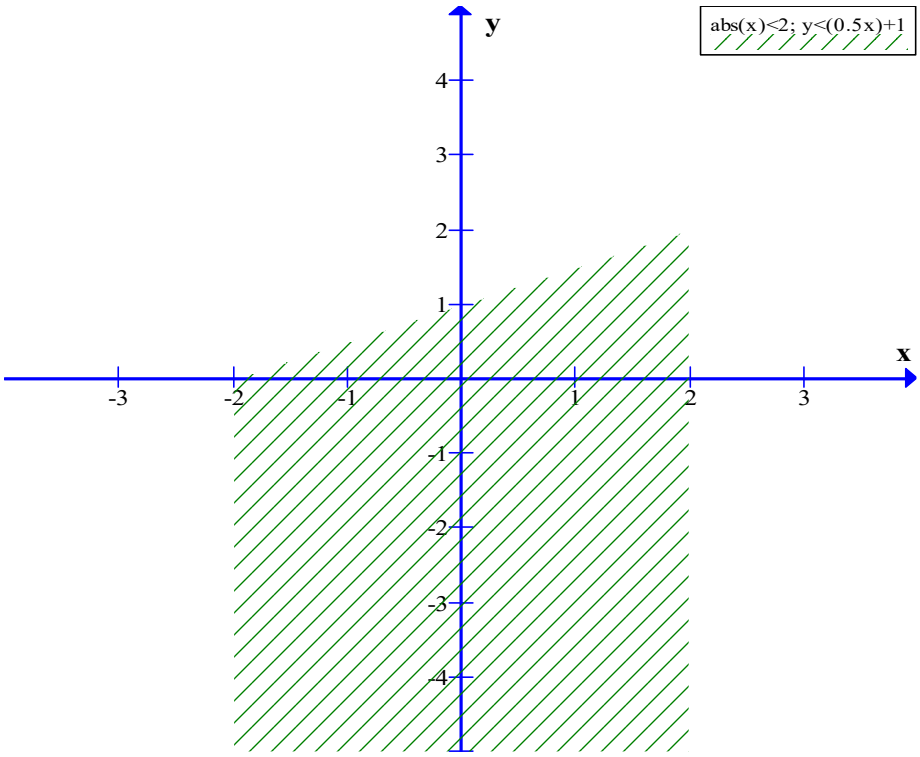


d) The inequalities $y < 4$ and $y > 0$

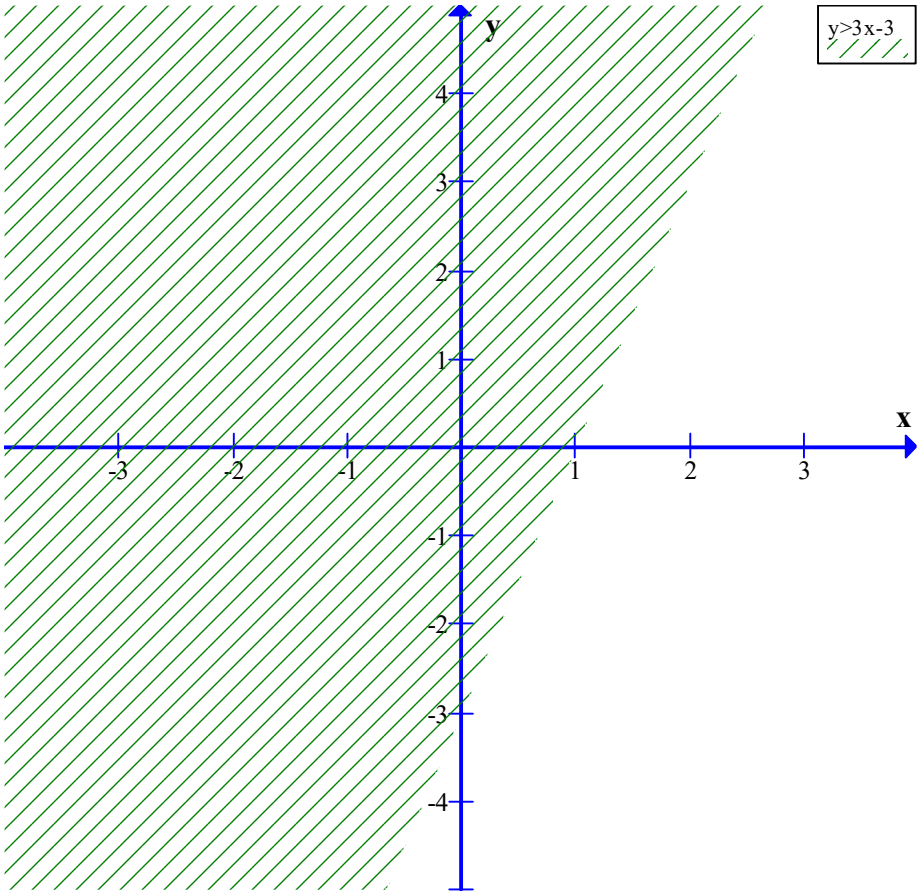


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e) The inequalities $|x| < 2$ and $y < \frac{1}{2}x + 1$



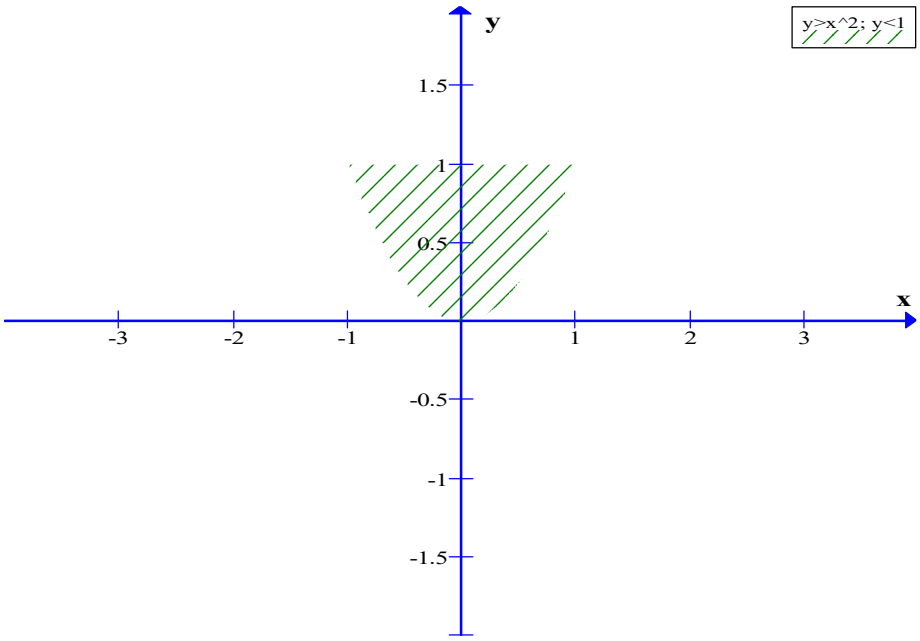
f) The inequality $y > 3x - 3$



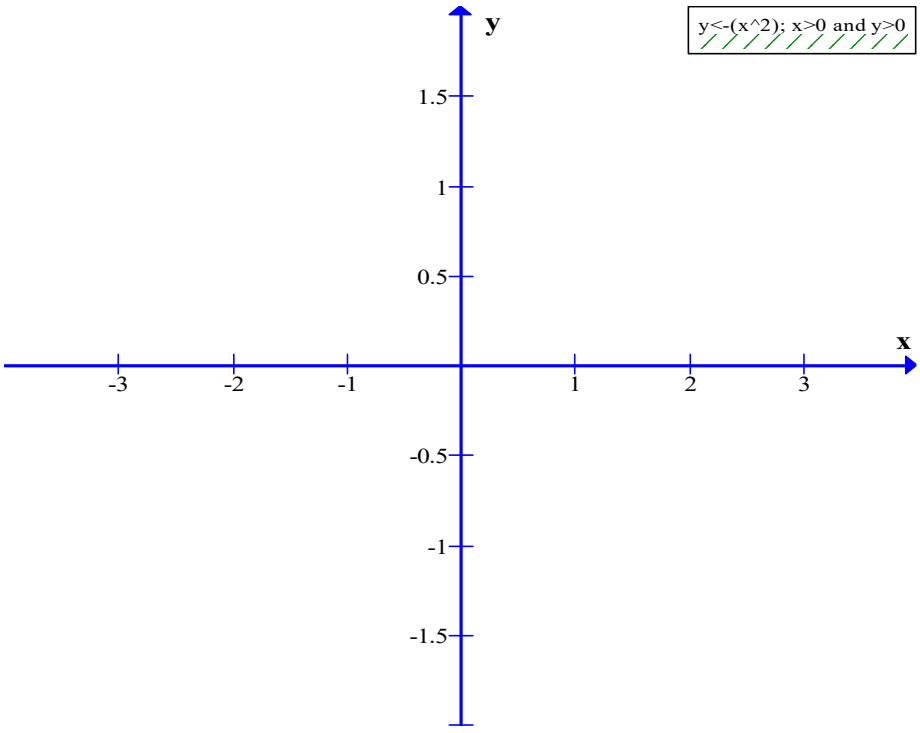
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2) Sketch and label the region bounded by

a) The inequalities $y > x^2$ and $y < 1$



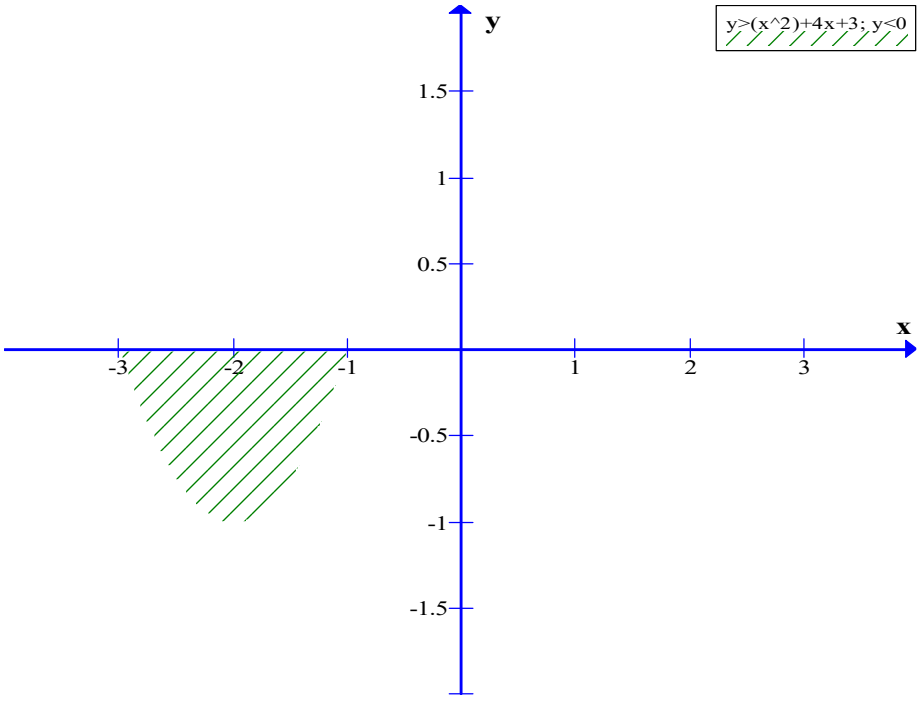
b) The inequalities $x > 0$, $y > 0$, and $y < -x^2$



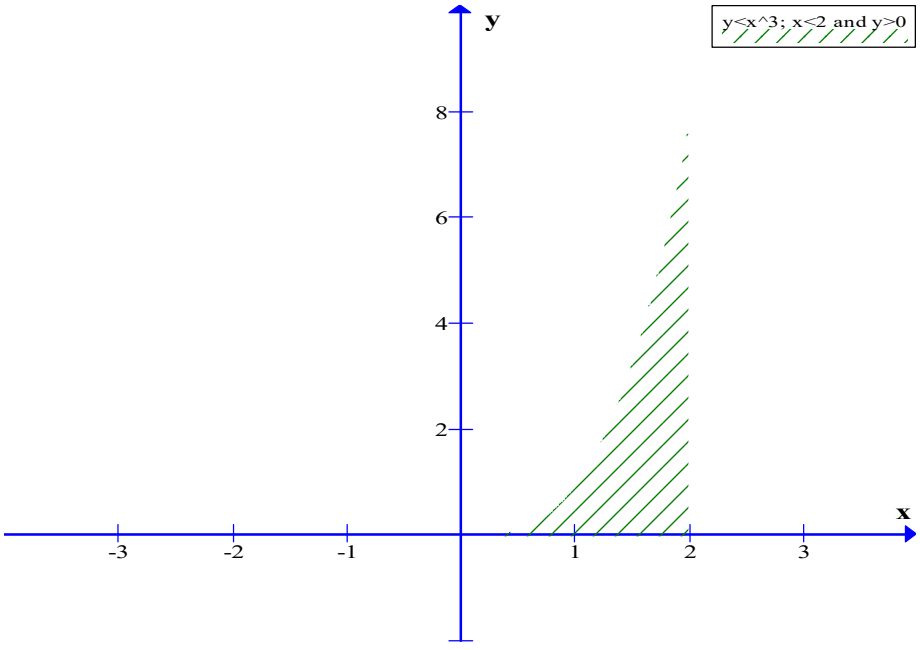
NO SOLUTION

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c) The inequalities $y < 0$ and $y > x^2 + 4x + 3$

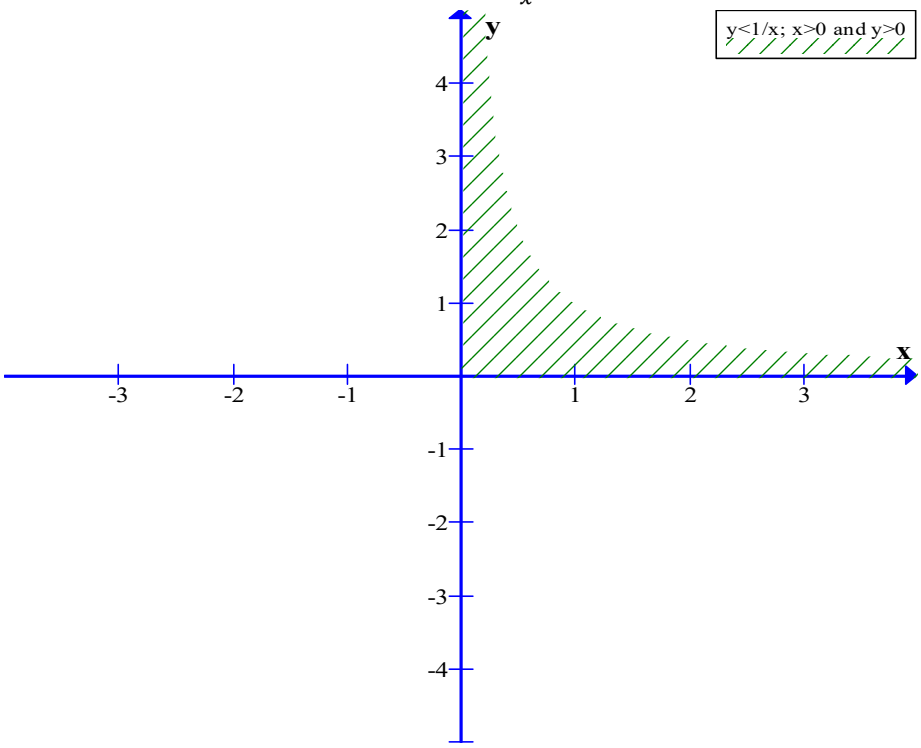


d) The x axis, and the inequalities $y < x^3$ and $x < 2$

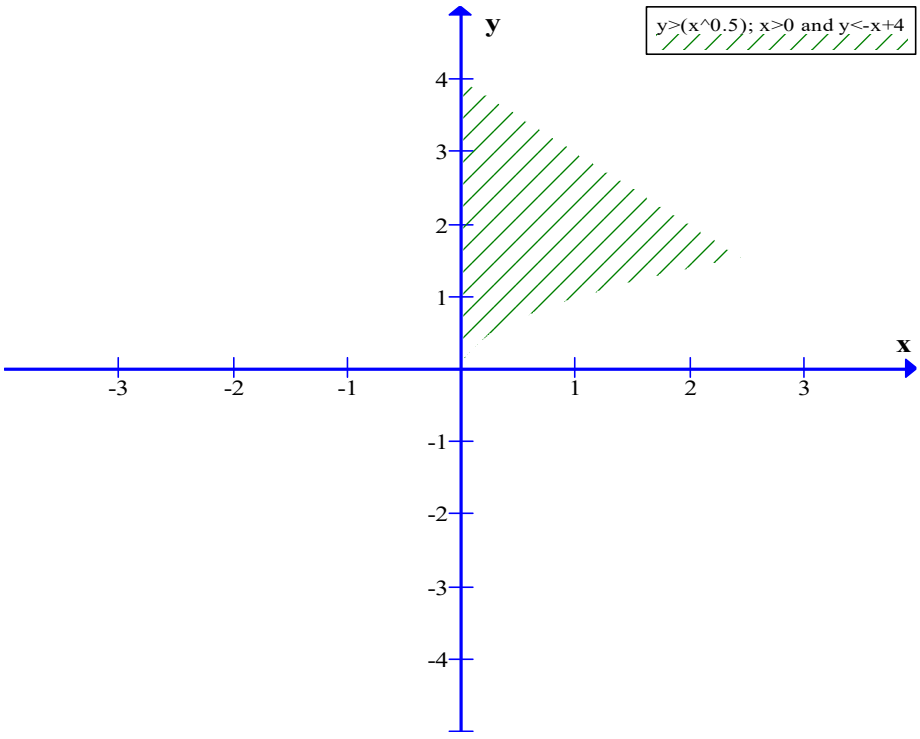


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e) The inequalities $x > 0$, $y > 0$, and $y < \frac{1}{x}$



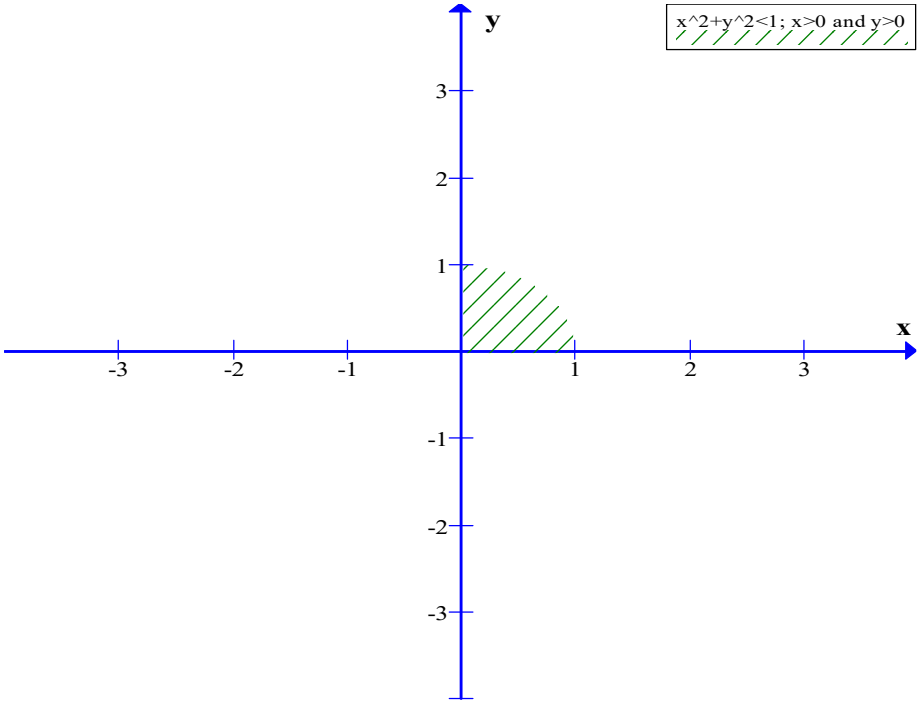
f) The inequalities $x > 0$, $y > \sqrt{x}$, and $y < -x + 4$



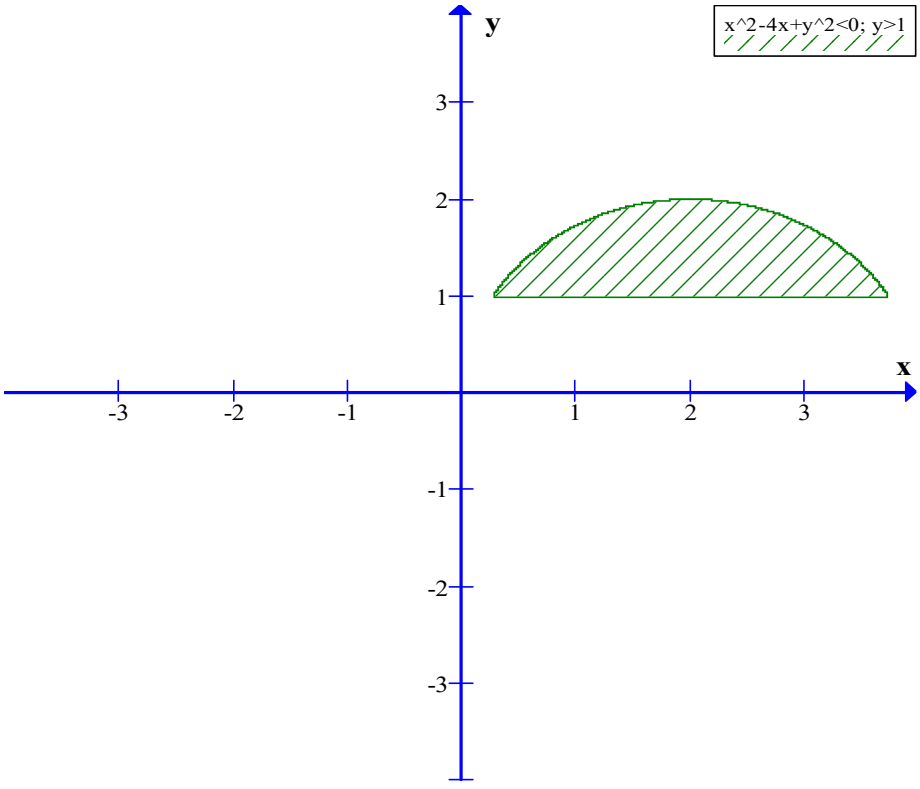
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3) Sketch and label the region bounded by

a) The inequalities $x^2 + y^2 < 1$, $x > 0$, and $y > 0$

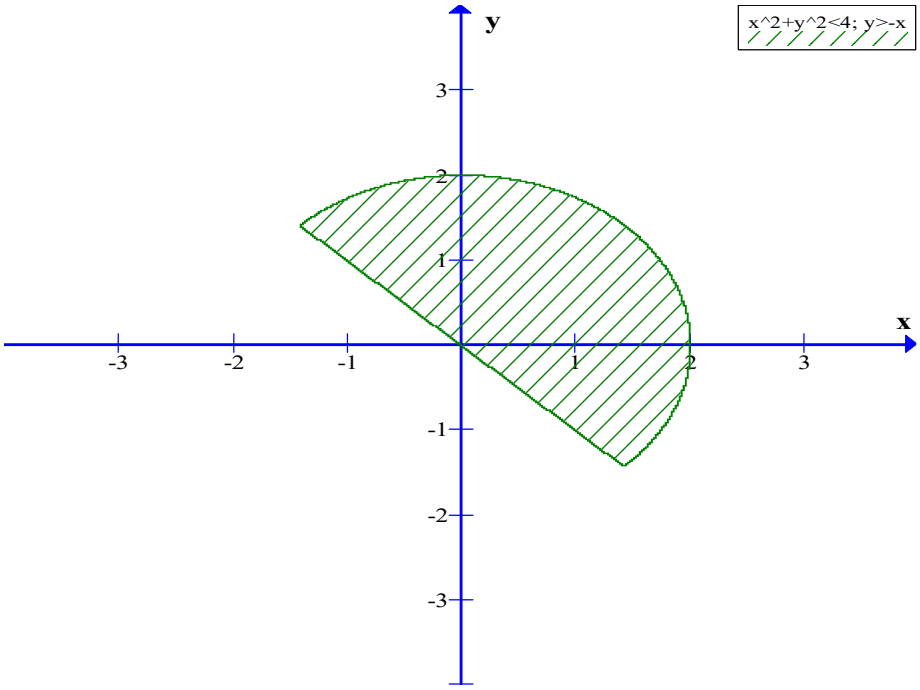


b) The inequalities $x^2 - 4x + y^2 < 0$ and $y > 1$

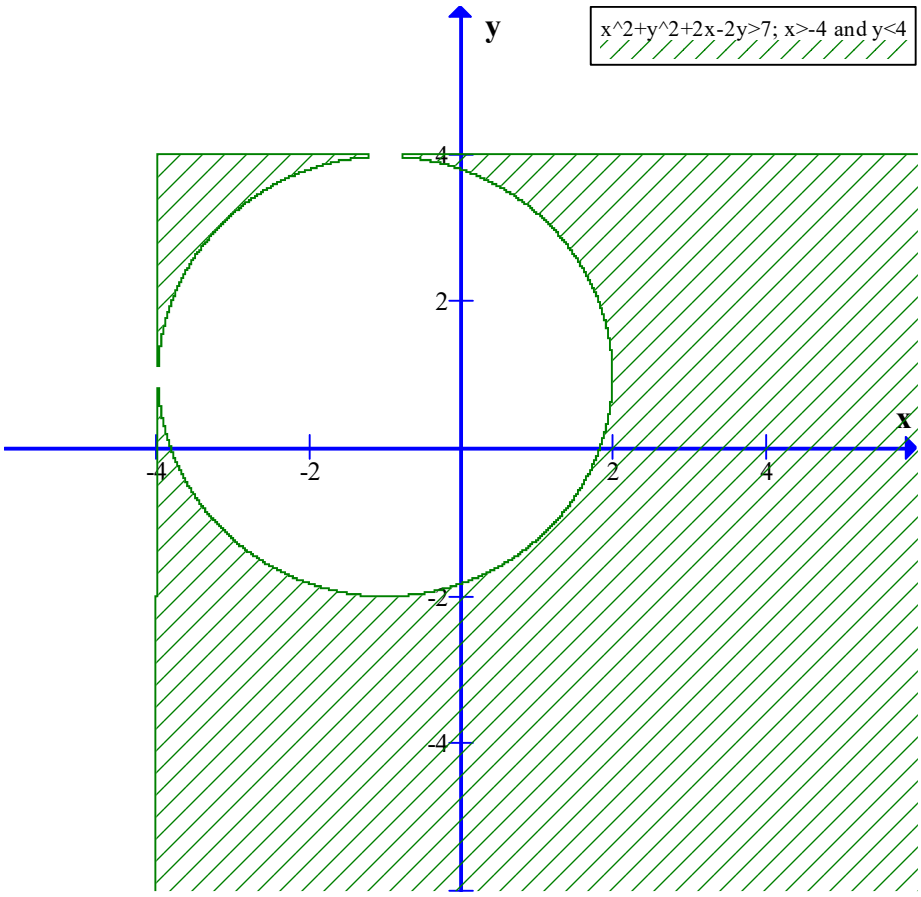


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c) The inequalities $x^2 + y^2 < 4$ and $y > -x$



d) The inequalities $x^2 + y^2 + 2x - 2y > 7$, $x > -4$, and $y < 4$



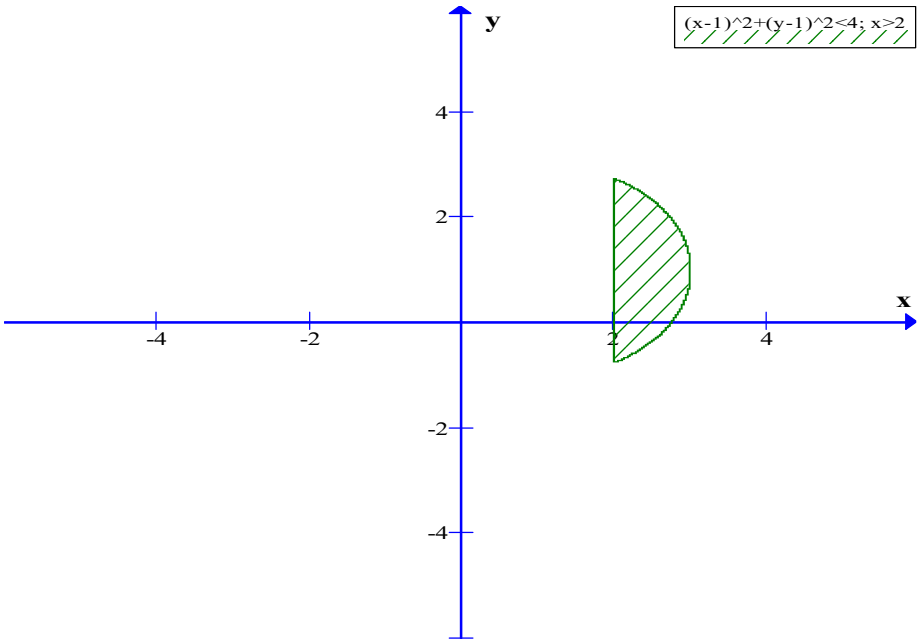
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4) Find a system of inequalities whose solutions correspond to the regions described; sketch the regions

a) The points lying inside the circle with centre (1, 1) and radius 2, but to the right of the line $x = 2$

$(x-1)^2 + (y-1)^2 < 4$

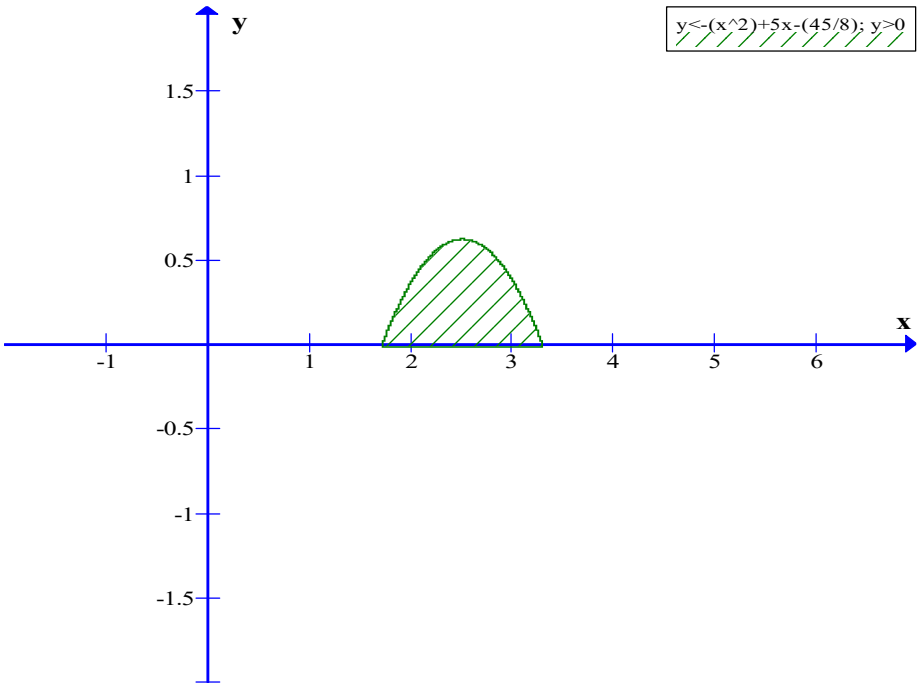
$x > 2$



b) The points whose boundary consists of portions of the x axis, the ordinates at $x = 2$, $x = 3$, and the curve having its turning point at $\left(\frac{5}{2}, 4\right)$, which is also its maximum

Equation is $y = -x^2 + 5x - \frac{45}{8}$

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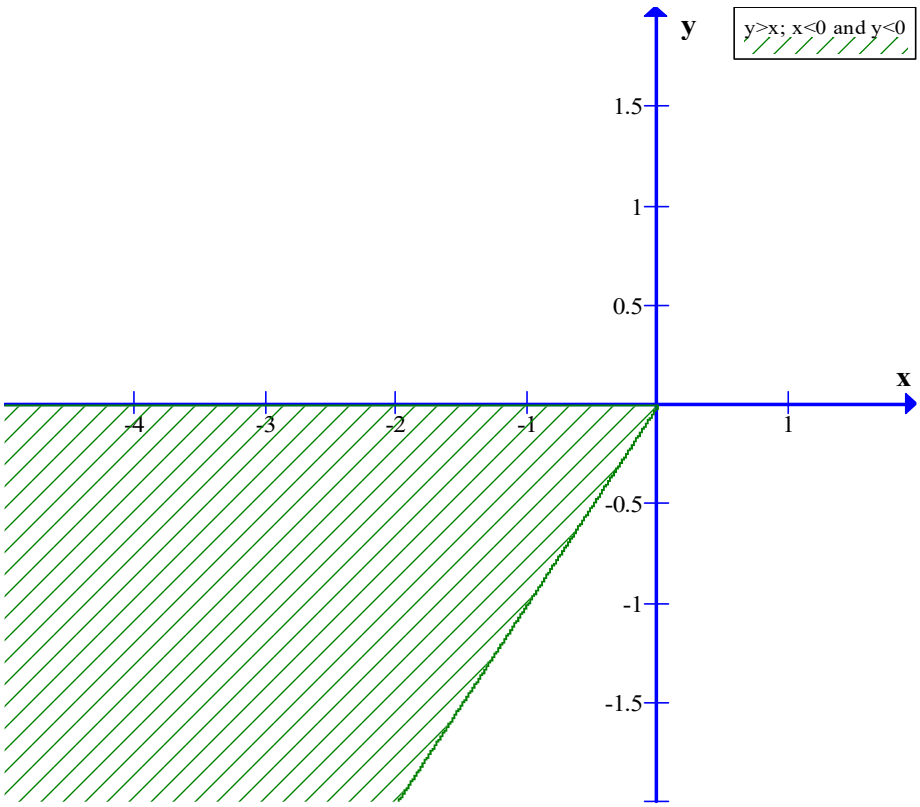
c) The points where y is greater than x and both x and y are negative

$y > x$

$x < 0$

$y < 0$

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d) The triangle bounded by the points (0, 2), (1, 1) and the origin

Equation 1; between points (0, 2) and (1, 1)

Gradient = $\frac{2-1}{0-1} = -1$

$y = -x + c$

Substituting either point gives $c = 2$

$y = -x + 2$

Equation 2; between points (0, 2) and (0, 0)

$x = 0$

Equation 3; between points (0, 0) and (1,1)

$y = x$

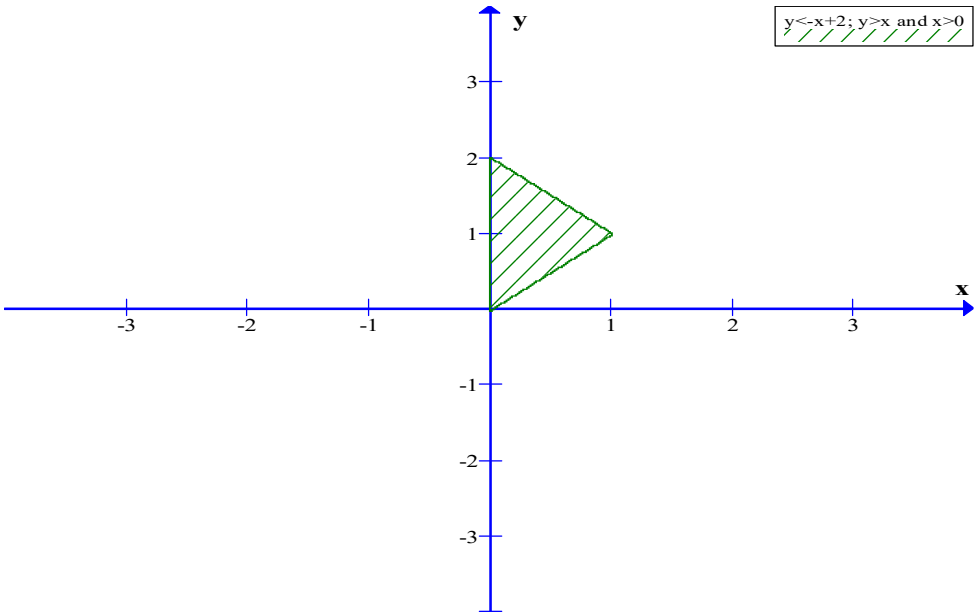
Inequalities are:

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$x > 0$

$y > x$

$y < -x + 2$

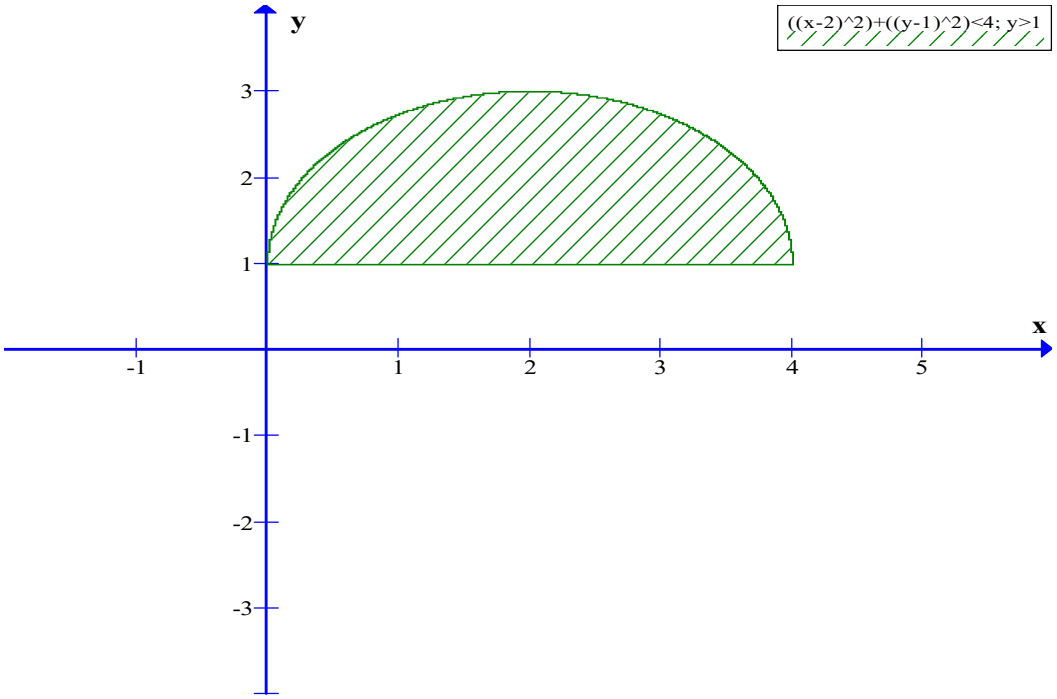


e) The region inside the circle of radius 2, centred at (2, 1) and the points for which y is greater than 1. Describe the shape formed

$(x-2)^2 + (y-1)^2 < 4$

$y > 1$

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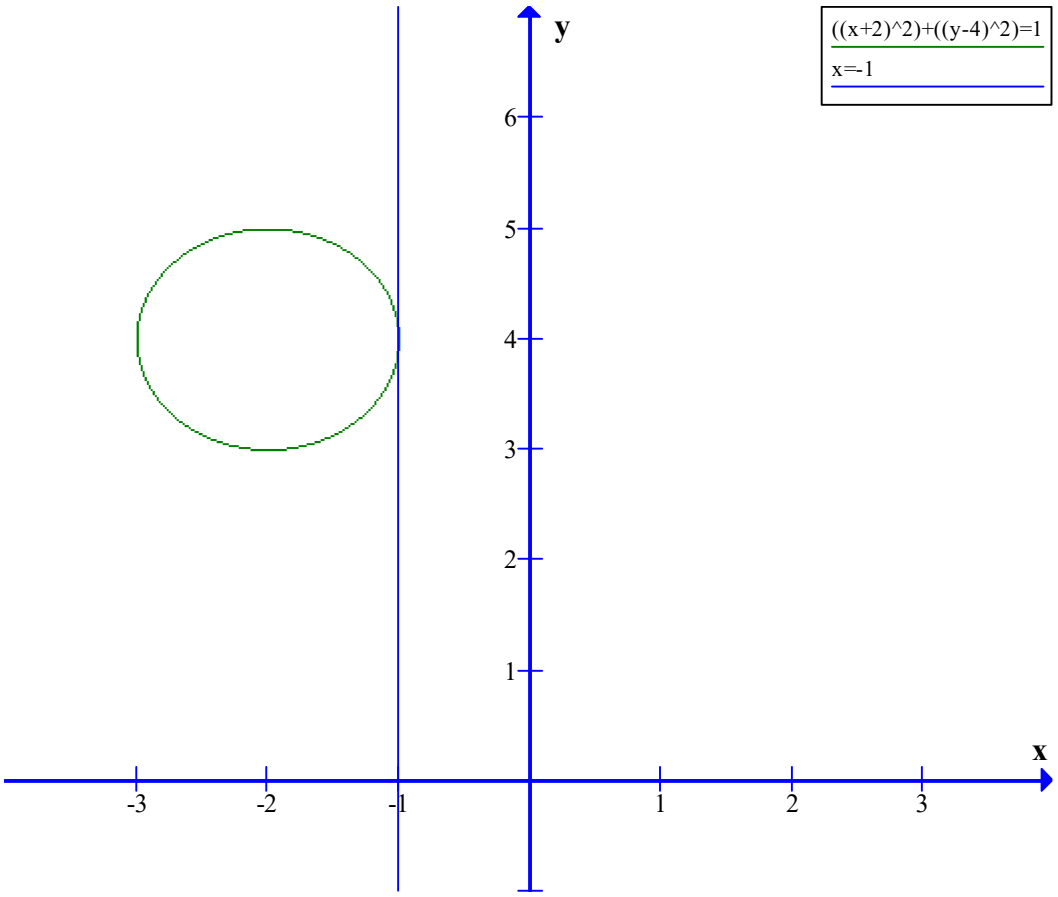


f) The region inside the circle of centre (-2, 4) with radius 1, and the points for which x is greater than -1

$(x+2)^2 + (y-4)^2 < 1$

$x > -1$

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NO SOLUTION

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Year 11 Unit 2 Mathematics

Basic Trigonometry

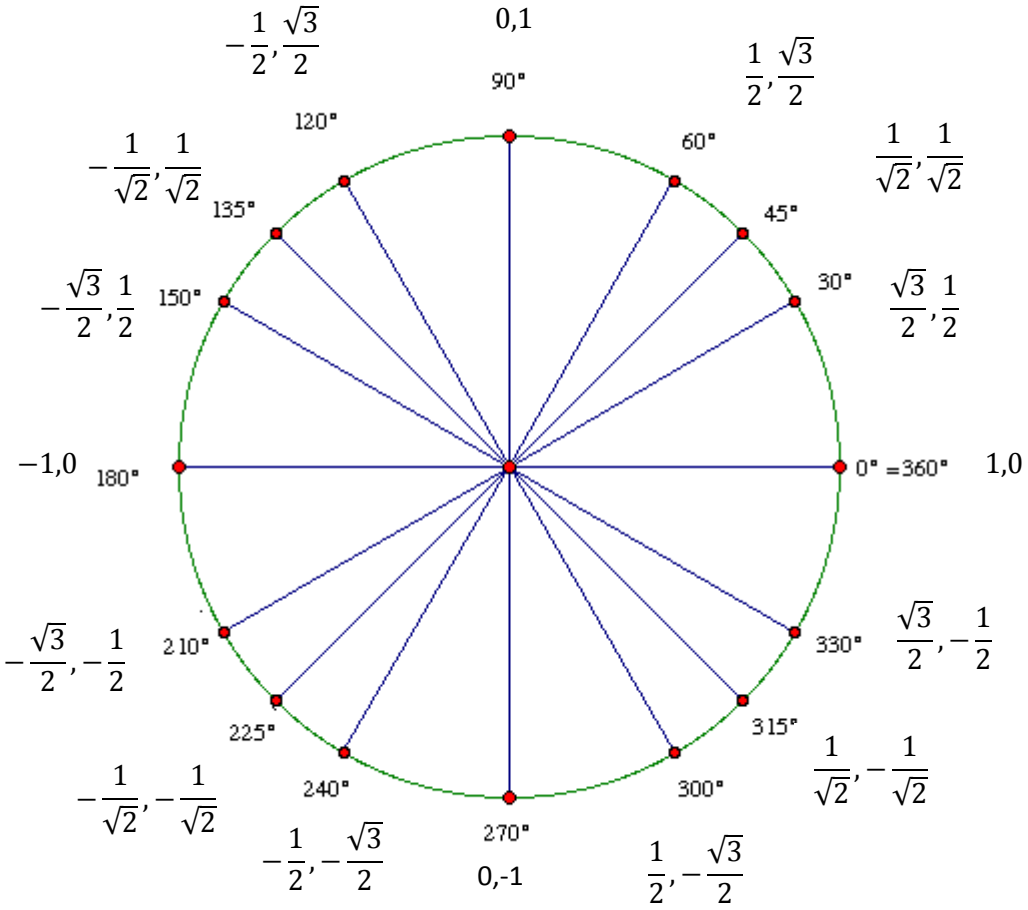
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Exercise 1

Trigonometric Ratios and Identities

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1)



2)

a) $\tan x = \frac{\sin x}{\cos x}$

b) $\csc x = \frac{1}{\sin x}$

c) $\sec x = \frac{1}{\cos x}$

d) $\cot x = \frac{\cos x}{\sin x}$

3) For what values of θ are the above trigonometric ratios not defined?

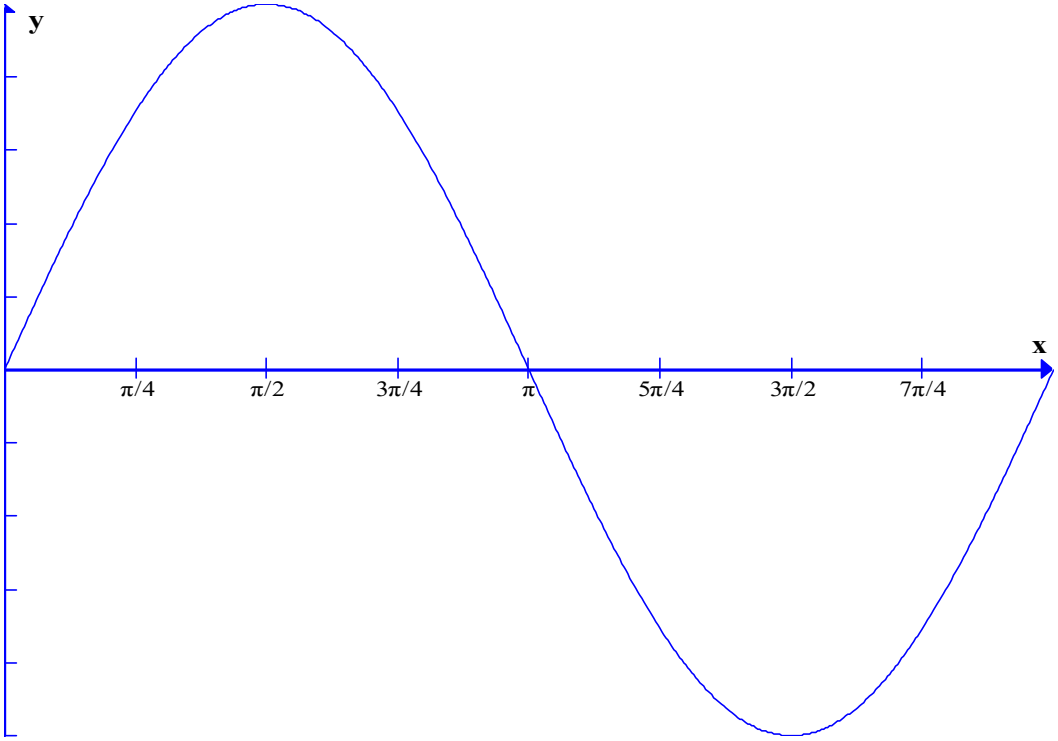
Tan and sec are not defined for θ equal to 90° ± 180k where k is an integer

Cosec and cot are not defined for θ equal to 0° ± 180k where k is an integer

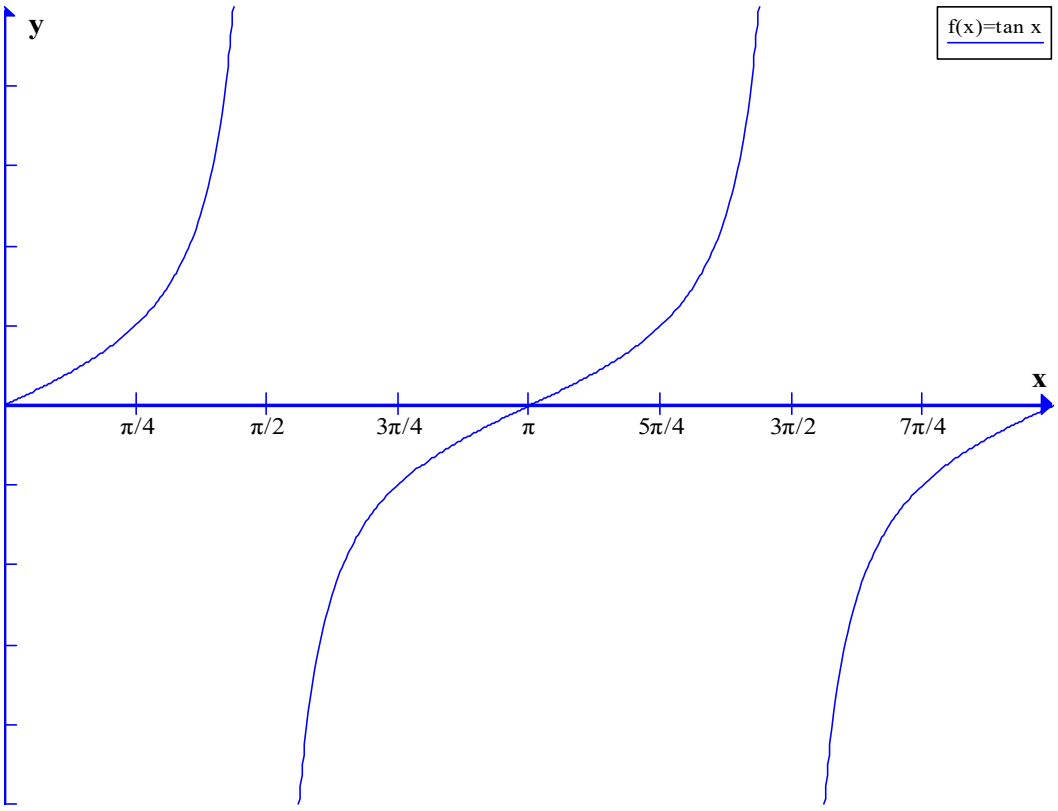
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4) Graph the following

a) $\sin x$ for x between 0 and 360°

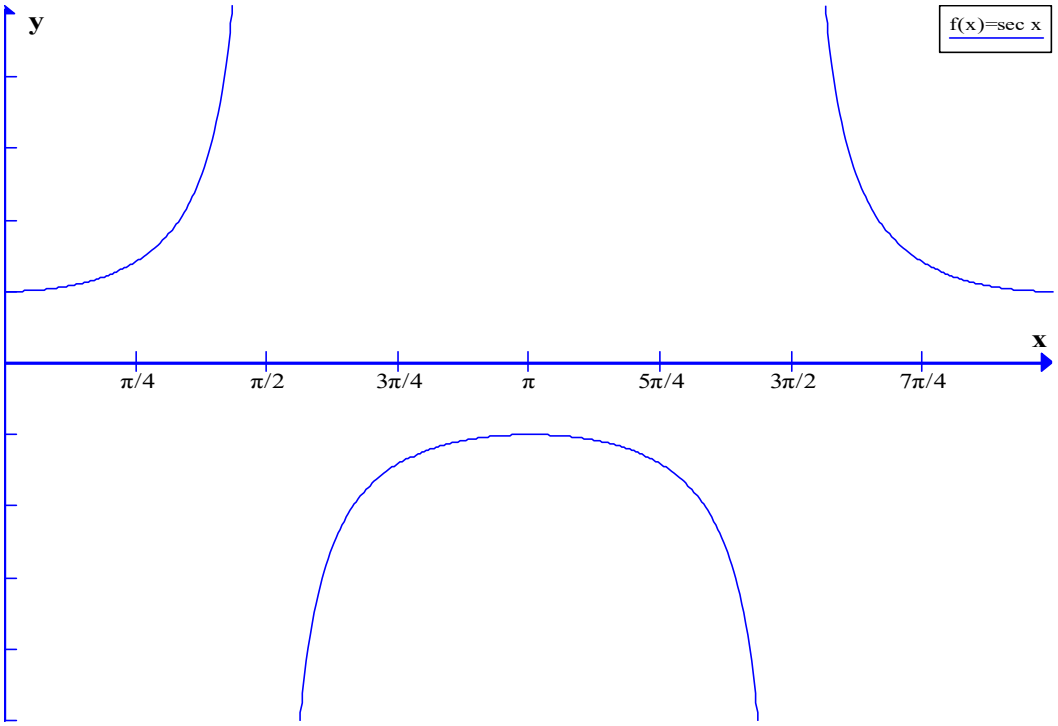


b) $\tan x$ for x between 0 and 360°



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c) $\sec x$ for x between 0 and 360°



5)

a) $\sin(-\theta) = -\sin\theta$

b) $\cos(90° - \theta) = \sin\theta$

c) $\tan(180° + \theta) = \tan\theta$

d) $\csc(90° - \theta) = \sec\theta = \frac{1}{\cos\theta}$

e) $\sec(-\theta) = \sec\theta$

6)

a) $\sin^2\theta + \cos^2\theta = 1$

b) $1 + \tan^2\theta = \sec^2\theta$

c) $1 + \cot^2\theta = \csc^2\theta$

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d) $\sin(2\theta) = 2\sin\theta\cos\theta$

e) $\cos(2\theta) = \cos^2\theta - \sin^2\theta$ or $1 - 2\sin^2\theta$

7)

a) $4\cos x = 1 + 2\cos x$, for $0° \le x \le 90°$

$2\cos x = 1$

$\cos x = \frac{1}{2}$

$x = 60°$

b) $\csc^2 x = 2$, for $0° \le x \le 180°$

$\frac{1}{\sin^2 x} = 2$

$\sin x = \pm\frac{1}{2}$

$x = 45°, 135°$

c) $4\sin x = 1 + 2\sin x$, for $-90° \le x \le 90°$

$2\sin x = 1$

$\sin x = \frac{1}{2}$

$x = 30°$

d) $\cot x = 2\cos x$, for $-180° \le x \le 180°$

$\frac{\cos x}{\sin x} = 2\cos x$

$\sin x = \frac{\cos x}{2\cos x}$

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$\sin x = \frac{1}{2}$

$x = 30°, 150°$

e) $10 \cos 2x = 4 \cos 60°$, for $0° \leq x \leq 360°$

$\cos 2x = \frac{2}{5} \cos 60°$

$\cos 2x = \frac{1}{5}$

$2x = 78.46°$

$x = 39.23°, 320.67°$

f) $\cot^2 x = \csc^2 x$, for $0° \leq x \leq 90°$

$\frac{\cos^2 x}{\sin^2 x} = \frac{1}{\sin^2 x}$

$\cos^2 x = 1$

$x = 0°$

g) $2 \sin x - \sin 30° = \cos 0°$, for $90° \leq x \leq 180°$

$2 \sin x - \frac{1}{2} = 1$

$2 \sin x = \frac{3}{2}$

$\sin x = \frac{3}{4}$

$x = 48.6°$

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8)

a) $\cos 30° \tan 30° = \cos 30° \frac{\sin 30°}{\cos 30°} = \sin 30° = \frac{1}{2}$

b) $\sec 45° - \sin 45° = \frac{1}{\cos 45°} - \sin 45° = \sqrt{2} - \frac{1}{\sqrt{2}} = \frac{2-1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$

c) $\csc 60° \sec 30° = \frac{1}{\sin 60°} \times \frac{1}{\cos 30°} = \frac{2}{\sqrt{3}} \times \frac{2}{\sqrt{3}} = \frac{4}{3}$

d) $\frac{\sec 30° \tan 60°}{\csc^2 45°} = \frac{\frac{1}{\cos 30°} \times \tan 60°}{1+\cot^2 45°} = \frac{\frac{2}{\sqrt{3}} \times \frac{1}{\sqrt{3}}}{2} = \frac{1}{3}$

e) $(\tan 30° + \csc 60°) \cos 30° = \left(\sqrt{3} + \frac{1}{\sin 60°}\right) \frac{\sqrt{3}}{2} = \left(\sqrt{3} + \frac{2}{\sqrt{3}}\right) \frac{\sqrt{3}}{2} = \frac{3}{2} + 1 = \frac{5}{2}$

f) $\sin^2 27° + \left(\frac{1}{\sec 27°} \times \cos 27°\right) = \sin^2 27° + (\cos 27° \times \cos 27°)$

$= \sin^2 27° + \cos^2 27° = 1$

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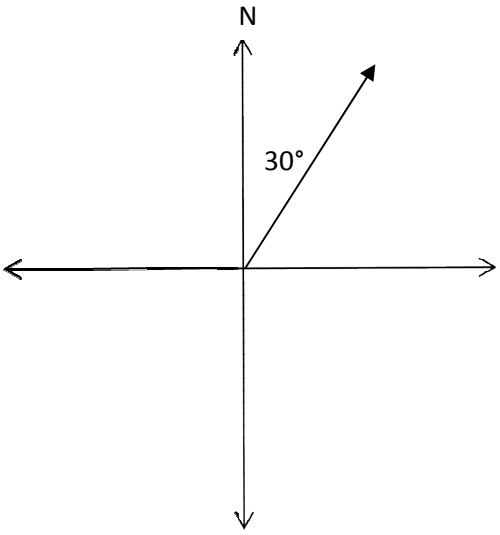
Exercise 2

Angles of Elevation & Bearings

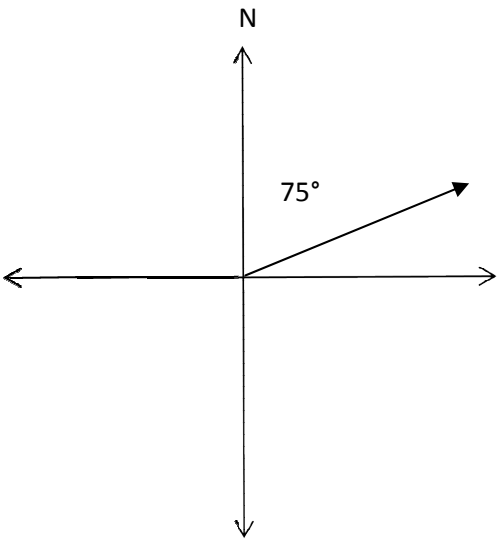
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1)

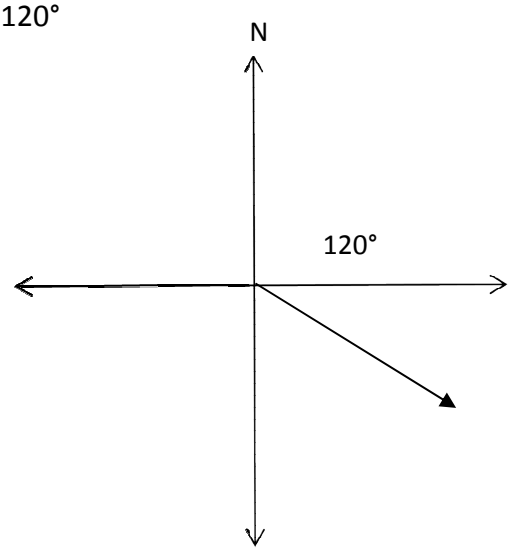
a) 030°



b) 075°

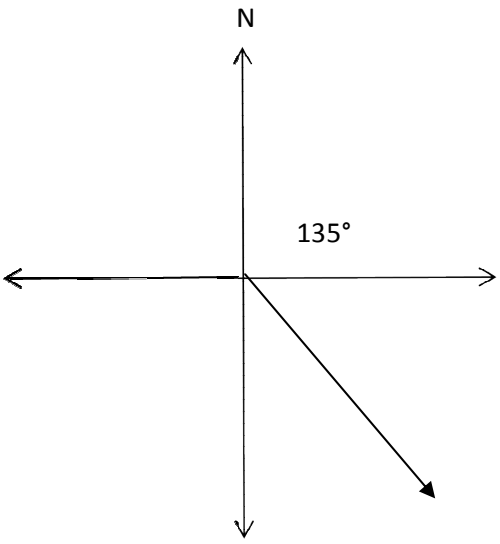


c) 120°

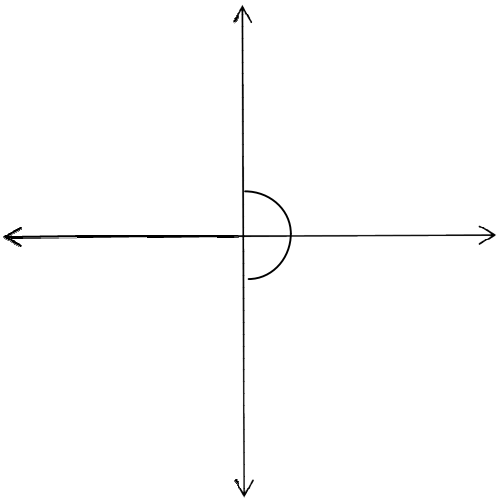


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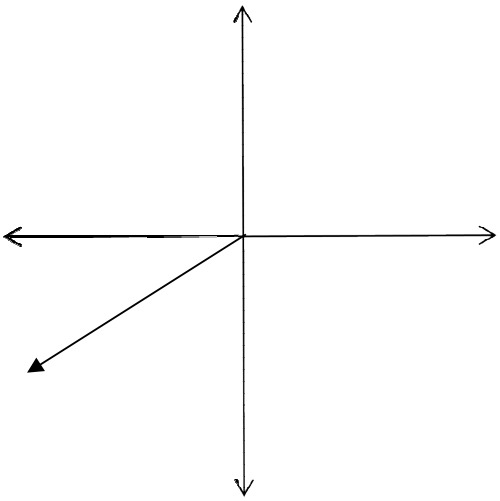
d) 135°



e) 180°

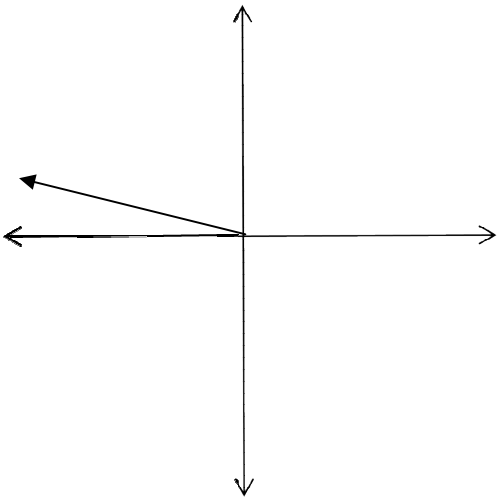


f) 240°

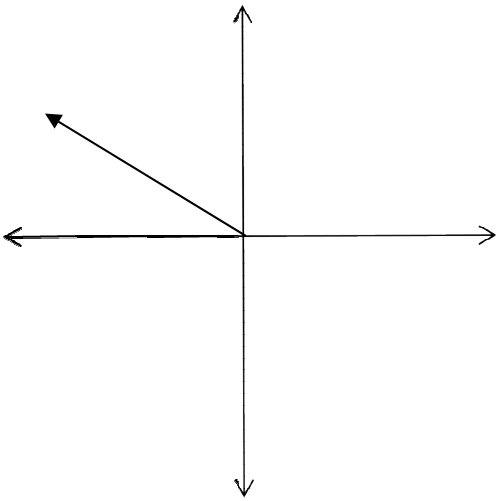


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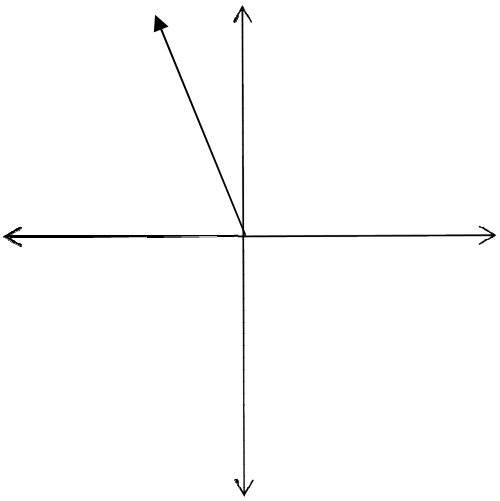
g) 280°



h) 300°



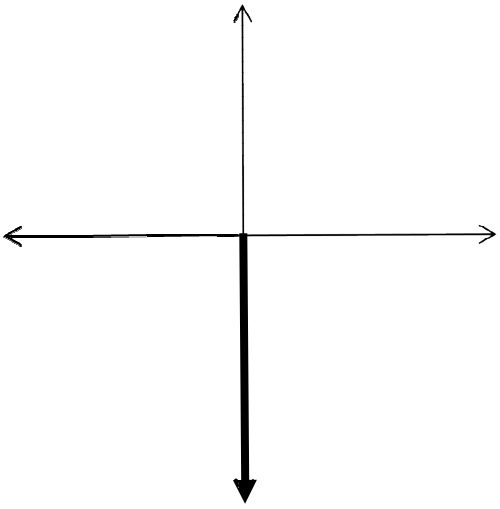
i) 345°



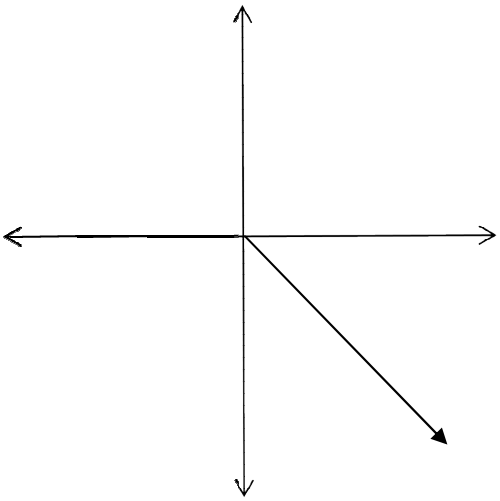
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2)

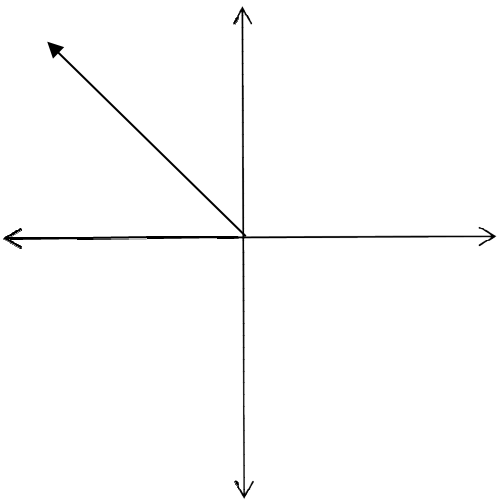
a) Due South



b) South-East

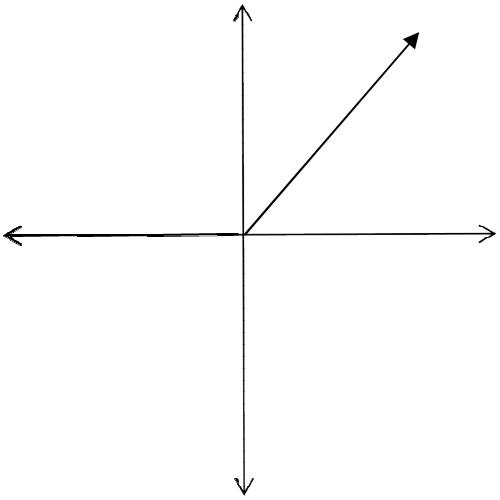


c) North-West

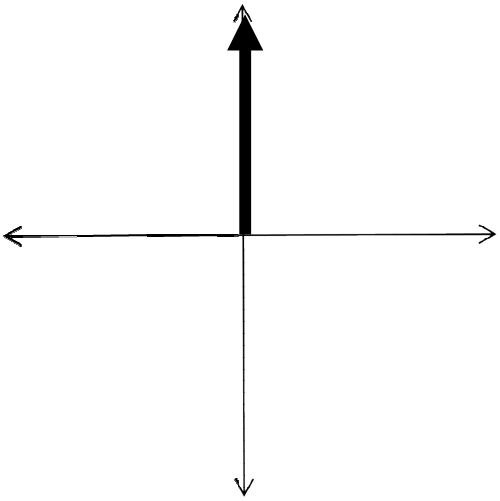


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d) North-East



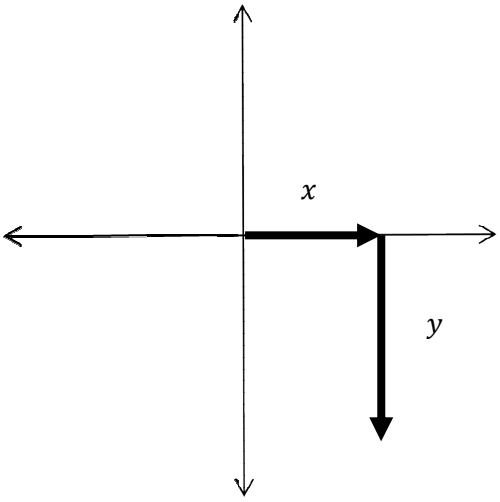
e) Due North



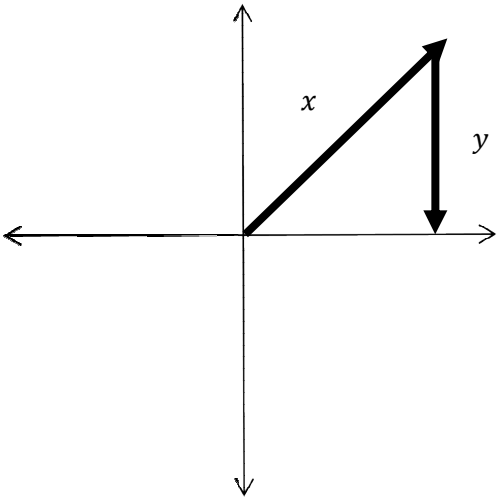
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3)

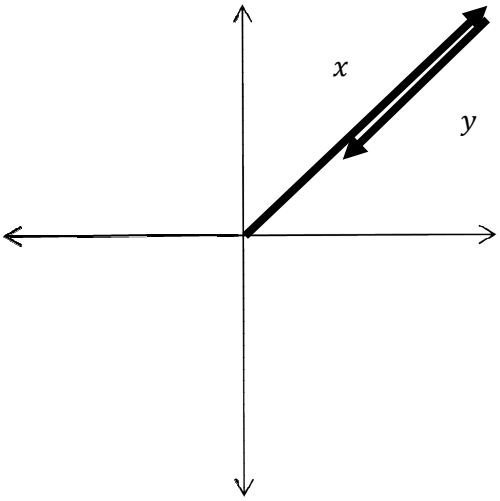
a) A man travels due East for x km then due South for y km



b) A man travels North-East for x km, then due South for y km

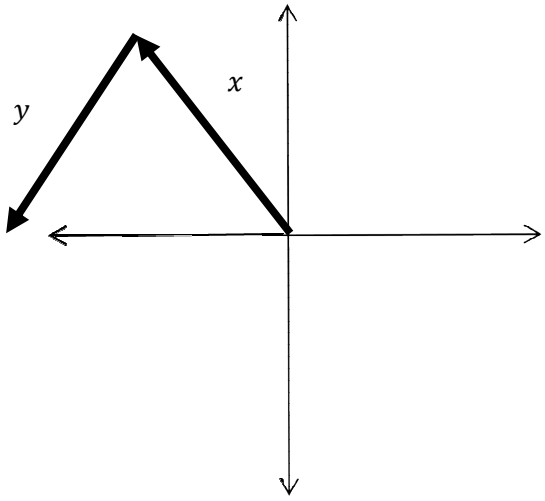


c) A man travels on a bearing of 45° for x km , then on a bearing of 225° for y km

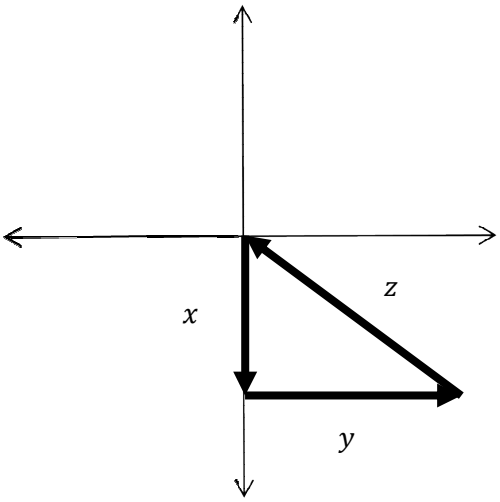


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d) A man travels on a bearing of 330° for x km, then on a bearing of 210° for y km



e) A man travels due South for x km, then travels due East for y km, he then walks back to his starting point for z km.



4) (Using diagrams from Q3)

a) A man travels due East for 3 km, then travels due South for 4 km. What is the shortest distance back to his original starting position?

Shortest distance is straight line along hypotenuse of right angle triangle formed.
By Pythagoras' $x^2 = 3^2 + 4^2; x = 5km$

b) A man travels North-East then turns and travels due South for 15 km until he is due East of his starting position. How far due East of his starting position is he?

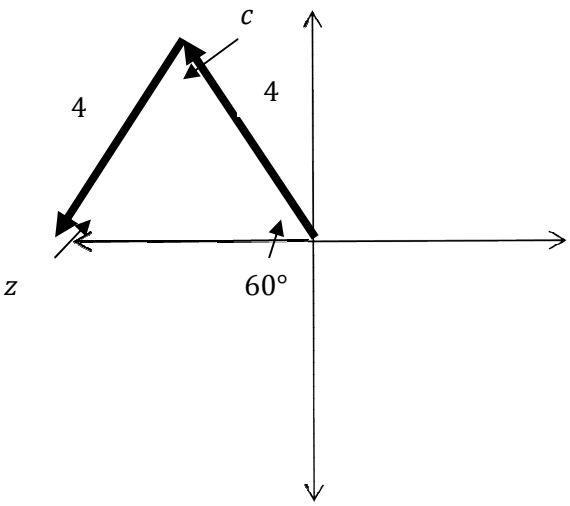
By tangent rule; $\tan 45° = \frac{15}{a}; a = 15km$

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c) A man travels on a bearing of 45° for 10 km; he then travels on a bearing of 225° for 12 km. What is the shortest distance back to his original starting position?

Since he has travelled back along same line, distance back to start is difference of the two distances; 2km

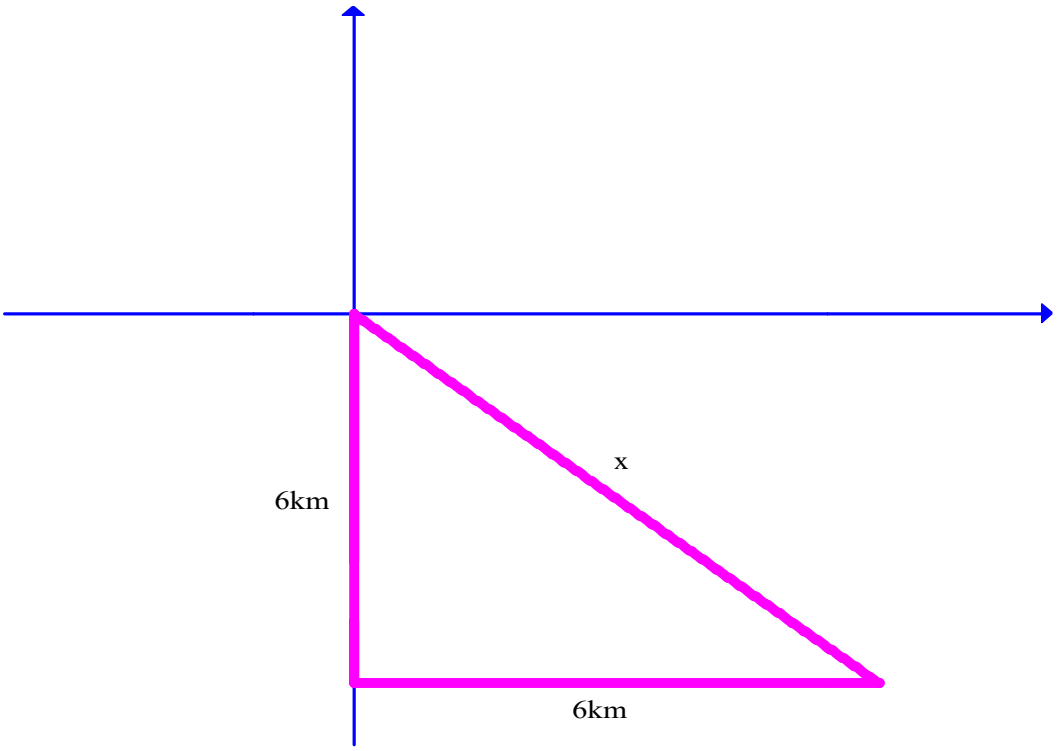
d) A man travels on a bearing of 330° for 4 km and then on a bearing of 210° for 4 km. How far and on what bearing is his shortest path back to his original starting position?



The two distances are equal; therefore the angles opposite them are equal. Since the bearing 330° is equal to 270° + 60°, angle z must be 60°. Therefore angle c must be 60° and the triangle equilateral. Therefore the distance back to the origin is also 4 km on a bearing of due east

e) A man travels due South for 6 km, then due East for 6 km. On what bearing must he travel and for what distance to take the shortest path back to his starting position?

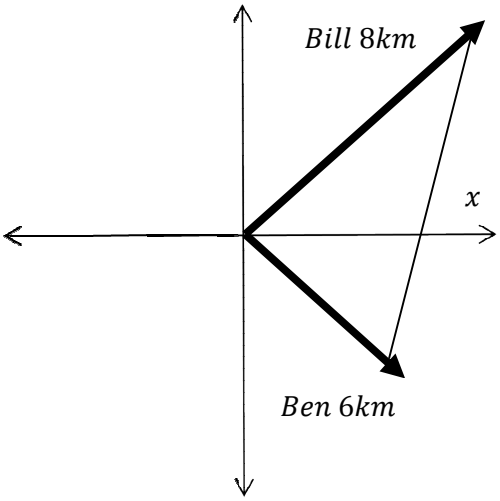
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The three paths form an isosceles right angle triangle of side length 6 km. Therefore the two non right angles must be each 45°; the path back to the origin is on an angle of 45° from west, which is north westerly or a bearing of 315°. The distance can be calculated by Pythagoras: $x^2 = 6^2 + 6^2 = 72; x = \sqrt{72}$

5)

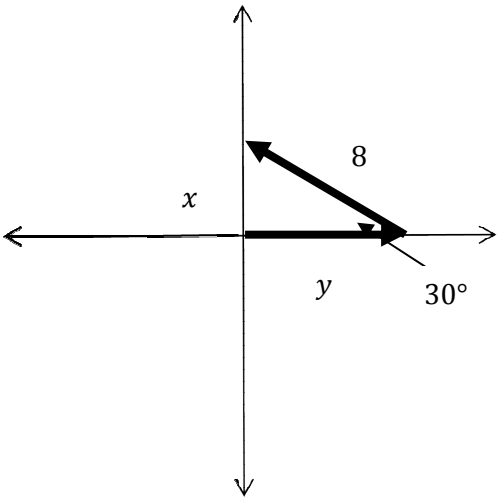
a) Two friends Bill and Ben leave from the same point at the same time. Bill walks North-East at 4 km per hour for 2 hours. Ben walks at a rate of 3 km per hour for 2 hours South-East. How far apart are they at this time?



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The directions north east and south east form a right angle; x is the distance of the hypotenuse. By Pythagoras' $x^2 = 6^2 + 8^2$, $x = 10\ km$

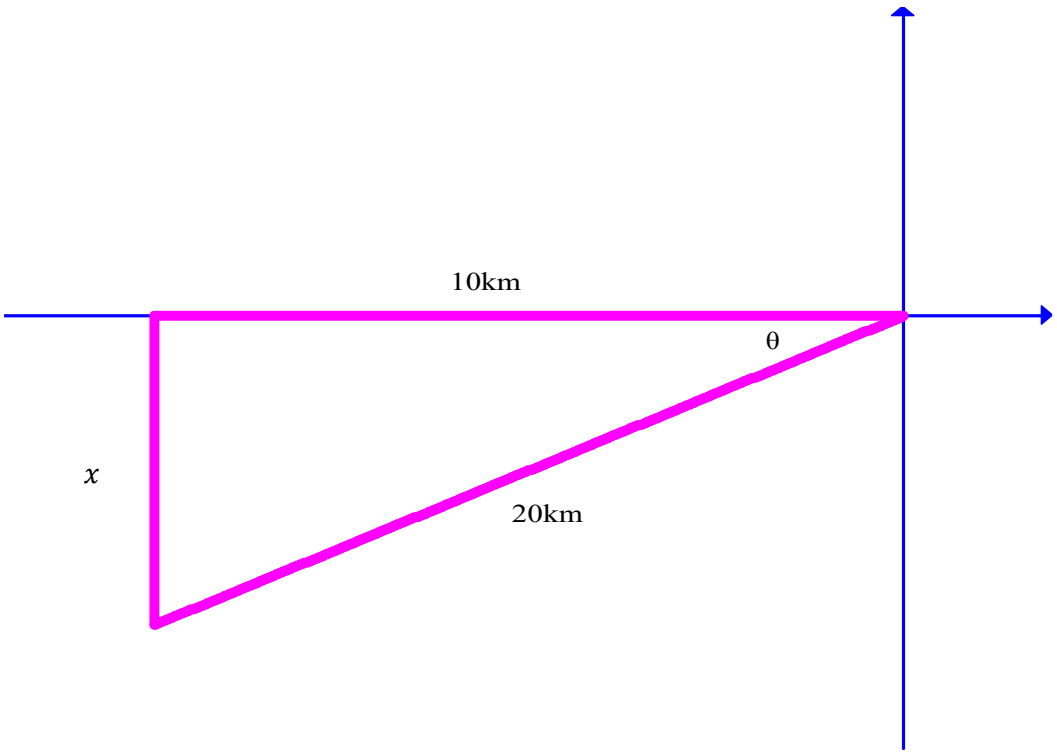
b) Fred travels due East then walks on a bearing of 300° for 8 km until he is due North of his original starting position. How far away from his original position is he? How far due East did he walk?



Fred is due north of his original position which he set out from at due east. Therefore a right angle triangle is formed of hypotenuse length 8. The bearing 300° is 270° (due west) plus 30°. Therefore $\sin 30° = \frac{x}{8}$, and $x = 4\ km$.
The value of y can be calculated using the cosine ratio or Pythagoras and is equal to $\sqrt{48}$

c) Alan and Ken each start rowing a boat from the same position. Alan rows due west for 10 km, whilst Ken rows for 20km at which time he is directly South of Alan. On what bearing did Ken row, and what distance was he away from Alan when he was due south of him?

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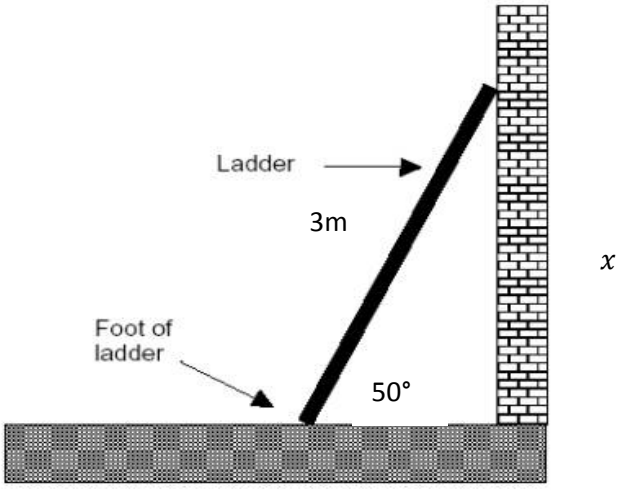


The paths form a right angled triangle of hypotenuse 20 and one side length 10. By using the relation $\cos\theta = \frac{adjacent}{hypotenuse} = \frac{1}{2}$, $\theta = 60°$.Therefore the bearing is $270° - 60° = 210°$

By using Pythagoras, $x^2 + 10^2 = 20^2$, and $x = \sqrt{300} \cong 17.32km$

6) Solve the following

a) A 3 meter ladder leans against a wall and makes an angle of 50° with the ground. How high up the wall does the ladder reach?

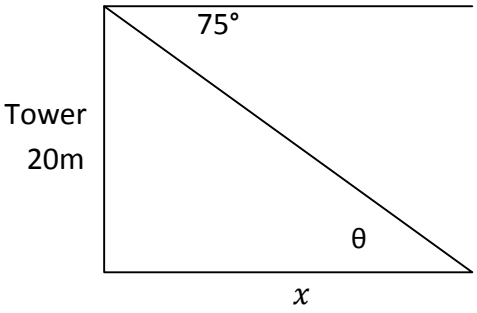


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$$\sin 50° = \frac{x}{3}$$

$$x = 3\sin 50° \cong 2.3m$$

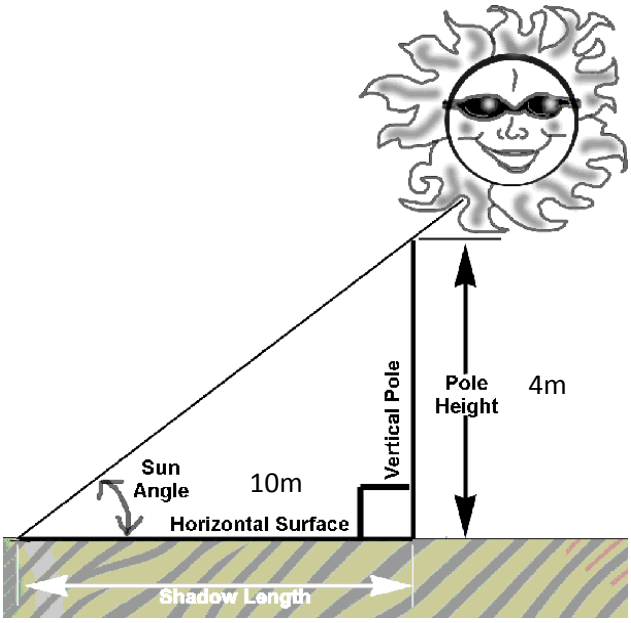
b) The light from a tower shines on an object on the ground. The angle of depression of the light is 75°. If the tower is 20 metres high, how far away is the object from the base of the tower?



The angle of elevation is equal to the angle of depression; therefore $\theta = 75°$

$$\tan 75° = \frac{20}{x}, x = \frac{20}{\tan 75°} \cong 5.36m$$

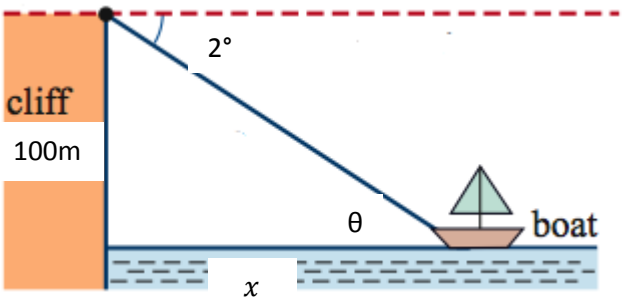
c) A 4 meter pole casts a 10 metre shadow. What is the angle of elevation of the pole from the end of the shadow?



$$\tan\theta = \frac{4}{10} = 0.25, \theta \cong 14°$$

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d) From the top of a cliff the angle of depression to a boat on the ocean is 2°. If the cliff is 100 metres high, how far out to sea is the boat?

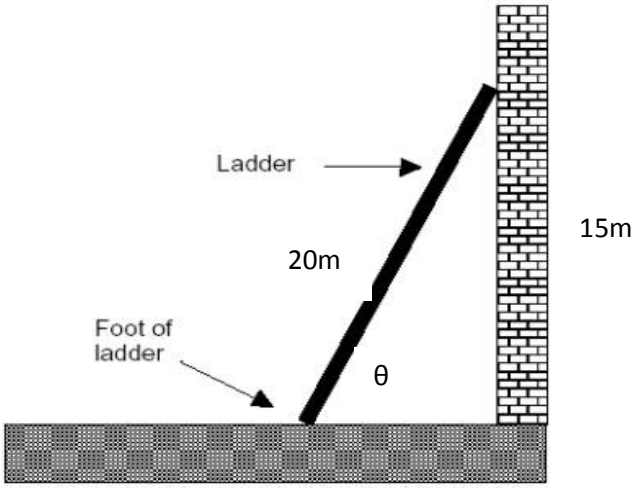


Angle of elevation = angle of depression=2°

$$\tan 2° = \frac{100}{x}$$

$$x = \frac{100}{\tan 2°} \cong 2857m = 2.857 \text{ km}$$

e) A fire fighter has to use his 20 metre ladder to reach the window of a burning apartment building. If the window is 15 meters from the ground, on what angle would the ladder be placed so it can be reached?



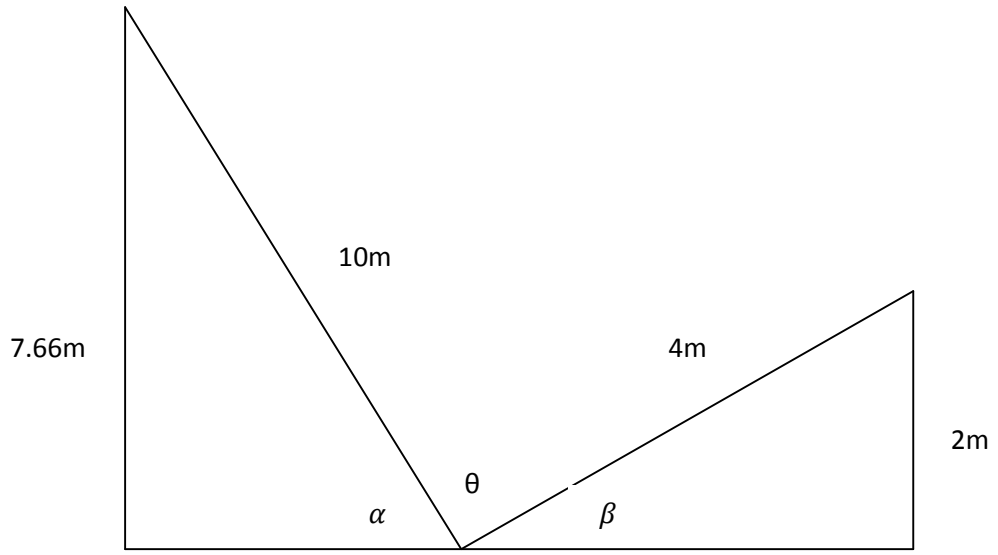
$$\sin\theta = \frac{15}{20} = 0.75$$

$$\theta \cong 48.6°$$

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f) A peg on the ground sits between two poles. The first pole is 2 metres high and the other is 7.66 metres high. From the peg a rope of length 4 metres is attached to the top of the first pole. Another rope of length 10 metres is attached to the top of the second pole. What angle is made between the two ropes?

(Diagram is not to scale)



$\alpha + \beta + \theta = 180°$

$\sin\alpha = \frac{7.66}{10} = 0.766$

$\alpha = 50°$

$\sin\beta = \frac{2}{4} = 0.5$

$\beta = 30°$

$\theta = 100°$

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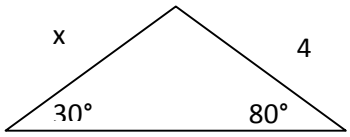
Exercise 3

Non-right Angled Triangles

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1) Sine rule is $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

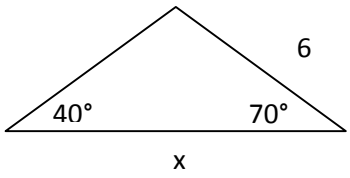
a)



$$\frac{x}{\sin 80°} = \frac{4}{\sin 30°}$$

$$x = \frac{4 \sin 80°}{\sin 30°} \cong \frac{4 \times 0.985}{0.5} = 7.88$$

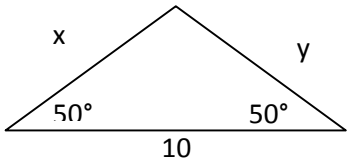
b)



$$\frac{x}{\sin 70°} = \frac{6}{\sin 40°}$$

$$x = \frac{6 \sin 70°}{\sin 40°} \cong 8.77$$

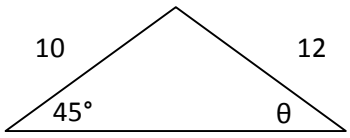
c)



$$\frac{x}{\sin 50°} = \frac{10}{\sin 80°}$$

$$x = \frac{10 \sin 50°}{\sin 80°} \cong 7.78$$

d)



$$\frac{\sin \theta}{10} = \frac{\sin 45}{12}$$

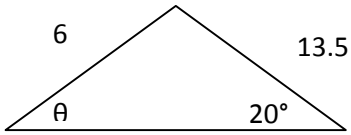
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$$\sin\theta = \frac{10\sin 45°}{12}$$

$$\sin\theta = 0.589$$

$$\theta = 36°$$

e)



$$\frac{\sin\theta}{13.5} = \frac{\sin 20°}{6}$$

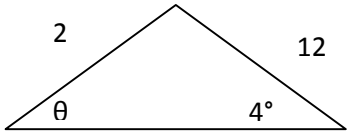
$$\sin\theta = \frac{13.5\sin 20°}{6}$$

$$\theta = 50.3°$$

$$\sin\theta = \sin(180 - \theta)$$

θ can also equal $129.7°$ which can also be a solution, since the third angle will equal 30.3°

f)



$$\frac{\sin\theta}{12} = \frac{\sin 4°}{2}$$

$$\sin\theta = \frac{12\sin 4°}{2}$$

$$\theta = 24.8°$$

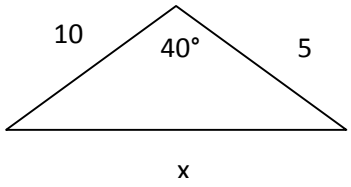
$$\sin\theta = \sin(180 - \theta)$$

θ can also be equal to 155.2°, since the third angle will equal 20.8°

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2) Cosine rule is $a^2 = b^2 + c^2 - 2bc\cos A$

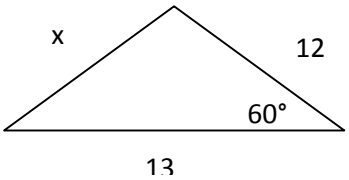
a)



$x^2 = 10^2 + 5^2 - 2 \times 10 \times 5\cos 40°$

$x = \sqrt{100 + 25 - 100 \times 0.766}$ ≅6.96

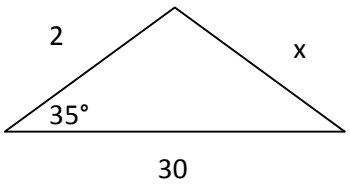
b)



$x^2 = 12^2 + 13 - 2 \times 12 \times 13\cos 60°$

$x = \sqrt{144 + 169 - 312 \times 0.5} \cong 12.53$

c)

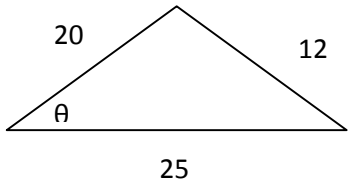


$x^2 = 2^2 + 30^2 - 2 \times 2 \times 30\cos 35°$

$x = \sqrt{4 + 900 - 120 \times 0.82} \cong 28.38$

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d)



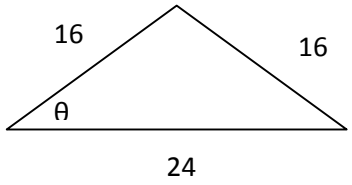
$12^2 = 20^2 + 25^2 - 2 \times 20 \times 25 \cos\theta$

$144 = 1025 - 1000 \cos\theta$

$\cos\theta = \dfrac{1025 - 144}{1000}$

$\theta \cong 28.24°$

e)



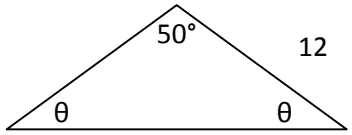
$16^2 = 16^2 + 24^2 - 2 \times 16 \times 24 \cos\theta$

$256 = 832 - 768 \cos\theta$

$\cos\theta = \dfrac{832 - 256}{768}$

$\theta = 41.4°$

f)



Side opposite other θ must also be 12, θ must be 65°, and cosine rule is not required

3) Area of triangle $= \frac{1}{2}ab\sin c$

a) Area$= \frac{1}{2} \times 10 \times 5 \times \sin 40° = 25 \sin 40° \cong 16.075$

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b) Area$= \frac{1}{2} \times 12 \times 13 \times \sin 60° = 78 \sin 60° \cong 67.55$

c) Area$= \frac{1}{2} \times 30 \times 2 \times \sin 35° = 30 \sin 35° \cong 17.22$

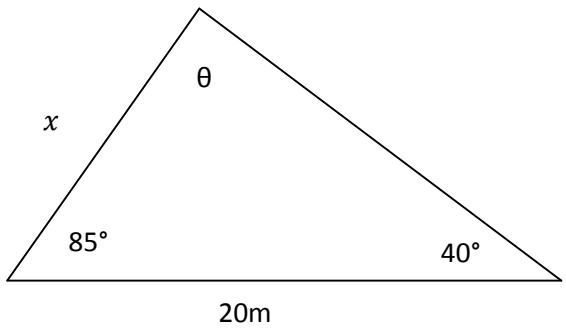
d) Area$= \frac{1}{2} \times 20 \times 25 \times \sin 28.24° = 250 \sin 28.24° \cong 118.25$

e) Area$= \frac{1}{2} \times 16 \times 24 \times \sin 41.4° = 192 \sin 41.4° \cong 126.91$

f) Area$= \frac{1}{2} \times 12 \times 12 \times \sin 50° = 72 \sin 50° \cong 55.15$

4)

a) A post has been hit by a truck and is leaning so it makes an angle of 85° with the ground. A surveyor walks 20 metres from the base of the pole and measures the angle of elevation to the top as 40°. How tall is the pole if it is leaning toward him? How tall is the pole if it is leaning away from him?



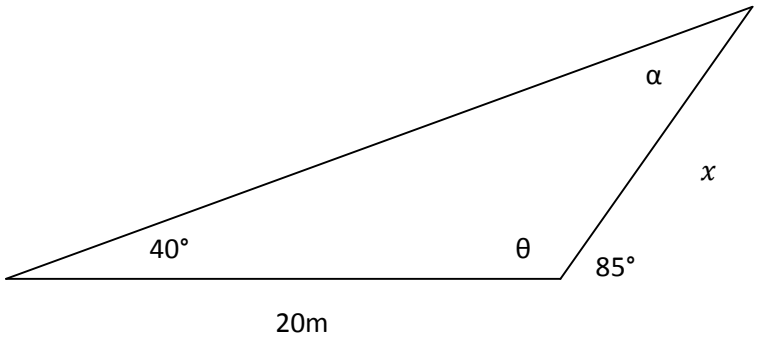
$$\frac{x}{\sin 40°} = \frac{20}{\sin \theta}$$

$$x = \frac{20 \sin 40°}{\sin \theta}$$

$$\theta = 55°$$

$$x \cong \frac{20 \times 0.643}{0.819} = 15.7m$$

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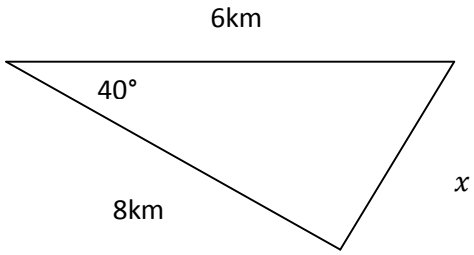


$\theta = 95°, \alpha = 45°$

$$\frac{20}{\sin \alpha} = \frac{x}{\sin 40°}$$

$$x = \frac{20 \sin 40}{\sin 45} \cong \frac{20 \times 0.643}{0.707} = 18.19m$$

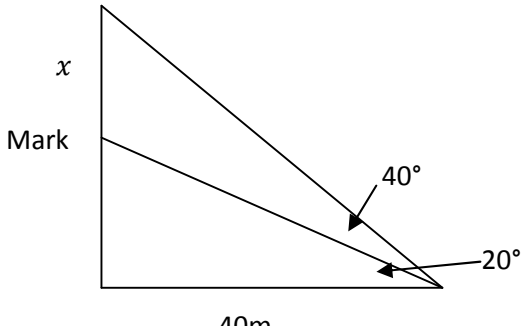
b) Boat A travels due east for 6 km. Boat B travels on a bearing of 130° for 8 km. How far apart are the boats?



$$x^2 = 6^2 + 8^2 - 2 \times 6 \times 8 \cos 40°$$

$$x \cong 100 - 96 \times 0.766 = 26.46km$$

c) A mark is made on the side of a wall. A man 40 metres from the base of the wall measures the angle of elevation to the mark as 20°, and the angle of elevation to the top of the wall as 60°. How far is the mark from the top of the wall?



Let d be the height of the wall, and r... ...eight of the mark

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Then $\tan 60° = \frac{d}{40}$

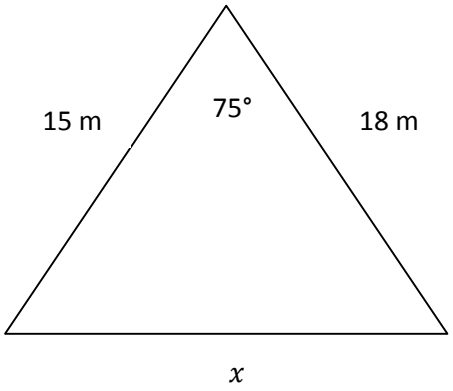
$d = 40 \tan 60° \cong 69.28$ m

$\tan 20° = \frac{m}{40}$

$m = 40 \tan 20° \cong 14.56$ m

Therefore the mark is 54.72 m from the top of the wall

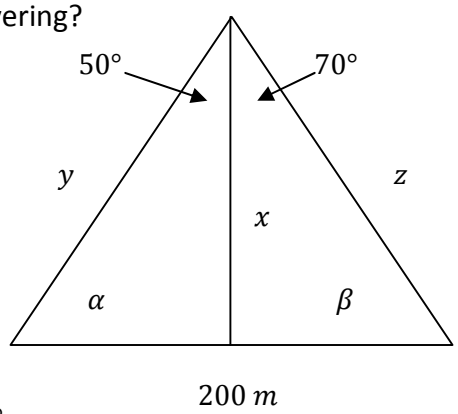
d) What is the perimeter of a triangle with two adjacent sides that measure 15 and 18 metres respectively, with the angle between them 75°?



$x^2 = 15^2 + 17^2 - 2 \times 15 \times 17 \cos 75°$

$x = \sqrt{514 - 510 \cos 75°} \cong 19.54$ m

e) The pilot of a helicopter hovering above the ocean measures the angle of depression to ship A on its left at 50°, and the angle of depression to ship B on its right at 70°. If the ships are 200 metres apart, how high above the ocean is the helicopter hovering?



Angle $\alpha = 40°$

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Angle $\beta = 20°$

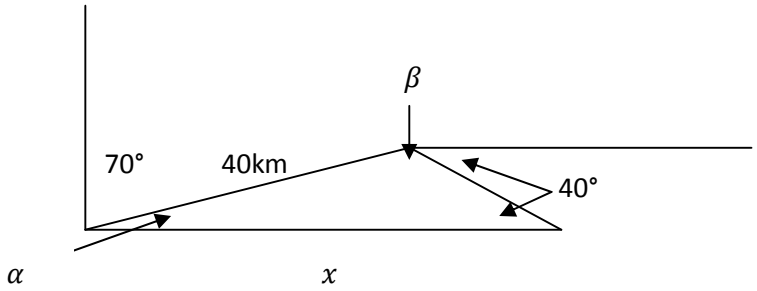
$$\frac{200}{\sin 120°} = \frac{y}{\sin 20°}$$

$$y = \frac{200 \sin 20°}{\sin 120°} \cong 79 \text{ m}$$

$$\sin 20° = \frac{x}{79}$$

$$x = 79 \sin 20° \cong 27 \text{ m}$$

f) A car travels 40 km on a bearing of 70°; then travels on a bearing of 130° until it is exactly due east of its starting position . What is the shortest distance back to its starting position?



Angle $\alpha = 20°$

Angle $\beta = 120°$

$$\frac{x}{\sin 120°} = \frac{40}{\sin 40°}$$

$$x = \frac{40 \sin 120°}{\sin 40°} \cong 53.87 \text{ km}$$

5) Find the areas of the triangles used in question 4, parts a, b and d

For part a:

First triangle: Area$= \frac{1}{2} \times 20 \times 15.7 \sin 85° \cong 156.37\ m^2$

Second triangle: Area $= \frac{1}{2} \times 20 \times 18.19 \sin 95° \cong 181.17\ m^2$

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For part b: Area $= \frac{1}{2} \times 6 \times 8 \sin 40° \cong 15.43\ km^2$

For part d: Area $= \frac{1}{2} \times 15 \times 18 \sin 75° \cong 130.4\ m^2$

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Year 11 Unit 2 Mathematics

Lines & Linear Functions

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Exercise 1

Algebraic Properties of Lines

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1) What is the root of each of the following linear equations?

a) $2x - 4 = 0$

$2x = 4$

$x = 2$

b) $3x - 3 = 0$

$3x = 3$

$x = 1$

c) $4x - 2 = 0$

$4x = 2$

$x = \frac{1}{2}$

d) $x + 5 = 0$

$x = -5$

e) $4x + 2 = 0$

$4x = -2$

$x = -\frac{1}{2}$

f) $3x + 1 = 0$

$3x = -1$

$x = -\frac{1}{3}$

g) $x + \frac{1}{2} = 0$

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$x = -\frac{1}{2}$

h) $2x - \frac{1}{8} = 0$

$2x = \frac{1}{8}$

$x = \frac{1}{16}$

i) $2x + 4 = 6$

$2x = 2$

$x = 1$

j) $3x - 2 = -3$

$3x = -1$

$x = -\frac{1}{3}$

2) Each equation in column 1 is parallel to one of the lines in column 2. Match the parallel lines

Column 1	**Column 2**
$y = 3x + 1$ D	$y = -x + 9$ A
$y = \frac{1}{2}x + 2$ C	$4y = 6x - 1$ B
$-4y = 2x - 1$ F	$-\frac{5}{2}y = -\frac{5}{4}x$ C
$-2y = 2x + 3$ A	$y = 3x + 10$ D
$-\frac{1}{2}y = x - 6$ E	$-3y = 6x - 4$ E
$\frac{1}{3}y = -\frac{1}{2}x + 4$ B	$6y = -3x + 2$ F

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3) Each equation in column 1 is perpendicular to one of the lines in column 2. Match the perpendicular lines

Column 1	Column 2
$y = x + 4$ C	$2y = -x + 3$ A
$y = 2x - 2$ A	$y = \frac{1}{2}x - 3$ B
$2y = 3x + 1$ F	$y = -x - 3$ C
$\frac{1}{2}y = 2x$ E	$3y = 3x + 2$ D
$-3y = 6x + 2$ B	$-4y = x - 5$ E
$-y = x - 8$ D	$3y = -\frac{2}{3}x + 2$ F

4)

a) Having a slope of 1 and passing through the point (2,4)

$y = 1x + c$

$4 = 1 \times 2 + c$

$c = 2$

$y = x + 2$

b) Having a slope of 2 and passing through the point (0,2)

$y = 2x + c$

$2 = 2 \times 0 + c$

$c = 2$

$y = 2x + 2$

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c) Having a slope of 4 and passing through the point (-2,-1)

$y = 4x + c$

$-1 = 4 \times -2 + c$

$c = 7$

$y = 4x + 7$

d) Having a slope of -1 and passing through the point (3,1)

$y = -1x + c$

$1 = -1 \times 3 + c$

$c = 4$

$y = -x + 4$

e) Having a slope of -2 and passing through the point (2,2)

$y = -2x + c$

$2 = -2 \times 2 + c$

$c = 6$

$y = -2x + 6$

f) Having a slope of -2 and passing through the point (-1,-3)

$y = -2x + c$

$-3 = -2 \times -1 + c$

$c = -5$

$y = -2x - 5$

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g) Having a slope of ½ and passing through the point (1,0)

$$y = \frac{1}{2}x + c$$

$$0 = \frac{1}{2} \times 1 + c$$

$$c = -\frac{1}{2}$$

$$2y = x - 1$$

h) Having a slope of $\frac{2}{3}$ and passing through the point (1,3)

$$y = -\frac{2}{3}x + c$$

$$3 = -\frac{2}{3} \times 1 + c$$

$$c = \frac{11}{3}$$

$$3y = -2x + 11$$

i) Having a slope of $-\frac{1}{2}$ and passing through the point (2,1)

$$y = -\frac{1}{2}x + c$$

$$1 = -\frac{1}{2} \times 2 + c$$

$$c = 2$$

$$y = -\frac{1}{2}x + 2$$

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j) Having a slope of $-\frac{3}{4}$ and passing through the point (3,0)

$$y = -\frac{3}{4}x + c$$

$$0 = -\frac{3}{4} \times 3 + c$$

$$c = \frac{9}{4}$$

$$4y = -3x + 9$$

k) Having a slope of $-\frac{2}{3}$ and passing through the point (-3,-2)

$$y = -\frac{2}{3}x + c$$

$$-2 = -\frac{2}{3} \times -3 + c$$

$$c = -4$$

$$y = -\frac{2}{3}x - 4$$

5) Write the equation of the lines passing through the following pairs of points

a) (1,1) and (2,2)

$$m = \frac{2-1}{2-1} = 1$$

$$y = x + c$$

$$2 = 2 + c$$

$$c = 0$$

$$y = x$$

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b) (1,4) and (3,6)

$$m = \frac{6-4}{3-1} = 1$$

$$y = x + c$$

$$6 = 3 + c$$

$$c = 3$$

$$y = x + 3$$

c) (2,0) and (4,4)

$$m = \frac{4-0}{4-2} = 2$$

$$y = 2x + c$$

$$0 = 4 + c$$

$$c = -4$$

$$y = 2x - 4$$

d) (-1,3) and (-3,6)

$$m = \frac{6-3}{-3-(-1)} = -\frac{3}{2}$$

$$y = -\frac{3}{2}x + c$$

$$6 = -\frac{3}{2} \times (-3) + c$$

$$c = \frac{3}{2}$$

$$2y = -3x + 3$$

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e) (2,-1) and (-2,5)

$$m = \frac{5-(-1)}{-2-2} = -\frac{3}{2}$$

$$y = -\frac{3}{2}x + c$$

$$-1 = -\frac{3}{2} \times 2 + c$$

$$c = 2$$

$$y = -\frac{3}{2}x + 2$$

f) (-3,-3) and (0,-1)

$$m = \frac{-1-(-3)}{0-(-3)} = \frac{2}{3}$$

$$y = \frac{2}{3}x + c$$

$$-1 = \frac{2}{3} \times 0 + c$$

$$c = -1$$

$$y = \frac{2}{3}x - 1$$

g) $(\frac{1}{2}, 2)$ and $(-\frac{1}{2}, 4)$

$$m = \frac{4-2}{-\frac{1}{2}-\frac{1}{2}} = -2$$

$$y = -2x + c$$

$$2 = -1 + c$$

$$c = 3$$

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$y = -2x + 3$

h) (-2,-6) and (-1,11)

$$m = \frac{11 - (-6)}{-1 - (-2)} = 17$$

$y = 17x + c$

$11 = -17 + c$

$c = 28$

$y = 17x + 28$

6) Find the equation of the following lines

a) Parallel to the line $y = 2x + 1$ and passing through the point (1,1)

$m = 2$

$y = 2x + c$

$1 = 2 \times 1 + c$

$c = -1$

$y = 2x - 1$

b) Parallel to the line $y = x - 4$ and passing through the point (0,3)

$m = 1$

$y = x + c$

$3 = 0 + c$

$c = 3$

$y = x + 3$

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c) Parallel to the line $2y = 3x + 1$ and passing through the point (-2,4)

$y = \frac{3}{2}x + \frac{1}{2}$

$m = \frac{3}{2}$

$y = \frac{3}{2}x + c$

$4 = \frac{3}{2} \times -2 + c$

$c = 7$

$y = \frac{3}{2}x + 7$

d) Parallel to the line $\frac{1}{2}y = x - 2$ and passing through the point (2,0)

$y = 2x - 4$

$m = 2$

$y = 2x + c$

$0 = 4 + c$

$c = -4$

$y = 2x - 4$

(The same line)

e) Parallel to the line $3x - 2y + 4 = 0$ and passing through the point (-1,-2)

$y = \frac{3}{2}x + 2$

$m = \frac{3}{2}$

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$y = \frac{3}{2}x + c$

$-2 = -\frac{3}{2} + c$

$c = -\frac{1}{2}$

$2y = 3x - 1$

f) Parallel to the line $x + 4y - 2 = 0$ and passing through the point (-2,0)

$4y = 2 - x$

$y = -\frac{1}{4}x + \frac{1}{2}$

$m = -\frac{1}{4}$

$y = -\frac{1}{4}x + c$

$0 = -\frac{1}{4} \times (-2) + c$

$c = -\frac{1}{2}$

g) Parallel to the line $2x - 2y + 3 = 0$ and passing through the point $(-\frac{1}{2}, -\frac{1}{2})$

$y = x + \frac{3}{2}$

$m = 1$

$y = x + c$

$\frac{1}{2} = -\frac{1}{2} + c$

$c = 1$

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$y = x + 1$

7) Find the equation of the following lines

a) Perpendicular to the line $y = -\frac{1}{2}x + 1$ and passing through the point (0,0)

Gradient of perpendicular line $= 2$

$y = 2x + c$

$c = 0$

$y = 2x$

b) Perpendicular to the line $y = -\frac{1}{4}x - 2$ and passing through the point (1,-1)

Gradient of perpendicular line $= 4$

$y = 4x + c$

$-1 = 4 + c$

$c = -5$

$y = 4x - 5$

c) Perpendicular to the line $y = 2x + 4$ and passing through the point (-2,-1)

Gradient of perpendicular line $= -\frac{1}{2}$

$$y = -\frac{1}{2}x + c$$

$$-1 = -\frac{1}{2} \times (-2) + c$$

$c = -2$

$$y = -\frac{1}{2}x - 2$$

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d) Perpendicular to the line $2y = -x - 3$ and passing through the point (3,1)

$y = -\frac{1}{2}x - \frac{3}{2}$

Gradient of perpendicular line $= 2$

$y = 2x + c$

$1 = 6 + c$

$c = -5$

$y = 2x - 5$

e) Perpendicular to the line $3x - 2y + 1 = 0$ and passing through the point (2,0)

$2y = 3x + 1$

$y = \frac{3}{2}x + \frac{1}{2}$

Gradient of perpendicular line $= -\frac{2}{3}$

$y = -\frac{2}{3}x + c$

$0 = -\frac{4}{3} + c$

$c = \frac{4}{3}$

$y = -\frac{2}{3}x + \frac{4}{3}$

$3y = -2x + 4$

f) Perpendicular to the line $4x + y + 2 = 0$ and passing through the point (-1, $\frac{1}{2}$)

$y = -4x - 2$

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Gradient of perpendicular line $= \frac{1}{4}$

$y = \frac{1}{4}x + c$

$\frac{1}{2} = -\frac{1}{4} + c$

$c = \frac{3}{4}$

$y = \frac{1}{4}x + \frac{3}{4}$

$4y = x + 3$

g) Perpendicular to the line $y = x$ and passing through the point (3,1)

Gradient of perpendicular line $= -1$

$y = -x + c$

$1 = -3 + c$

$c = 4$

$y = -x + 4$

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Exercise 2

Intersection of Lines

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1) Which of the following pairs of lines intersect? Give your reasons.

a) $2x + 3y + 2 = 0$ and $y = -\frac{3}{2}x - 2$

$y = -\frac{2}{3}x - \frac{2}{3}$

The lines have different gradients therefore they intersect

b) $y - 2 = 2x + 4$ and $y = 2x$

The lines have the same gradient therefore they are parallel

c) $y - x = 0$ and $y + x = 0$

$y = x$ and $y = -x$

The lines have different gradients therefore they intersect

d) $2y + 4 - 3x = -x + 5$ and $x - y = 0$

$2y = 2x + 1$

$y = x + \frac{1}{2}$ and $y = x$

The lines have the same gradient therefore they are parallel

e) $\frac{1}{2}x - 4y + 3 = -\frac{7}{2}y$ and $\frac{1}{2}x - 4y + 3 = \frac{7}{2}y$

$\frac{1}{2}y = \frac{1}{2}x + 3$ and $\frac{15}{2}y = \frac{1}{2}x + 3$

$y = x + 6$ and $y = \frac{1}{15}x + \frac{2}{5}$

The lines have different gradients therefore they intersect

f) $\sqrt{2}y = \sqrt{x}$ and $y = \frac{\sqrt{2x}}{2}$

$y = \sqrt{\frac{x}{2}}$ and $y = \sqrt{\frac{2x}{4}}$

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$y = \sqrt{\frac{x}{2}}$ and $y = \sqrt{\frac{x}{2}}$

These are not linear equations as they are not in the form $y = mx + c$

However they are the same equation and hence intersect

2) Give example equations of each of the following pairs of lines

a) Two lines that intersect at a point

Any pair of linear equations having different gradients

b) Two lines that intersect at an infinite number of points

Two equations that reduce to the same equation

Example: $2y = 4x + 6$ and $3y = 6x + 9$

c) Two lines that intersect at two points

Two lines cannot intersect at two points

d) Two lines that never intersect

Any pair of lines that have the same gradient, and do not reduce to the same equation

3) At what point(s) do the following pairs of lines intersect?

a) $y = x + 2$ and $2y = x - 4$

Equation 1 subtract equation 2 gives

$-y = 6$

$y = -6$

Substituting gives $x = -8$

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b) $2y - x + 4 = 0$ and $x + 4y + 2 = 0$

Equation 1 plus equation gives

$6y + 6 = 0$

$y = -1$

Substituting gives $x = 2$

c) $\frac{1}{2}x + 3 = y - 3$ and $y = x + 1$

Equation 1 becomes $y = \frac{1}{2}x + 6$

Equation 1 subtract equation 2 gives

$-\frac{1}{2}x + 5 = 0$

$x = 10$

Substituting gives $y = 11$

d) $2y = 4x - 6$ and $3y - 6x + 9 = 0$

Multiply equation 1 by 3 to give

$6y = 12x - 18$

Multiply equation 2 by 2 and rearrange to give

$6y = 12x - 18$

The equations and hence the two lines are the same and intersect at all points

e) $2x - y + 1 = 0$ and $y - 3x = 4$

Rearranging equation 1 gives

$y = 2x + 1$

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Rearranging equation 2 gives

$y = 3x + 4$

Equation 1 subtract equation 2 gives

$0 = -x - 3$

$x = -3$

Substituting gives

$y = -5$

f) $y = \frac{1}{2}x + 5$ and $2y - x + 4 = 0$

Double equation 1 gives $2y = x + 10$

Rearranging equation 2 gives $y = x - 4$

The two lines are parallel and hence never intersect

g) $y = x$ and $-x = y$

Equation 1 subtract equation 2 gives

$2x = 0$

Solution is $x = 0, y = 0$

h) $x - 2 = y + 3x$ and $\frac{1}{2}y - \frac{5}{2} = 6x$

Double and rearranging equation 2 gives

$y = 12x - 5$

Rearranging equation 1 gives

$y = -2x - 2$

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Equation 2 subtract equation 1 gives

$14x - 3 = 0$

$x = \frac{3}{14}$

Substituting gives

$y = -\frac{34}{14}$

4) Find the equation of the following

a) The line that has a slope of -2, and passes through the point of intersection of the lines $y = 2x - 1$ and $y = 3x - 2$

Intersection of lines is point $(1,1)$

$y = -2x + c$

$1 = -2 \times 1 + c$

$c = 3$

$y = -2x + 3$

b) The line that passes through the origin, and also passes through the intersection of the lines $2x - y = 2$ and $\frac{1}{2}y + 1 = x$

The two lines are the same; therefore any line that passes through the origin and intersects the line (is not parallel to it) satisfies the condition

c) The line that passes through the intersection of the lines $2y + x = 5$ and $-x + y = 4$, and is also perpendicular to the second line

Intersection of lines is point $(-1,3)$

Second line has gradient of 1

Therefore required gradient is -1

$y = -x + c$

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$3 = -1 \times -1 + c$

$c = 2$

$y = -x + 2$

d) The line that passes through the point (-2,-1) and also passes through the intersection of the lines $y = \frac{1}{2}x + 2$ and $\frac{1}{2}y - x = -1$

Intersection of lines is point $\left(\frac{8}{3}, \frac{10}{3}\right)$

$$m = \frac{\frac{10}{3} - (-1)}{\frac{8}{3} - (-2)} = \frac{13}{14}$$

$y = \frac{13}{14}x + c$

$-1 = \frac{13}{14} \times (-2) + c$

$c = \frac{12}{14}$

$y = \frac{13}{14}x + \frac{12}{14}$

$y = 13x + 12$

e) The line that passes through the intersection of $y = 2x$ and $y = -3x + 5$, and is also parallel to the first line

Intersection of lines is point $(1,2)$

Gradient of $y = 2x$ is 2

Gradient of parallel line is 2

$y = 2x + c$

$2 = 2 \times 1 + c$

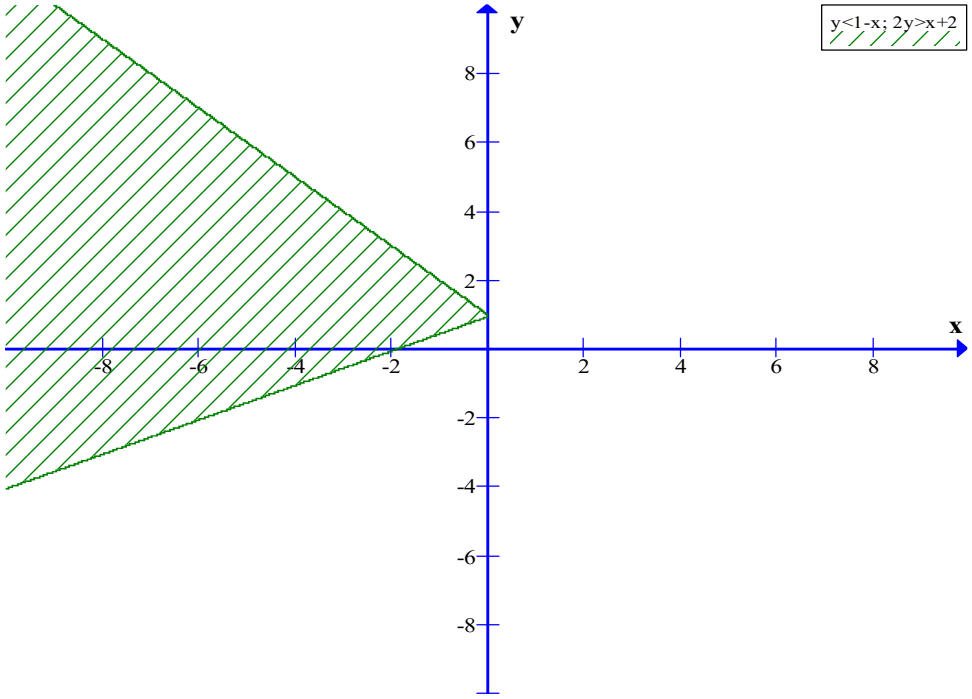
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$c = 0$

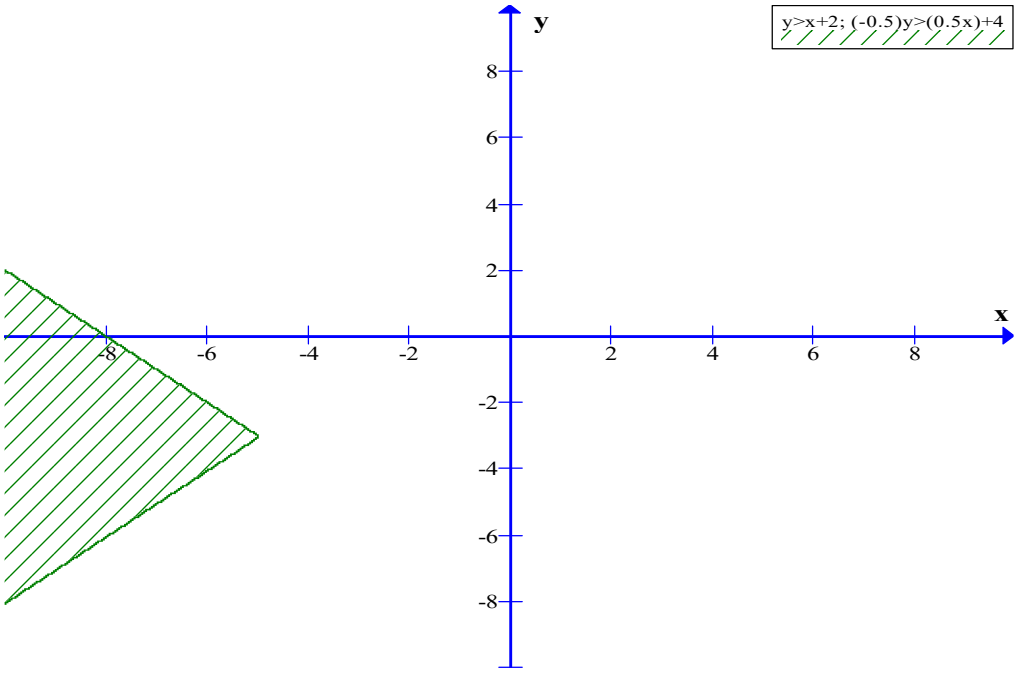
$y = 2x$

5) Shade the region(s) of the number plane as defined in the following questions

a) The region where $y < 1 - x$ and $2y > x + 2$

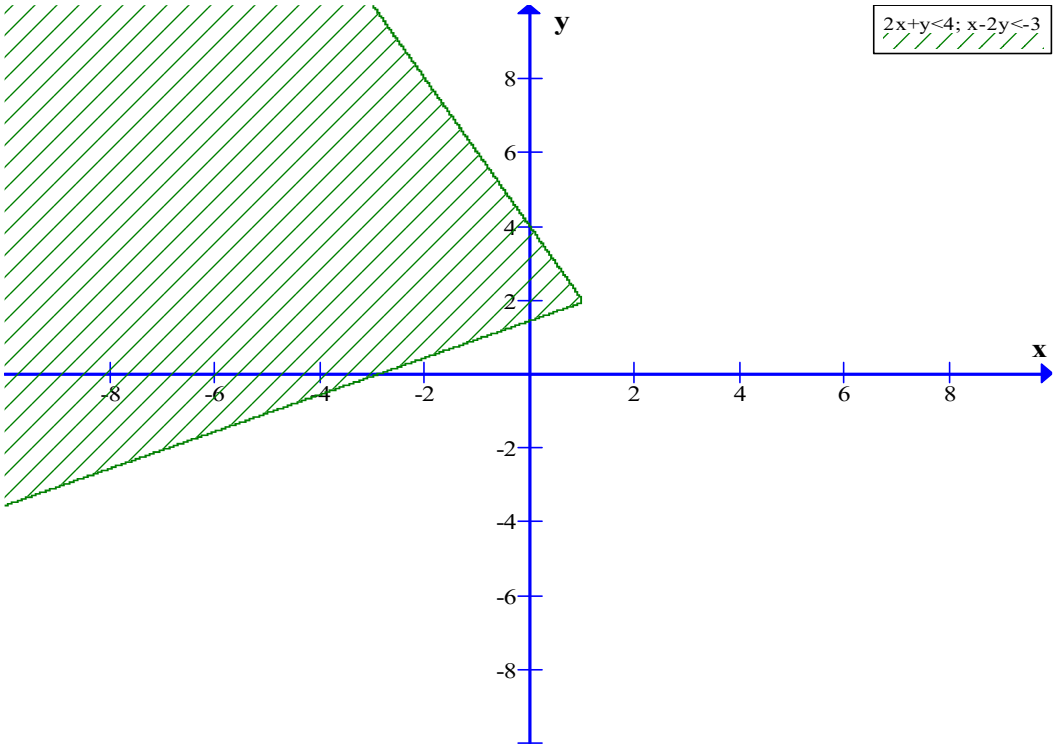


b) The region where $y > x + 2$ and $-\frac{1}{2}y > \frac{1}{2}x + 4$

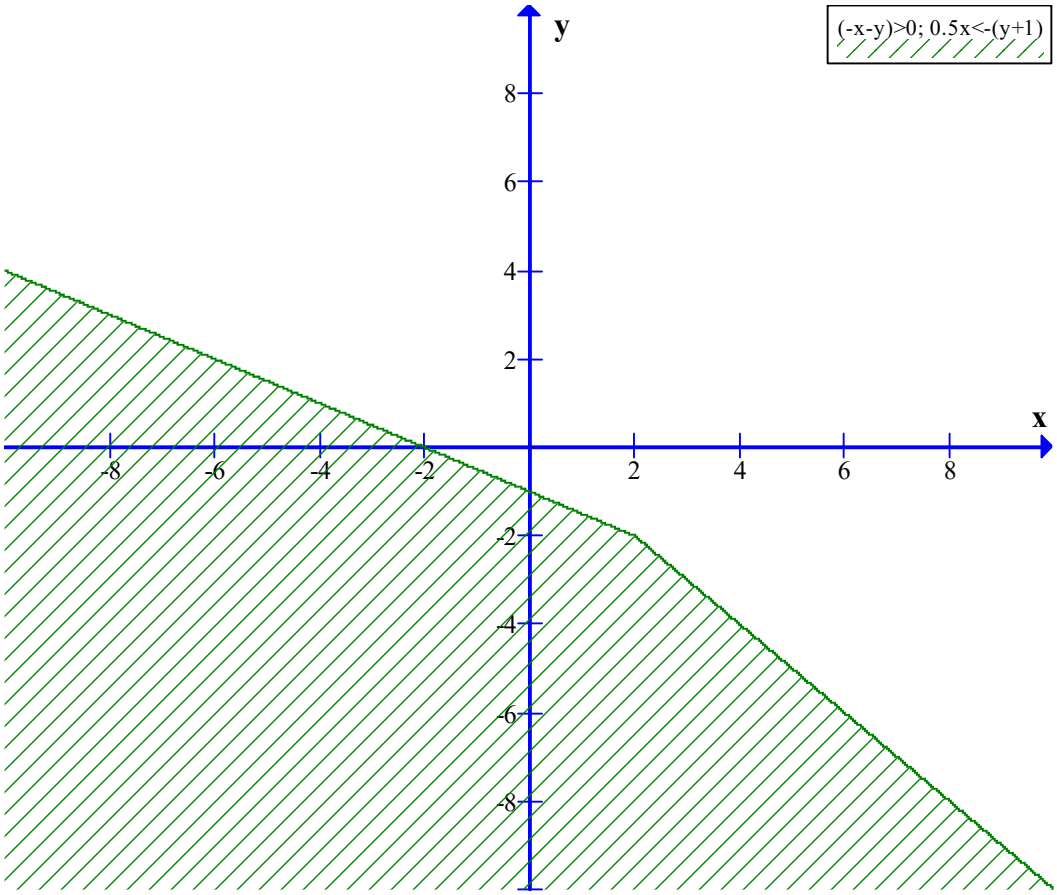


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c) The region where $2x + y < 4$ and $x - 2y < -3$



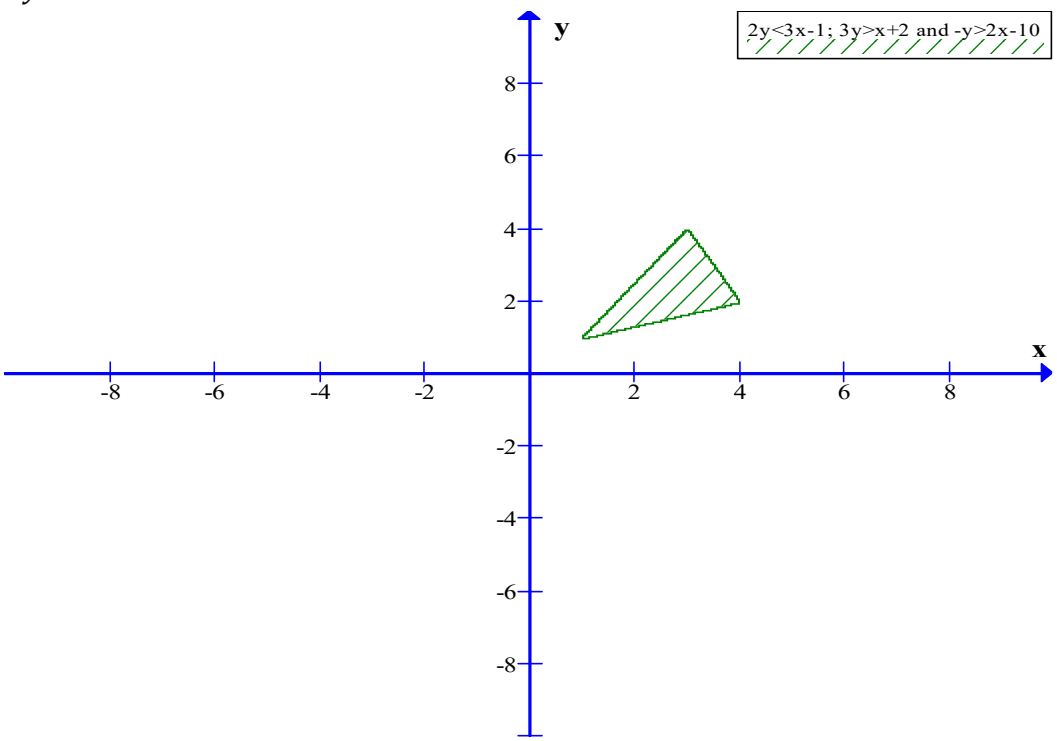
d) The region where$(-x - y) > 0$ and $\frac{1}{2}x < -(y + 1)$



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6) Draw and describe

a) The region bounded by the inequalities $2y < 3x - 1,\ -y > 2x - 10$ and $3y > x + 2$



b) The equations of the lines that pass through each of the following pairs of points

i. (-2,1) and (0,0)

$$m = -\frac{1}{2}$$

$$1 = -\frac{1}{2} \times -2 + c$$

$$c = 0$$

$$y = -\frac{1}{2}x$$

ii. (-4,-4) and (-2,1)

$$m = \frac{5}{2}$$

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$$1 = \frac{5}{2} \times -2 + c$$

$$c = 6$$

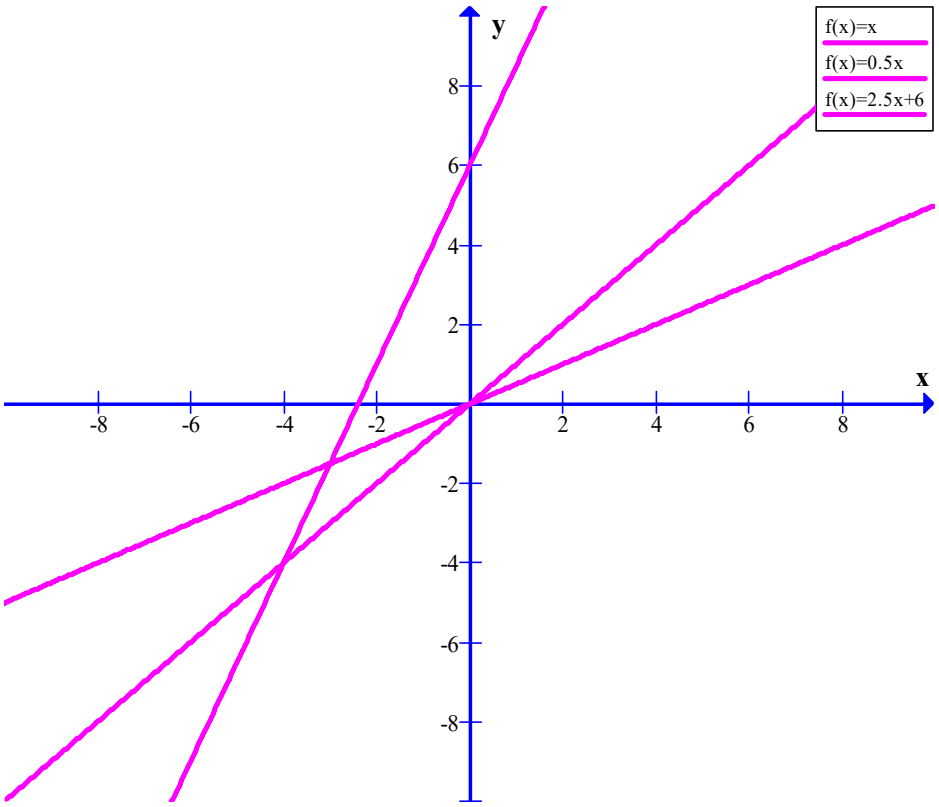
$$y = \frac{5}{2}x + 6$$

iii. (-4,-4) and (0,0)

$$m = 1$$

$$c = 0$$

$$y = x$$



c) The inequalities that form a triangle bounded by the lines in part b

$$y > x$$

$$y < \frac{5}{2}x + 6$$

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$$y < \frac{1}{2}x$$

d) Show in your diagram and by substitution into the inequalities that the point (-3,-2) lies within the triangle.

$-2 > -3$

$-2 < \frac{5}{2} \times (-3) + 6$

$-2 < \frac{1}{2} \times -3$

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Exercise 3

Distance & Midpoints

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1) Find the distance between the following pairs of points. Leave answer in surd form if necessary.

Distance $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

a) (2,2) and (1,1)

$d = \sqrt{(2-1)^2 + (2-1)^2}$

$= \sqrt{2}$

b) (3,4) and (0,2)

$d = \sqrt{(3-0)^2 + (4-2)^2}$

$= \sqrt{13}$

c) (2,6) and (1,3)

$d = \sqrt{(2-1)^2 + (6-3)^2}$

$= \sqrt{10}$

d) (1,4) and (3,3)

$d = \sqrt{(1-3)^2 + (4-3)^2}$

$= \sqrt{5}$

e) (0,2) and (2,1)

$d = \sqrt{(0-2)^2 + (2-1)^2}$

$= \sqrt{5}$

f) (4,5) and (6,2)

$d = \sqrt{(4-6)^2 + (5-2)^2}$

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$= \sqrt{13}$

2) Find the distance between the following pairs of points. Leave answer in surd form if necessary

a) (-3,-1) and (1,-2)

$d = \sqrt{(-3-1)^2 + (-1--2)^2}$

$= \sqrt{17}$

b) (0,-3) and (-2,1)

$d = \sqrt{(0--2)^2 + (-3-1)^2}$

$= \sqrt{18}$

c) (-1,-2) and (3,-4)

$d = \sqrt{(-1-3)^2 + (-2--4)^2}$

$= \sqrt{20}$

d) (4,-1) and (0,-3)

$d = \sqrt{(4-0)^2 + (-1--3)^2}$

$= \sqrt{20}$

e) (2,2) and (-1,1)

$d = \sqrt{(2--1)^2 + (2-1)^2}$

$= \sqrt{10}$

f) (1,1) and (-3,3)

$d = \sqrt{(1--3)^2 + (1-3)^2}$

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$= \sqrt{20}$

3) Find the distance between the following points. Leave answer in surd form if necessary

a) $(\frac{1}{2}, \frac{1}{2})$ and $(\frac{3}{2}, 0)$

$$d = \sqrt{\left(\frac{1}{2} - \frac{3}{2}\right)^2 + \left(\frac{1}{2} - 0\right)^2}$$

$$= \frac{\sqrt{5}}{2}$$

b) $(\frac{5}{2}, \frac{3}{2})$ and $(6, \frac{1}{2})$

$$d = \sqrt{\left(\frac{5}{2} - 6\right)^2 + \left(\frac{3}{2} - \frac{1}{2}\right)^2}$$

$$= \frac{\sqrt{53}}{2}$$

c) $(0, -\frac{1}{2})$ and $(\frac{3}{2}, 4)$

$$d = \sqrt{\left(0 - \frac{3}{2}\right)^2 + \left(-\frac{1}{2} - 4\right)^2}$$

$$= \frac{\sqrt{90}}{2}$$

d) $(-\frac{3}{2}, \frac{1}{2})$ and (2, -2)

$$d = \sqrt{\left(-\frac{3}{2} - 2\right)^2 + \left(\frac{1}{2} - -2\right)^2}$$

$$= \frac{\sqrt{74}}{2}$$

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e) $(\frac{1}{2}, -\frac{1}{2})$ and $(-\frac{1}{2}, \frac{1}{2})$

$$d = \sqrt{\left(\frac{1}{2} - -\frac{1}{2}\right)^2 + \left(-\frac{1}{2} - \frac{1}{2}\right)^2}$$

$$= \sqrt{2}$$

f) $(\frac{3}{2}, \frac{1}{2})$ and $(-\frac{3}{2}, -\frac{1}{2})$

$$d = \sqrt{\left(\frac{3}{2} - -\frac{3}{2}\right)^2 + \left(\frac{1}{2} - -\frac{1}{2}\right)^2}$$

$$= \sqrt{10}$$

4) Find the midpoints of the line segments joining the following pairs of points

a) (2,2) and (1,1)

$$\left(\frac{3}{2}, \frac{3}{2}\right)$$

b) (3,4) and (0,2)

$$\left(\frac{3}{2}, 3\right)$$

c) (2,6) and (1,3)

$$\left(\frac{3}{2}, \frac{9}{2}\right)$$

d) (1,4) and (3,3)

$$\left(2, \frac{7}{2}\right)$$

e) (0,2) and (2,1)

$$\left(1, \frac{3}{2}\right)$$

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f) (4,5) and (6,2)

$\left(5, \frac{7}{2}\right)$

5) Find the midpoints of the line segments joining the following pairs of points

a) (-3,-1) and (1,-2)

$\left(-1, -\frac{1}{2}\right)$

b) (0,-3) and (-2,1)

$(-1, -1)$

c) (-1,-2) and (3,-4)

$(1, -3)$

d) (4,-1) and (0,-3)

$(2, -2)$

e) (2,2) and (-1,1)

$\left(\frac{1}{2}, \frac{1}{2}\right)$

f) (1,1) and (-3,3)

$(-1,2)$

6) Find the midpoints of the line segments joining the following pairs of points

a) $(\frac{1}{2}, \frac{1}{2})$ and $(\frac{3}{2}, 0)$

$\left(1, \frac{1}{4}\right)$

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b) $(\frac{5}{2}, \frac{3}{2})$ and $(6, \frac{1}{2})$

$\frac{17}{4}, 1$

c) $(0, -\frac{1}{2})$ and $(\frac{3}{2}, 4)$

$\left(\frac{3}{4}, \frac{7}{4}\right)$

d) $(-\frac{3}{2}, \frac{1}{2})$ and (2, -2)

$\left(\frac{1}{4}, -\frac{1}{4}\right)$

e) $(\frac{1}{2}, -\frac{1}{2})$ and $(-\frac{1}{2}, \frac{1}{2})$

$(0,0)$

f) $(\frac{3}{2}, \frac{1}{2})$ and $(-\frac{3}{2}, -\frac{1}{2})$

$(0,0)$

7) Find the perpendicular distance from each line to the point given

a) $y = 2x + 2$ and the point (1,2)

Gradient of perpendicular line is $-\frac{1}{2}$

$y = -\frac{1}{2}x + c$

$2 = -\frac{1}{2} \times 1 + c$

$c = \frac{5}{2}$

$2y = -x + 5$

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Intersects $y = 2x + 2\ at\ \left(\frac{1}{5}, \frac{12}{5}\right)$

Distance from $(1,2)$ to $\left(\frac{1}{5}, \frac{12}{5}\right)$ is

$$d = \sqrt{\left(1 - \frac{1}{5}\right)^2 + \left(2 - \frac{12}{5}\right)^2} = \frac{2}{\sqrt{5}}$$

b) $3y - x = 1$ and the point (-1,3)

Gradient of perpendicular line is -3

$y = -3x + c$

$3 = -3 \times -1 + c$

$c = -1$

$y = -3x - 1$

Intersects $3y - x = 1$ at $\left(-\frac{2}{5}, \frac{1}{5}\right)$

Distance from $(-1,3)$ to $\left(-\frac{2}{5}, \frac{1}{5}\right)$ is

$$d = \sqrt{\left(-1 - -\frac{2}{5}\right)^2 + \left(3 - \frac{1}{5}\right)^2} = \sqrt{\frac{41}{5}}$$

c) $y = -x$ and the point (2,0)

Gradient of perpendicular line is 1

$y = x + c$

$0 = 2 + c$

$c = -2$

$y = x - 2$

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Intersects $y = -x$ at $(1, -1)$

Distance from $(2,0)$ to $(1, -1)$ is

$d = \sqrt{(2-1)^2 + (0--1)^2} = \sqrt{2}$

d) $2y + x - 2 = 0$ and the point (-2,1)

Gradient of perpendicular line is 2

$y = 2x + c$

$1 = 2 \times -2 + c$

$c = 5$

$y = 2x + 5$

Intersects $2y + x - 2 = 0$ at $\left(-\frac{8}{5}, \frac{9}{5}\right)$

Distance from $(-2,1)$ to $\left(-\frac{8}{5}, \frac{9}{5}\right)$ is

$$d = \sqrt{\left(-2 - -\frac{8}{5}\right)^2 + \left(1 - \frac{9}{5}\right)^2} = \frac{2}{\sqrt{5}}$$

e) $\frac{1}{2}y = x - 2$ and the point (1,-1)

Gradient of perpendicular line is $-\frac{1}{2}$

$$y = -\frac{1}{2}x + c$$

$$-1 = -\frac{1}{2} \times 1 + c$$

$$c = -\frac{1}{2}$$

$2y = -x - 1$

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Intersects $\frac{1}{2}y = x - 2$ at $\left(\frac{7}{5}, -\frac{6}{5}\right)$

Distance from $(1, -1)$ to $\left(\frac{7}{5}, -\frac{6}{5}\right)$ is

$$d = \sqrt{\left(1 - \frac{7}{5}\right)^2 + \left(-1 - -\frac{6}{5}\right)^2} = \frac{1}{\sqrt{5}}$$

f) $y = 4$ and the point (2,4)

The point $(2,4)$ lies on the line $y = 4$, therefore the distance $= 0$

8) Draw the line segment (A) connecting the points (1, 2) and (3, 8). Also draw the line segment (B) connecting the points (-2,-10) and (1,-1). Find the midpoint of each line segment, the length of each line segment, and the equation of the line joining the midpoint of A to the midpoint of B.

Midpoint of A is $(2,5)$

Midpoint of B is $\left(-\frac{1}{2}, -\frac{11}{2}\right)$

Length of A = $\sqrt{(3-1)^2 + (8-2)^2} = \sqrt{40}$

Length of B $= \sqrt{(-2 - -1)^2 + (-10 - -1)^2} = \sqrt{82}$

Equation of midpoints

$$m = \frac{5 - -\frac{11}{2}}{2 - -\frac{1}{2}} = \frac{21}{5}$$

$$y = \frac{21}{5}x + c$$

$$5 = \frac{21}{5} \times 2 + c$$

$$c = -\frac{17}{5}$$

$$5y = 21x - 17$$

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Year 11 Unit 2 Mathematics

Quadratic Polynomials

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Exercise 1

Graphical Representation of Properties

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1) Factorize and hence solve the following quadratic equations

a) $x^2 = 0$

$x = 0$

b) $x^2 - 4 = 0$

$(x - 2)(x + 2) = 0$

$x = 2 \; or \; x = -2$

c) $x^2 + x - 6 = 0$

$(x + 3)(x - 2) = 0$

$x = -3 \; or \; x = 2$

d) $x^2 - 6x + 9 = 0$

$(x - 3)^2 = 0$

$x = 3$

e) $x^2 - 4x + 3 = 0$

$(x - 3)(x - 1) = 0$

$x = 3 \; or \; x = 1$

f) $-x^2 - 5x - 6 = 0$

$(-x - 2)(x + 3)$

$x = -2 \; or \; x = -3$

g) $2x^2 + 8x + 8 = 0$

$(2x + 4)(x + 2) = 0$

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$x = -2$

h) $3x^2 - x - 10 = 0$

$(3x + 5)(x - 2) = 0$

$x = -\frac{5}{3}$ *or* $x = 2$

i) $x^2 + x + 8 = 0$

Cannot be factorised

j) $4x^2 + 4x + 1 = 0$

$(2x + 1)(2x + 1) = 0$

$(2x + 1)^2 = 0$

$x = -\frac{1}{2}$

k) $x^2 + 2x + 3 = 0$

Cannot be factorised

2) Complete the square and hence identify the turning point of the following functions

a) $y = x^2$

$y = (x - 0)^2 + 0$

Turning point is $(0,0)$

b) $y = x^2 - 4$

$y = (x - 0)^2 - 4$

Turning point is $(0, -4)$

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c) $y = x^2 + x - 6$

$y = \left(x + \frac{1}{2}\right)^2 - \frac{25}{4}$

Turning point is $\left(-\frac{1}{2}, -\frac{25}{4}\right)$

d) $y = x^2 - 6x + 9$

$y = (x - 3)^2 + 0$

Turning point is $(3,0)$

e) $y = x^2 - 4x + 3$

$y = (x - 2)^2 - 1$

Turning point is $(2, -1)$

f) $y = -x^2 - 5x - 6$

$y = -(x^2 + 5x + 6)$

$y = -1\left(\left(x + \frac{5}{2}\right)^2 - \frac{1}{4}\right)$

$y = -\left(x + \frac{5}{2}\right)^2 + \frac{1}{4}$

Turning point is $\left(-\frac{5}{2}, 6\right)$

g) $y = 2x^2 + 8x + 8$

$y = 2(x^2 + 4x + 4)$

$y = 2(x + 2)^2 + 0$

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Turning point is $(-2,0)$

h) $y = 3x^2 - x - 10$

$$y = 3\left(x^2 - \frac{1}{3}x - \frac{10}{3}\right)$$

$$y = 3\left(\left(x - \frac{1}{6}\right)^2 - \frac{121}{36}\right)$$

$$y = 3\left(x - \tfrac{1}{6}\right)^2 - \tfrac{121}{12}$$

Turning point is $\left(\frac{1}{6}, -\frac{121}{12}\right)$

i) $y = x^2 + x + 8$

$$y = \left(x + \frac{1}{2}\right)^2 + \frac{31}{4}$$

Turning point is $\left(-\frac{1}{2}, \frac{31}{4}\right)$

j) $y = 4x^2 + 4x + 1$

$$y = 4\left(x^2 + x + \frac{1}{4}\right)$$

$$y = 4\left(\left(x + \frac{1}{2}\right)^2 + 0\right)$$

$$y = 4\left(x + \frac{1}{2}\right)^2 + 0$$

Turning point is $\left(-\frac{1}{2}, 0\right)$

k) $y = x^2 + 2x + 3$

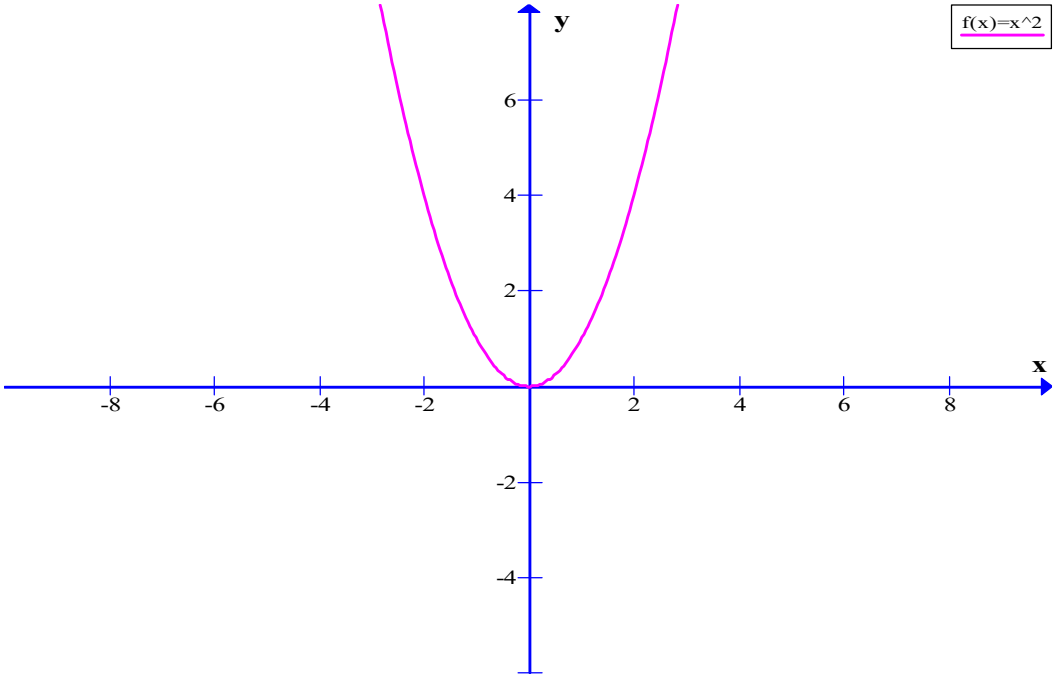
$$y = (x+1)^2 + 2$$

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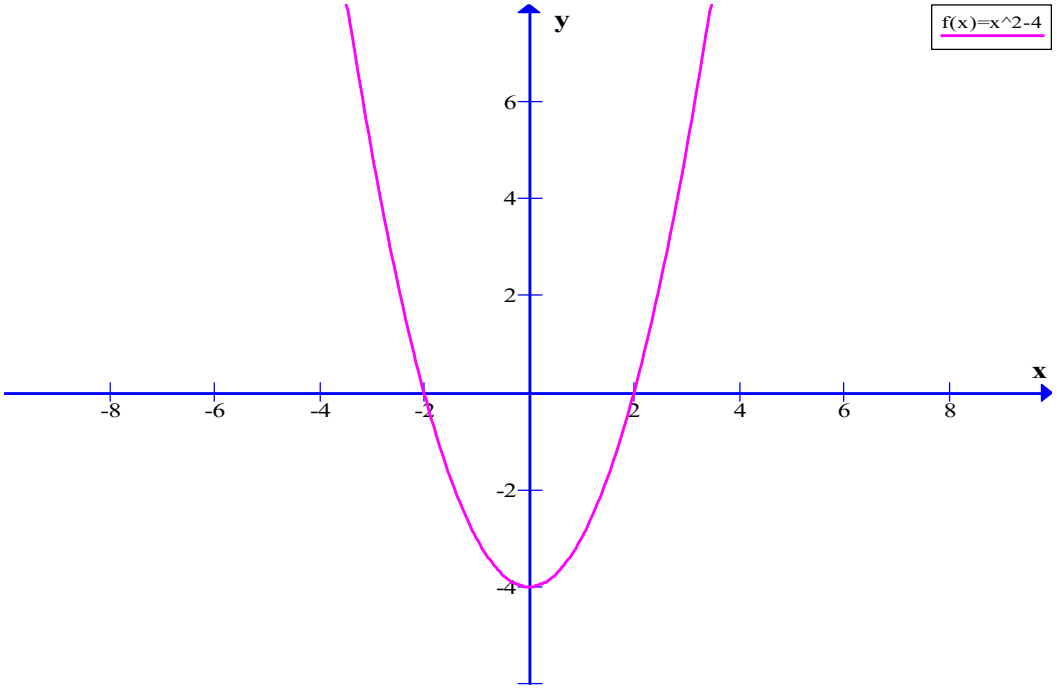
Turning point is $(-1,2)$

3) Using your answers to questions 1 and 2, graph the following functions

a) $y = x^2$

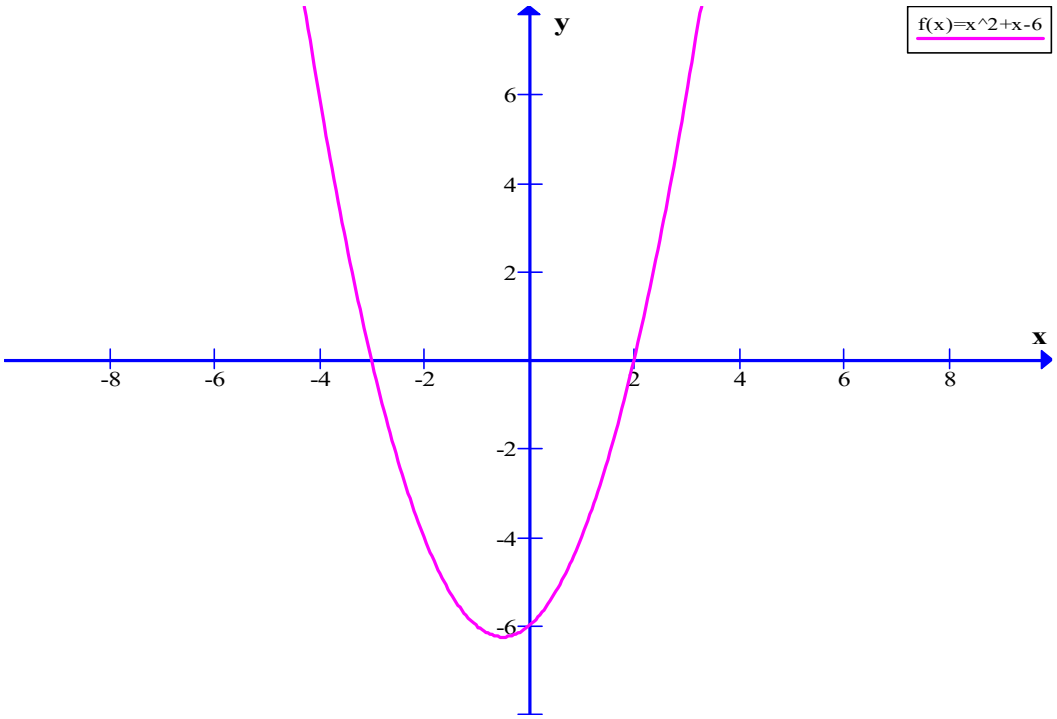


b) $y = x^2 - 4$



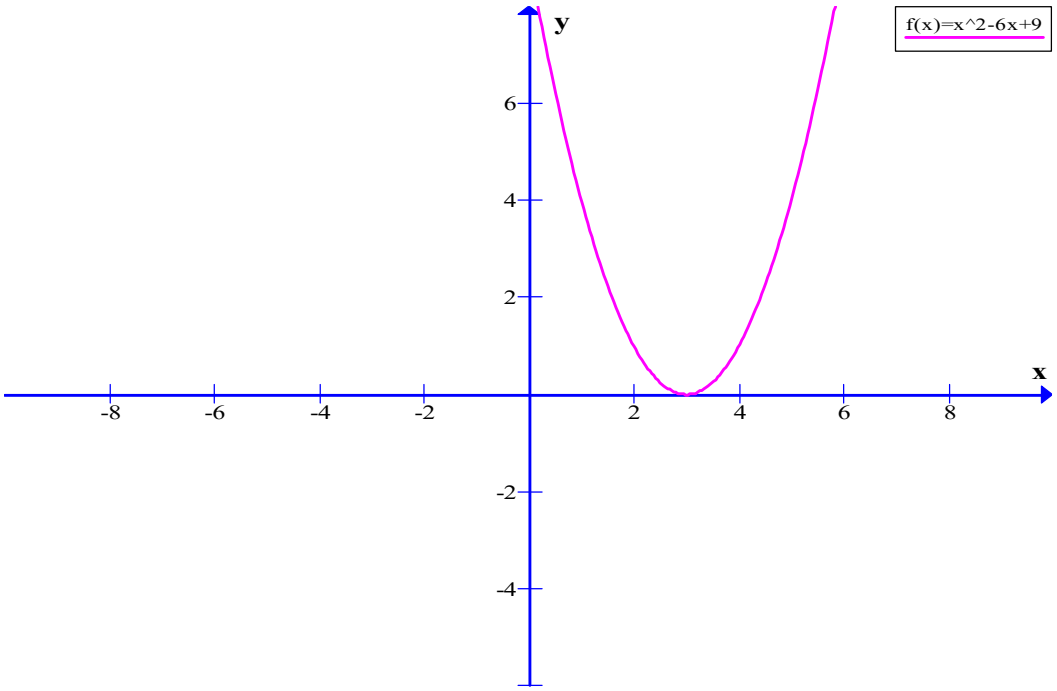
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c) $y = x^2 + x - 6$

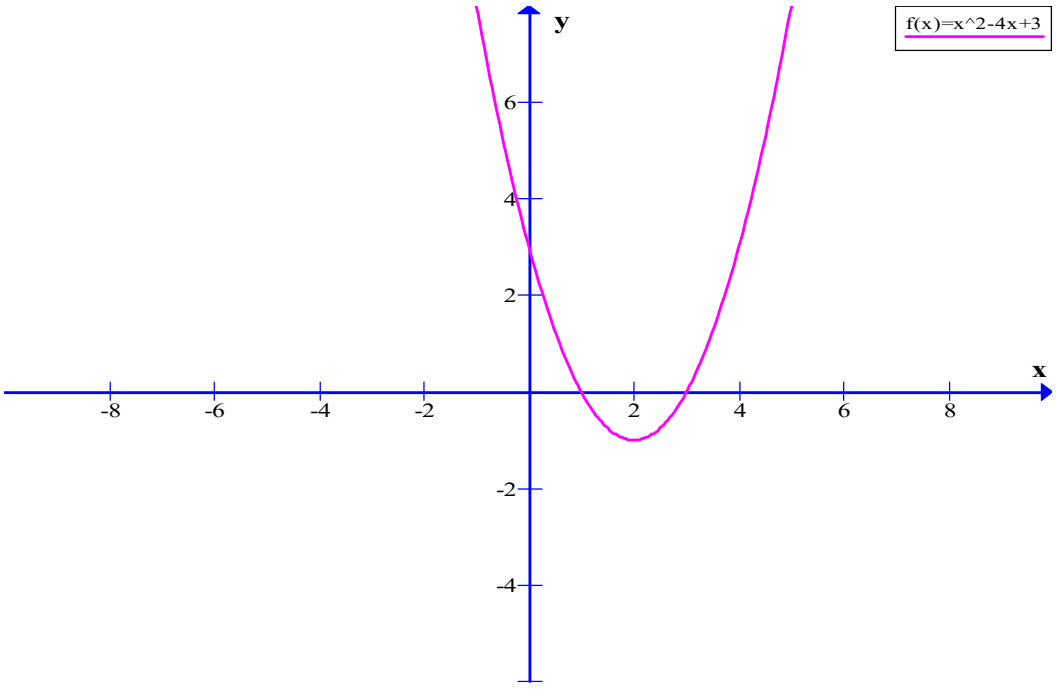


d) $y = x^2 - 6x + 9$

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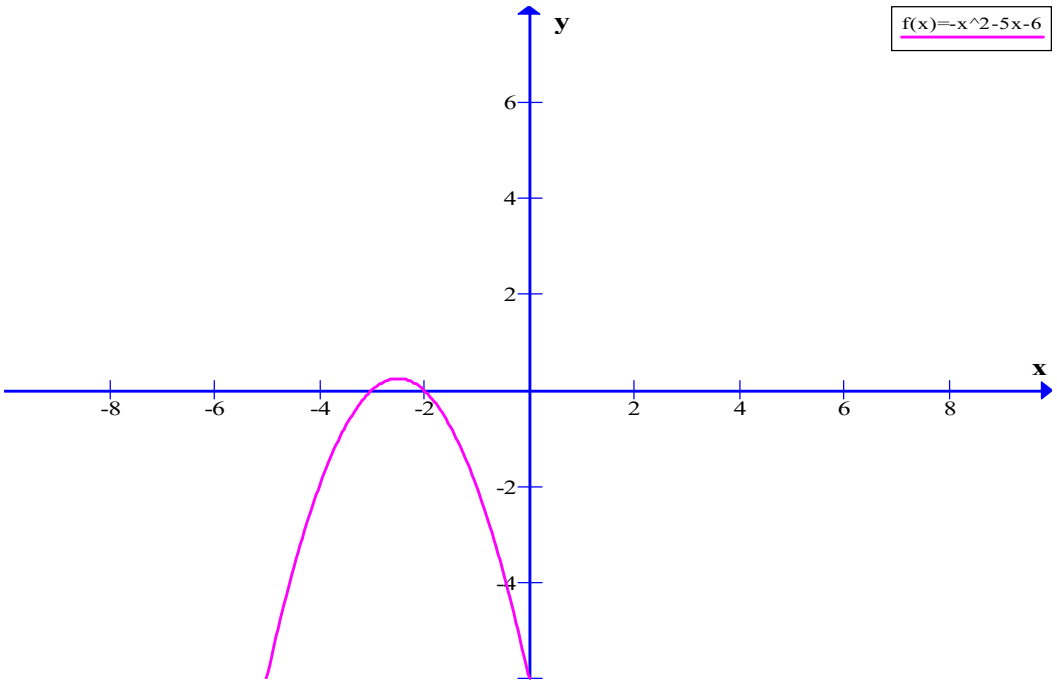


e) $y = x^2 - 4x + 3$

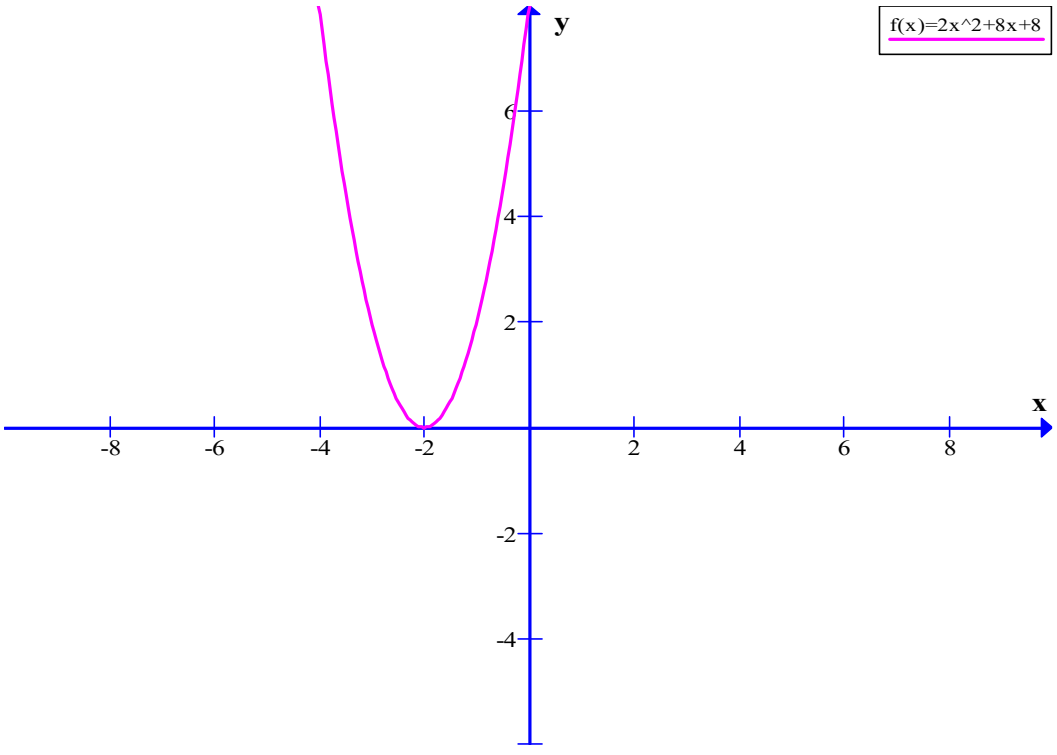


f) $y = -x^2 - 5x - 6$

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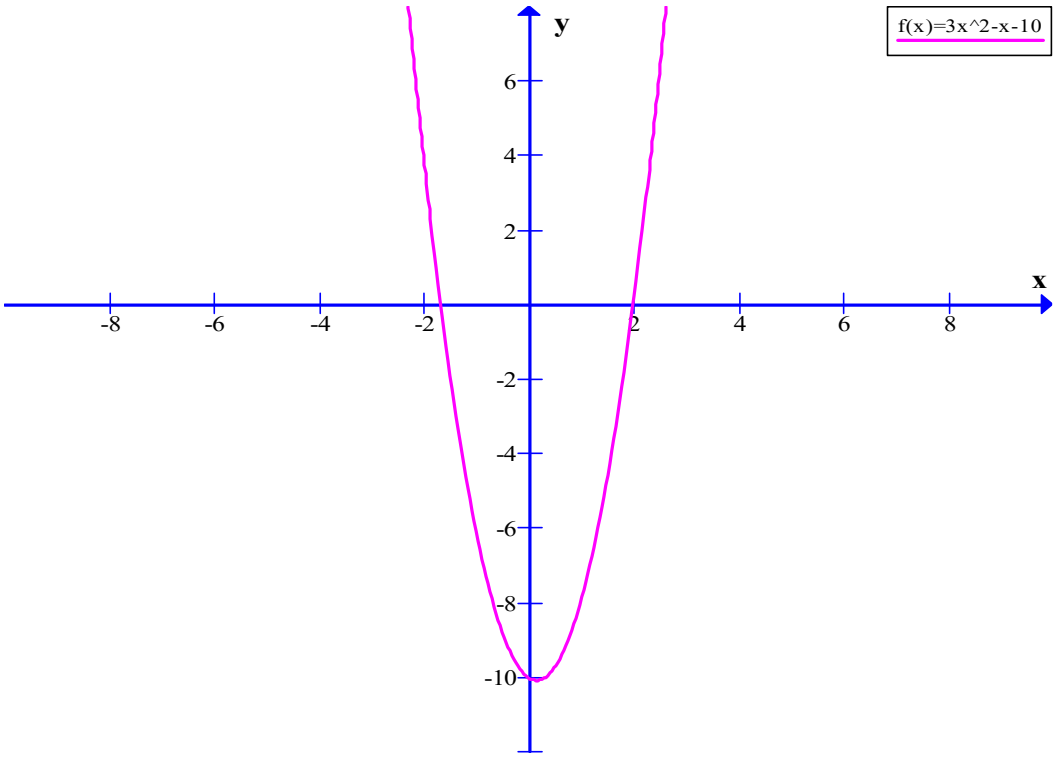


g) $y = 2x^2 + 8x + 8$

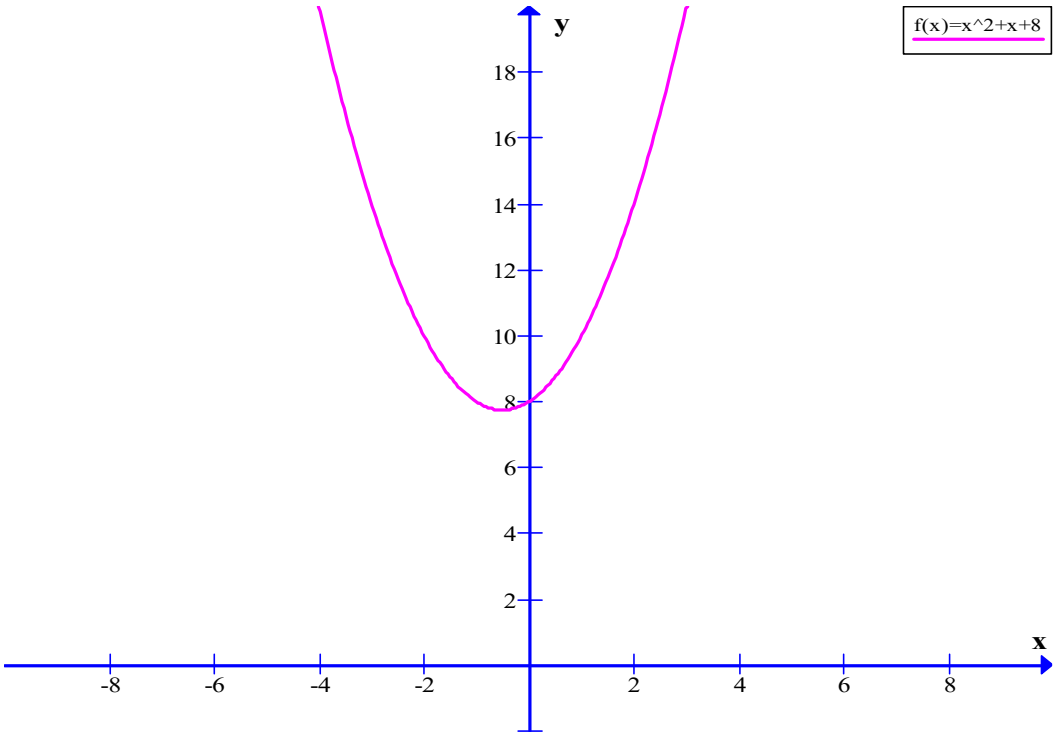


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h) $y = 3x^2 - x - 10$

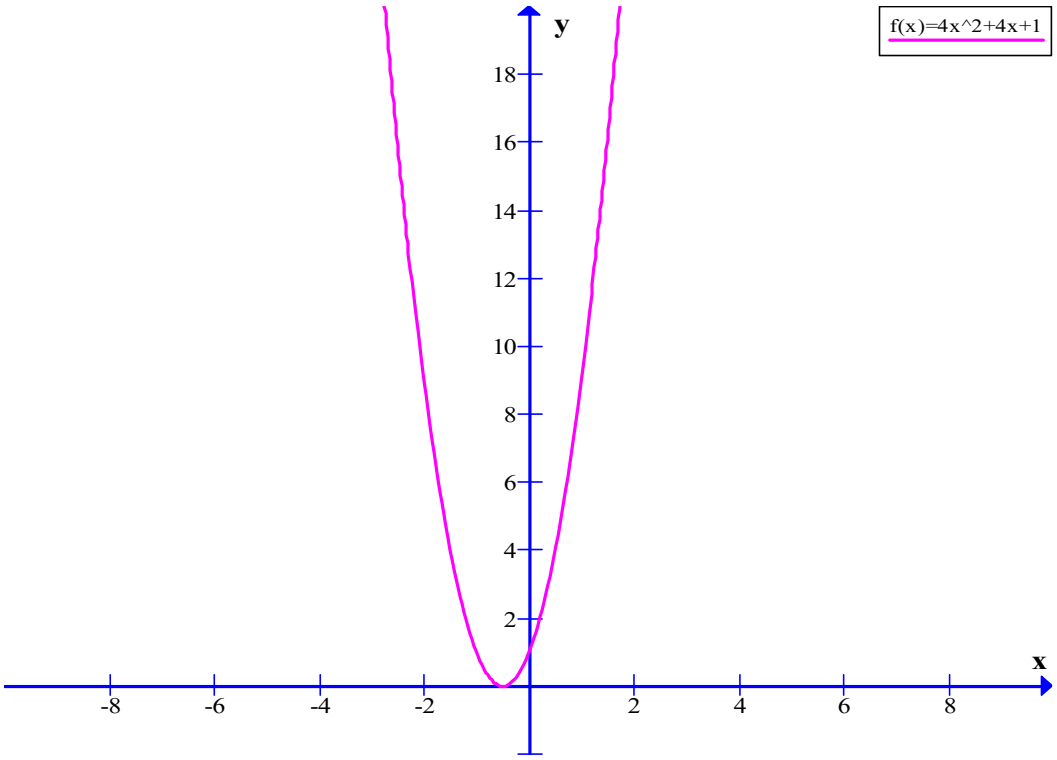


i) $y = x^2 + x + 8$



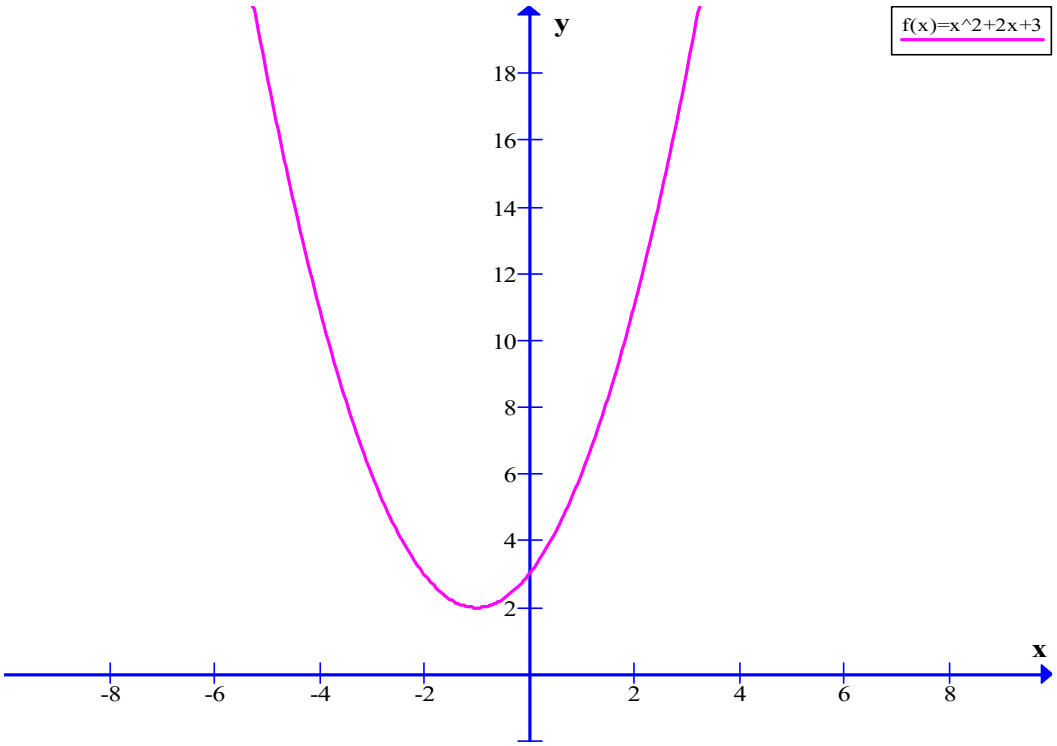
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j) $y = 4x^2 + 4x + 1$



k) $y = x^2 + 2x + 3$

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4) Using your graphs from question 3, what value(s) of x (if any) make the following inequalities true?

a) $x^2 \leq 0$

No values

b) $x^2 - 4 < 0$

$|x| < 2$

c) $x^2 + x - 6 > 0$

$x > 2 \; and \; x < -3$

d) $x^2 - 6x + 9 < 0$

No values

e) $x^2 - 4x + 3 < 0$

$x > 1 \; and \; x < 3$

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f) $-x^2 - 5x - 6 \geq 0$

$x \leq -2 \; and \; x \geq -3$

g) $2x^2 + 8x + 8 < 0$

No values

h) $3x^2 - x - 10 > 0$

$x < -\frac{5}{3} \; and \; x > 2$

i) $x^2 + x + 8 < 0$

No values

j) $4x^2 + 4x + 1 > 0$

$x > -\frac{1}{2}$

k) $x^2 + 2x + 3 > 0$

All values of x

5)

a) From your previous answers, what is the relationship between the solutions to a quadratic equation and the point(s) where the graph of the equation intersects the x axis?

They are the same values

b) From your previous answers, what is the relationship between the solutions to an inequality and the graph of the equation?

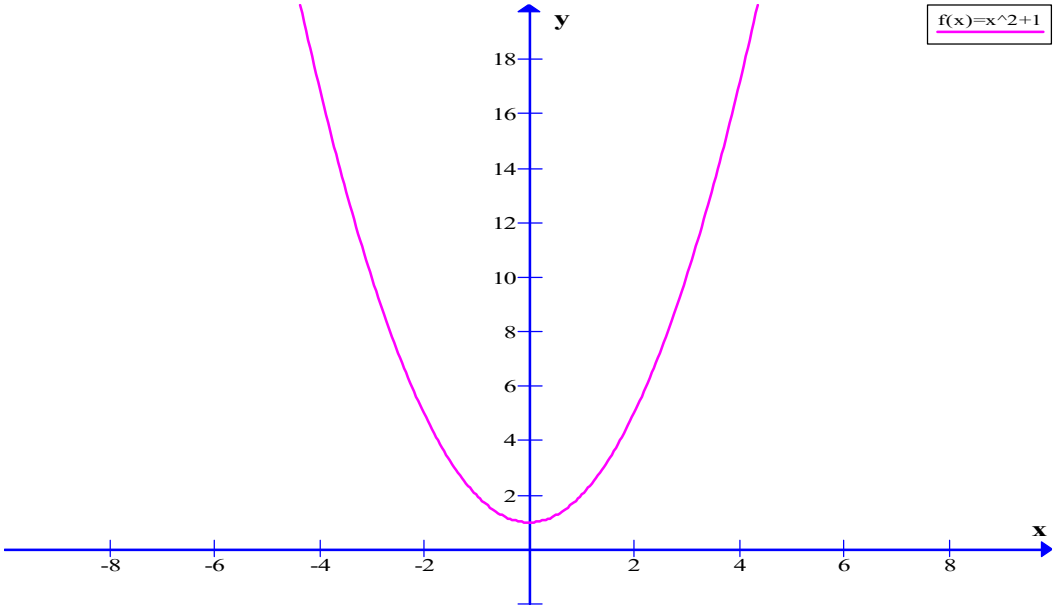
The solutions are all points for which
$x > or \; x < the \; roots \; of \; the \; equation, depending \; on \; the \; inequality$

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If an equation has no roots, an inequality can still be valid, but it is either for all x or for no values of x

6) By graphing the quadratic equations determine which values of x makes the following inequalities true

a) $x^2 + 1 \leq 0$

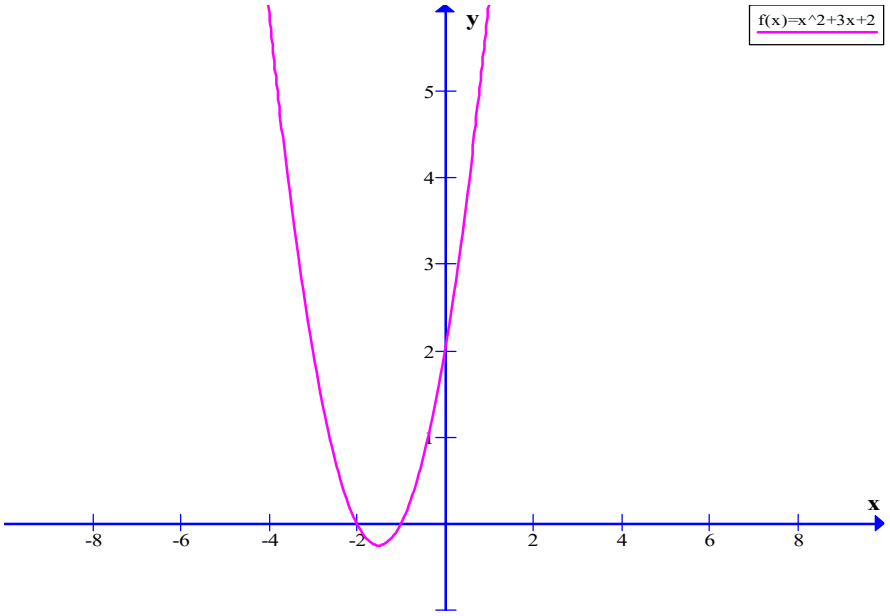


No values

b) $x^2 + 3x < -2$

$x^2 + 3x + 2 < 0$

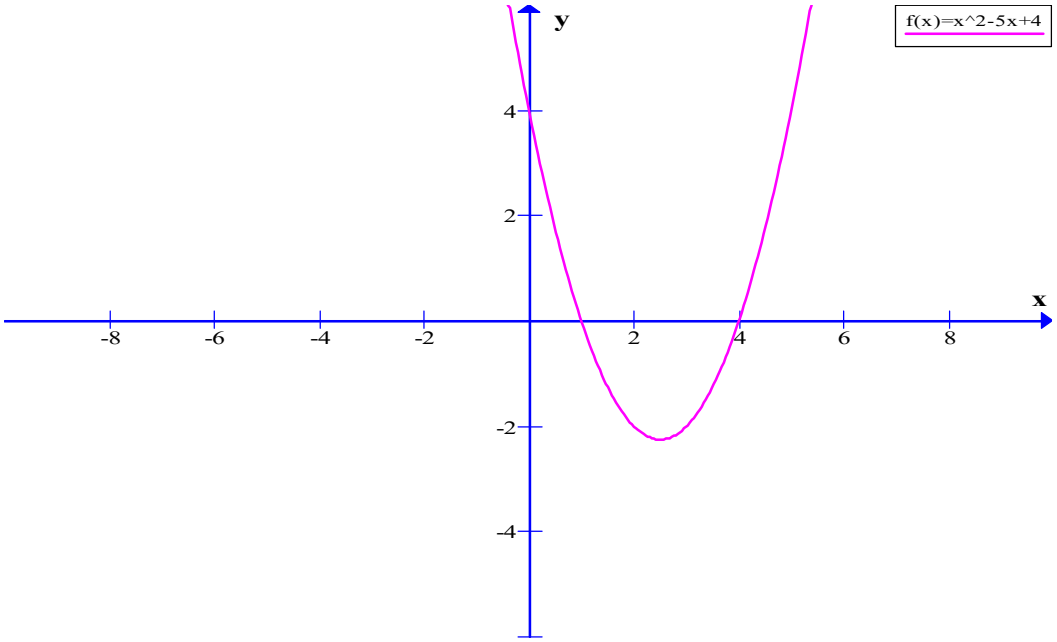
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$x < -1, x > -2$

c) $x^2 - 5x + 7 > 3$

$x^2 - 5x + 4 < 0$

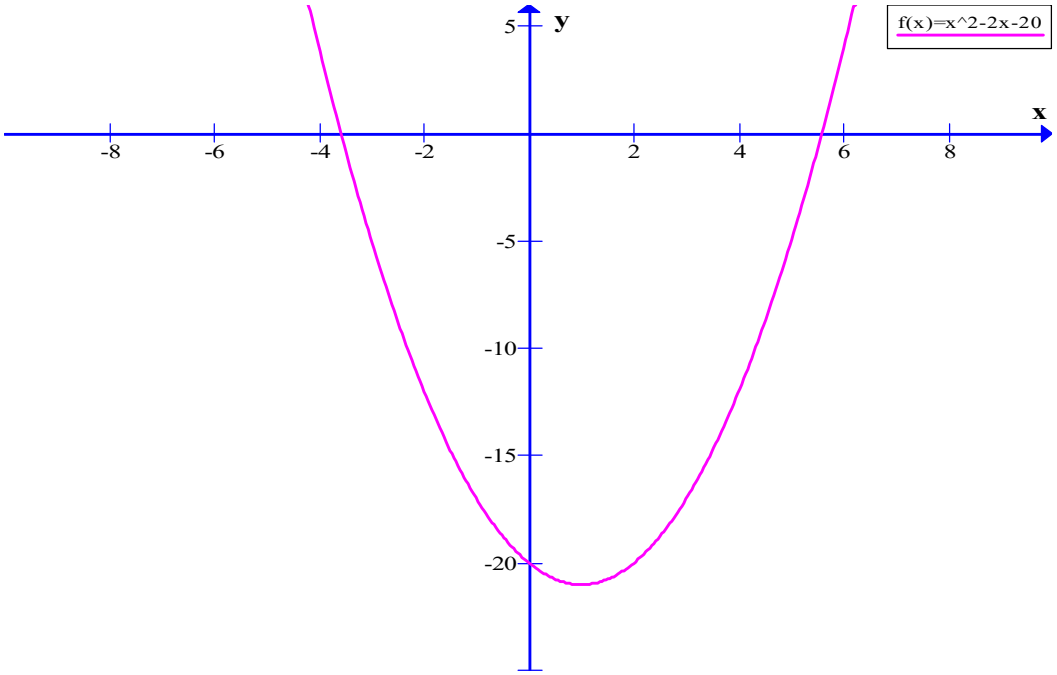


$x > 1, x < 4$

d) $x^2 - 2x - 8 < 12$

$x^2 - 2x - 20 < 0$

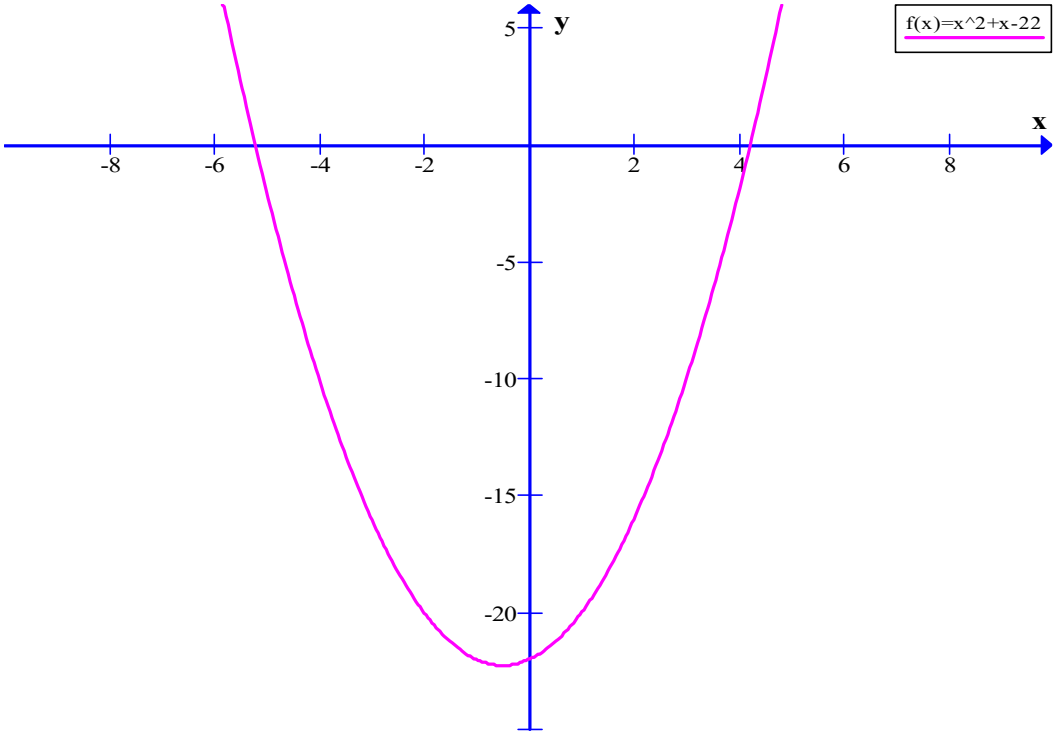
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$x < 5.58 \; and \; x > -3.58$

e) $x^2 + x - 17 > 5$

$x^2 + x - 22 > 0$

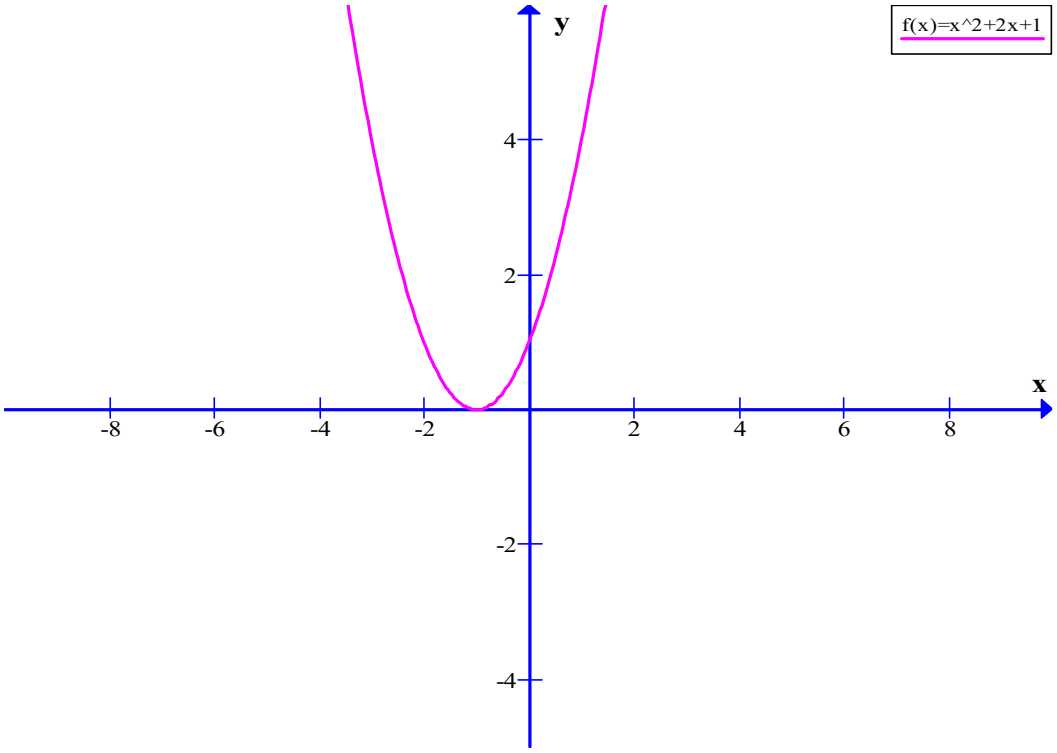


$x > -5.215 \; and \; x < 4.215$

f) $x^2 + 2x + 3 < 2$

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$x^2 + 2x + 1 < 0$

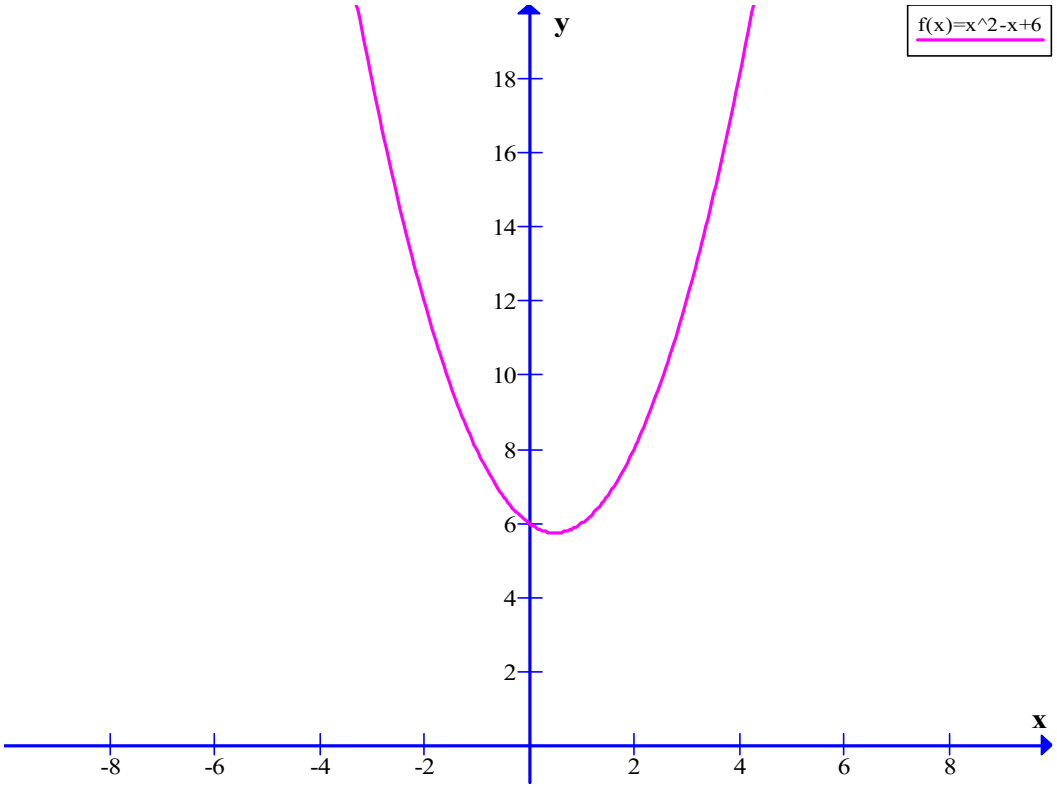


No values

g) $x^2 - x + 8 > 2$

$x^2 - x + 6 > 0$

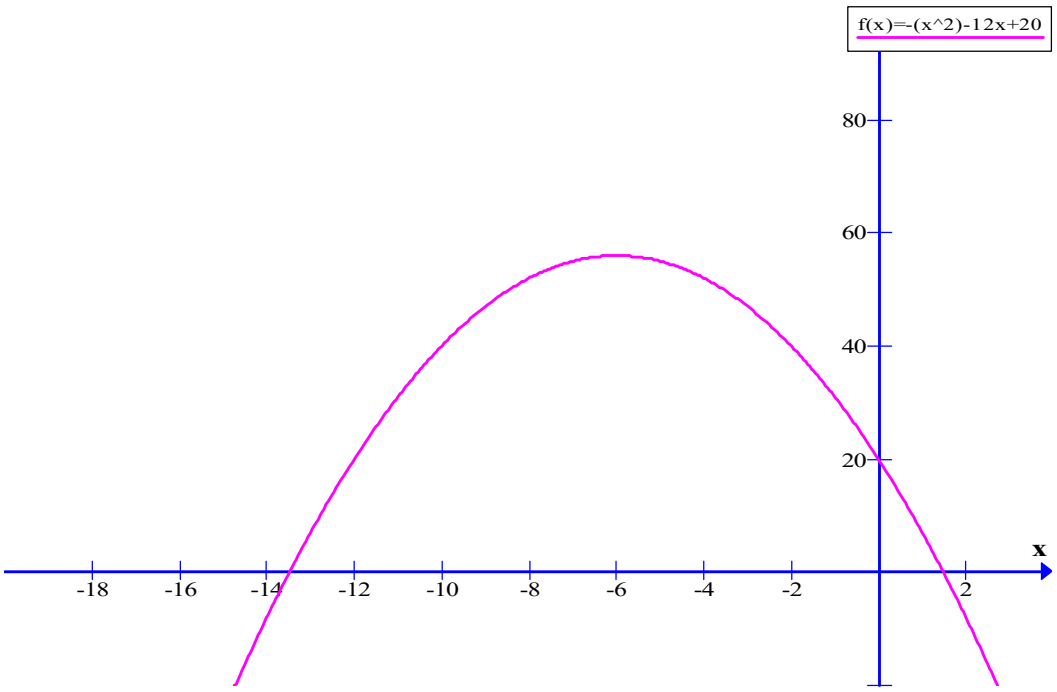
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All values of x

h) $-x^2 - 12x + 10 > -10$

$-x^2 - 12x + 20 > 0$



$x > -13.48 \; and \; x < 1.48$

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Exercise 2

Identities & Determinants

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1) Calculate the determinant of the following quadratic functions, and hence determine how many solutions exist for each

When equation is in the form

$y = ax^2 + bx + c$

Determinant $= b^2 - 4ac$

a) $y = x^2 - 3x + 2$

Det$= 9 - 4 \times 1 \times 2 = 1$

Therefore 2 solutions

b) $y = 2x^2 + 4x - \frac{3}{2}$

Det $= 16 - 4 \times 2 \times \frac{3}{2} = 4$

Therefore 2 solutions

c) $y = x^2 + 6x - 9$

Det $= 36 - 4 \times 1 \times -9 = 72$

Therefore 2 solutions

d) $y = 3x^2 + 3x + 1$

Det $= 9 - 4 \times 3 \times 1 = -3$

Therefore 0 solutions

e) $y = 4x^2 - 8x + 4$

Det $= 64 - 4 \times 4 \times 4 = 0$

Therefore 1 solution

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f) $y = 3x^2 + 5x - \frac{5}{4}$

Det $= 25 - 4 \times 3 \times -\frac{5}{4} = 40$

Therefore 2 solutions

g) $x^2 - \frac{4}{3}x - \frac{1}{3}$

Det $= \frac{16}{9} - 4 \times 1 \times -\frac{1}{3} = \frac{20}{9}$

Therefore 2 solutions

h) $x^2 - \sqrt{2}x + 1$

Det $= 2 - 4 \times 1 \times \sqrt{2} = 2 - 4\sqrt{2} < 0$

Therefore 0 solutions

i) $-2x^2 - 6x - 5$

Det $= 36 - 4 \times -2 \times -5 = -4$

Therefore 0 solutions

2) Express each of the following in the form $Ax(x-1) + Bx + C$; where: $A = a, C = c, and\ B = (a + b)$

a) $y = x^2 + 5x + 6$

$y = x(x-1) + 6x + 6$

b) $y = x^2 - 2x + 8$

$y = x(x-1) - x + 8$

c) $y = x^2 - 2$

$y = x(x-1) + x - 2$

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d) $y = 2x^2 - 3x + 6$

$y = 2x(x-1) - x + 6$

e) $y = 4x^2 + 3x - 5$

$y = 4x(x-1) + 7x - 5$

f) $y = x^2$

$y = x(x-1) + 1$

g) $y = \frac{1}{2}x^2 + \frac{1}{3}x - \frac{1}{4}$

$y = \frac{1}{2}x(x-1) + \frac{5}{6}x - \frac{1}{4}$

h) $y = -x^2 - x - 1$

$y = -x(x-1) - 2x - 1$

i) $y = -3x^2 - 3x - 3$

$y = -3x(x-1) - 6x - 3$

3) Find the quadratic equation that fits each of the three sets of points below

Substitute the three values in to the general quadratic formula and solve for a, b and c

a) (1,2) (0,6) (3,0)

$2 = a + b + c$

$6 = 0 + 0 + c$

$0 = 9a + 3b + c$

$c = 6$

$a + b = -4$

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$9a + 3b = -6$

Solving for a and b gives $a = 1, b = -5$

$y = x^2 - 5x + 6$

b) (2,8) (1,5) (-1,5)

$8 = 4a + 2b + c$

$5 = a + b + c$

$5 = a - b + c$

$a + b + c = a - b + c$

Therefore $2b = 0$

$b = 0$

$a + c = 5$

$4a + c = 8$

Solving for a and c gives

$a = 1, c = 4$

$y = x^2 + 4$

c) (1,3) (-2,18) (-1,9)

$3 = a + b + c, c = 3 - a - b$

$18 = 4a - 2b + c$

$9 = a - b + c, c = 9 - a + b$

$3 - a - b = 9 - a + b$

$2b = -6$

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$b = -3$

$c = 3 - a + 3$

$18 = 4a + 6 + c$

Solving for a and c gives

$a = 2, c = 4$

$y = 2x^2 - 3x + 4$

d) (2,-2) (-1,9) (0,6)

$-2 = 4a + 2b + c, c = -4a - 2b - 2$

$9 = a - b + c, \quad c = -a + b + 9$

$6 = 0 + 0 + c$

$c = 6$

$-4a - 2b - 2 = 6$

$-a + b + 9 = 6$

Solving for a and b gives

$a = -\frac{1}{3}, b = -\frac{10}{3}$

$3y = -x^2 - 10x + 18$

e) (1,1) (-2,-8) (-1,1)

$1 = a + b + c, \qquad c = 1 - a - b$

$-8 = 4a - 2b + c$

$1 = a - b + c, \qquad c = 1 - a + b$

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$1 - a - b = 1 - a + b$

$2b = 0, \qquad b = 0$

$-8 = 4a + c$

$c = 1 - a$

Solving for a and c gives

$a = -3, c = 4$

$y = -3x^2 + 4$

$a = -\frac{3}{2}$

$y = -\frac{3}{2}x^2 + \frac{3}{2} + 1$

$2y = -3x^2 + 3x + 2$

f) ($\frac{1}{2}$,-1) (1,0) (2,6)

$-1 = \frac{1}{4}a + \frac{1}{2}b + c$

$0 = a + b + c, c = -a - b$

$6 = 4a + 2b + c, c = 6 - 4a - 2b$

$6 - 4a - 2b = -a - b$

$3a + b = 6$

$-1 = \frac{1}{4}a + \frac{1}{2}b + (-a - b)$

$1 = \frac{3}{4}a + \frac{1}{2}b$

Solving for a and b gives

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$a = \frac{8}{3}, b = -2$

Substituting gives $c = -\frac{2}{3}$

$y = \frac{8}{3}x^2 - 2x - \frac{2}{3}$

$3y = 8x^2 - 6x - 2$

g) (2,4) ($\frac{3}{2}$,$\frac{9}{4}$) (-3,9)

$4 = 4a + 2b + c, \qquad c = 4 - 4a - 2b$

$\frac{9}{4} = \frac{9}{4}a + \frac{3}{2}b + c$

$9 = 9a - 3b + c, \qquad c = 9 - 9a + 3b$

$9 - 9a + 3b = 4 - 4a - 2b$

$5a - 5b = 5$

$a = 1 + b$

$\frac{9}{4} = \frac{9}{4}a + \frac{3}{2}b + (4 - 4a - 2b) = -\frac{7}{4}a - \frac{1}{2}b + 4$

Solving for a and b gives

$b = 0, a = 1, c = 0$

$y = x^2$

h) (1,2) (-2,20) (0,2)

$2 = a + b + c, \qquad c = 2 - a - b$

$20 = 4a - 2b + c$

$2 = 0 + 0 + c$

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$c = 2$

$a + b = 0$

$20 = 4a - 2b + 2$

Solving for a and b gives

$a = 3, b = -3$

$y = 3x^2 - 3x + 2$

i) (1,-5) (2,7) ($\frac{1}{2}$, -8)

$-5 = a + b + c, \quad c = -5 - a - b$

$7 = 4a + 2b + c, \quad c = 7 - 4a - 2b$

$-8 = \frac{1}{4}a + \frac{1}{2}b + c$

$-5 - a - b = 7 - 4a - 2b$

$3a + b = 12$

$b = 12 - 3a$

$-8 = \frac{1}{4}a + \frac{1}{2}(12 - 3a) + c = -\frac{5}{4}a + c + 6$

Solving for a and c gives

$a = 4, c = -9$

$b = 0$

$y = 4x^2 - 9$

j) (1,64) (-1,4) ($\frac{1}{3}$, 36)

$64 = a + b + c, c = 64 - a - b$

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$4 = a - b + c, c = 4 - a + b$

$64 - a - b = 4 - a - b$

$2b = 60$

$b = 30$

$36 = \frac{1}{9}a + \frac{1}{3}b + c = \frac{1}{9}a + \frac{1}{3}b + (4 - a + b) = -\frac{8}{9}a + 4 + 40$

$-\frac{8}{9}a = -8$

$a = 9$

$c = 25$

$y = 9x^2 + 30x + 25$

4) Solve the following by first reducing them to quadratic equations of the form $ax^2 + bx + c = 0$

a) $x^4 + x^2 - 6 = 0$

$(x^2)^2 + (x^2) - 6 = 0$

$(x^2 + 3)(x^2 - 2) = 0$

Real solutions are $x = \pm\sqrt{2}$

b) $x^6 - 4x^3 + 4 = 0$

$(x^3)^2 - 4(x^3) + 4 = 0$

$(x^3 - 2)(x^3 - 2) = 0$

Solution is $x = \sqrt[3]{2}$

c) $4x^4 + 2x^2 - 8 = 0$

$(2x^2)^2 + 2(x^2) - 8 = 0$

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$(2x^2 + 4)(2x^2 - 2) = 0$

Real solutions are $x = \pm 1$

d) $8x + 2\sqrt{x} - 1 = 0$

$2\left(2\sqrt{x}\right)^2 + \left(\sqrt{x}\right) - 1 = 0$

$\left(4\sqrt{x} - 1\right)\left(2\sqrt{x} + 1\right) = 0$

Real solutions are $x = \frac{1}{16}$

e) $(x + 2)^2 = 4x^2 + 1$

$x^2 + 4x + 4 = 4x^2 + 1$

$3x^2 - 4x - 3 = 0$

Solutions are $x = 1.867$ *or* $x = -0.5333$

f) $(x - 3)^2 + 2 = (x + 1)^2 - 1$

$x^2 - 6x + 9 + 2 = x^2 + 2x + 1 - 1$

$-6x + 11 = 2x$

$x = \frac{11}{8}$

g) $(x - 4)^2 - 12 = x + 1$

$x^2 - 8x + 16 - 12 = x + 1$

$x^2 - 9x + 3 = 0$

Solutions are $x = 8.65, x = 0.65$

h) $4^x - 2(2)^x + 1 = 0$

$2^{2x} - 2(2)^x + 1 = 0$

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$(2^x - 1)(2^x - 1) = 0$

Solution is $2^x = 1$

$x = 0$

i) $16^x - 6(4)^x + 8 = 0$

$(4)^{2x} - 6(4)^x + 8 = 0$

$(4^x - 4)(4^x - 2) = 0$

$4^x = 4, 4^x = 2$

$x = 1, x = \frac{1}{2}$

j) $81^x - 4(3)^{2x} + 3 = 0$

$3^{4x} + 4(3)^{2x} + 3 = 0$

$(3^{2x} - 3)(3^{2x} - 1) = 0$

$3^{2x} = 3, 3^{2x} = 1$

$2x = 1, x = 0$

$x = \frac{1}{2}, x = 0$

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Exercise 3

Equations of Parabolas

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1) Find the equations of the parabolas defined by the given focus, axis and directrix.

a) Focus at (0,1), axis $x = 0$, directrix $y = -1$

$x_0 = 0, y_0 = 0, A = 1$

$(x - 0)^2 = 4 \times 1\ (y - 0)$

$x^2 = 4y$

$y = \frac{x^2}{4}$

b) Focus at $(0,\frac{1}{2})$, axis $x = 0$, directrix $y = -\frac{1}{2}$

$x_0 = 0, y_0 = 0, A = \frac{1}{2}$

$(x - 0)^2 = 4 \times \frac{1}{2}\ (y - 0)$

$x^2 = 2y$

$y = \frac{x^2}{2}$

c) Focus at $(0,\frac{1}{4})$, axis $x = 0$, directrix $y = -\frac{1}{4}$

$x_0 = 0, y_0 = 0, A = \frac{1}{4}$

$(x - 0)^2 = 4 \times \frac{1}{4}\ (y - 0)$

$y = x^2$

d) Focus at (0,4), axis $x = 0$, directrix $y = -4$

$x_0 = 0, y_0 = 0, A = 4$

$(x - 0)^2 = 4 \times 4\ (y - 0)$

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$x^2 = 16y$

$y = \frac{x^2}{16}$

2) Find the equations of the parabolas defined by the given focus, axis and directrix.

a) Focus at (2,1), axis $x = 2$, directrix $y = -1$

$x_0 = 2, y_0 = 0, A = 1$

$(x-2)^2 = 4 \times 1\ (y-0)$

$x^2 - 4x + 4 = 4y$

$y = \frac{x^2}{4} - x + 1$

b) Focus at (3,-3), axis $x = 3$, directrix $y = 3$

$x_0 = 3, y_0 = 0, A = -3$

$(x-3)^2 = 4 \times -3\ (y-0)$

$x^2 - 6x + 9 = -12y$

c) Focus at (-2,-2), axis $x = -2$, directrix $y = 2$

$x_0 = -2, y_0 = 0, A = -2$

$(x+2)^2 = 4 \times -2\ (y-0)$

$x^2 + 4x + 4 = -8y$

d) Focus at $(1,\frac{1}{2})$, axis $x = 1$, directrix $y = -\frac{1}{2}$

$x_0 = 1, y_0 = 0, A = \frac{1}{2}$

$(x-1)^2 = 4 \times \frac{1}{2}\ (y-0)$

$x^2 - 2x + 1 = 2y$

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3) Find the equations of the parabolas defined by the given focus, axis and directrix.

a) Focus at (0,-4), axis $x = 0$, directrix $y = 6$

$x_0 = 0, y_0 = 1, A = -5$

$(x - 0)^2 = 4 \times -5\ (y - 1)$

$x^2 = -20y + 20$

$20y = -x^2 + 20$

b) Focus at (0,-2), axis $x = 0$, directrix $y = 2$

$x_0 = 0, y_0 = 0, A = -2$

$(x - 0)^2 = 4 \times -2\ (y - 0)$

$x^2 = -8y$

$y = -\frac{x^2}{8}$

c) Focus at (0,1), axis $x = 0$, directrix $y = -3$

$x_0 = 0, y_0 = -1, A = 2$

$(x - 0)^2 = 4 \times -2\ (y + 1)$

$x^2 = -8y - 8$

$8y = -x^2 - 8$

d) Focus at (0,3), axis $x = 0$, directrix $y = -1$

$x_0 = 0, y_0 = 1, A = 2$

$(x - 0)^2 = 4 \times 2\ (y - 1)$

$x^2 = 8y - 8$

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$8y = x^2 + 8$

4) Find the equations of the parabolas defined by the given focus, axis and directrix.

a) Focus at (3,1), axis $x = 3$, directrix $y = 0$

$x_0 = 3, y_0 = \frac{1}{2}, A = \frac{1}{2}$

$(x-3)^2 = 4 \times \frac{1}{2}\left(y - \frac{1}{2}\right)$

$x^2 - 6x + 9 = 2y - 1$

$2y = x^2 - 6x + 10$

b) Focus at (2,-4), axis $x = 2$, directrix $y = -6$

$x_0 = 2, y_0 = -5, A = 1$

$(x-2)^2 = 4 \times 1\ (y - -5)$

$x^2 - 4x + 4 = 4y + 20$

$4y = x^2 - 4x - 16$

c) Focus at $(1,\frac{1}{2})$, axis $x = 1$, directrix $y = 1$

$x_0 = 1, y_0 = \frac{3}{4}, A = -\frac{1}{4}$

$(x-1)^2 = 4 \times -\frac{1}{4}\left(y - \frac{3}{4}\right)$

$x^2 - 2x + 1 = -y + \frac{3}{4}$

$y = -x^2 + 2x - \frac{1}{4}$

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d) Focus at (-2,-1), axis $x = -2$, directrix $y = 5$

$x_0 = -2, y_0 = 2, A = -3$

$(x+2)^2 = 4 \times -3\,(y-2)$

$x^2 + 4x + 4 = -12y + 24$

$12y = -x^2 - 4x + 20$

5) By rewriting the following in parabolic form, find the focus, vertex, axis and directrix

a) $y = x^2$

$4 \times \frac{1}{4}(y-0) = (x-0)^2$

Vertex is $(0,0)$

Focus is $\left(0, \frac{1}{4}\right)$

Axis is $x = 0$

Directrix is $y = -\frac{1}{4}$

b) $y = x^2 + 4$

$y - 4 = (x-0)^2$

$4 \times \frac{1}{4}(y-4) = (x-0)^2$

Vertex is $(0,4)$

Focus is $\left(0, \frac{17}{4}\right)$

Axis is $x = 0$

Directrix is $y = \frac{15}{4}$

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c) $y = x^2 - 3x + 2$

$$y = \left(x - \frac{3}{2}\right)^2 - \frac{1}{4}$$

$$y + \frac{1}{4} = \left(x - \frac{3}{2}\right)^2$$

$$4 \times \frac{1}{4}\left(y + \frac{1}{4}\right) = \left(x - \frac{3}{2}\right)^2$$

Vertex is $\left(\frac{3}{2}, -\frac{1}{4}\right)$

Focus is $\left(\frac{3}{2}, 0\right)$

Axis is $x = \frac{3}{2}$

Directrix is $y = -\frac{1}{2}$

d) $y = 2x^2 + 3x - 2$

$$\frac{1}{2}y = x^2 + \frac{3}{2}x - 1$$

$$\frac{1}{2}y = \left(x + \frac{3}{4}\right)^2 - \frac{25}{16}$$

$$\frac{1}{2}y + \frac{25}{16} = \left(x + \frac{3}{4}\right)^2$$

$$4 \times \frac{1}{8}\left(y + \frac{25}{8}\right) = \left(x + \frac{3}{4}\right)^2$$

Vertex is $\left(-\frac{3}{4}, -\frac{25}{8}\right)$

Focus is $\left(-\frac{3}{4}, -3\right)$

Axis is $x = -\frac{3}{4}$

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Directrix is $y = -\frac{13}{4}$

e) $y = \frac{1}{2}x^2 - \frac{1}{4}x + 1$

$$2y = x^2 - \frac{1}{2}x + 2$$

$$2y = \left(x - \frac{1}{4}\right)^2 + \frac{31}{16}$$

$$2y - \frac{31}{16} = \left(x - \frac{1}{4}\right)^2$$

$$2\left(y - \frac{31}{32}\right) = \left(x - \frac{1}{4}\right)^2$$

$$4 \times \frac{1}{2}\left(y - \frac{31}{32}\right) = \left(x - \frac{1}{4}\right)^2$$

Vertex is $\left(\frac{1}{4}, \frac{31}{32}\right)$

Focus is $\left(\frac{1}{4}, \frac{47}{32}\right)$

Axis is $x = \frac{1}{4}$

Directrix is $y = \frac{15}{32}$

f) $y = 4x^2 - 6x + 2$

$$\frac{1}{4}y = x^2 - \frac{3}{2}x + \frac{1}{2}$$

$$\frac{1}{4}y = \left(x - \frac{3}{4}\right)^2 - \frac{1}{16}$$

$$\frac{1}{4}y + \frac{1}{16} = \left(x - \frac{3}{4}\right)^2$$

$$\frac{1}{4}\left(y + \frac{1}{4}\right) = \left(x - \frac{3}{4}\right)^2$$

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$$4 \times \frac{1}{16}\left(y + \frac{1}{4}\right) = \left(x - \frac{3}{4}\right)^2$$

Vertex is $\left(\frac{3}{4}, -\frac{1}{4}\right)$

Focus is $\left(\frac{3}{4}, -\frac{3}{16}\right)$

Axis is $x = \frac{3}{4}$

Directrix is $y = -\frac{5}{16}$

6) Find the general equation of the parabola with axis $x = 2$, and vertex at the point $(2, y_0)$ by considering the values of $y_0$to be

General equation will be $(x-2)^2 = 4A(y - y_0)$, where A is the distance from the vertex to the focus

a) -1

$(x-2)^2 = 4A(y+1)$

b) -4

$(x-2)^2 = 4A(y+4)$

c) 1

$(x-2)^2 = 4A(y-1)$

d) 0

$(x-2)^2 = 4Ay$

e) 3

$(x-2)^2 = 4A(y-3)$

f) $\frac{1}{2}$

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$$(x-2)^2 = 4A\left(y-\frac{1}{2}\right)$$

7) Find the general equation of the parabola with axis $x = -3$, having a focal length of A by considering the values of A to be

General equation will be: $(x+3)^2 = 4A(y-y_0)$, where y_0 is the y co-ordinate of the vertex

a) 2

$$(x+3)^2 = 8(y-y_0)$$

b) 4

$$(x+3)^2 = 16(y-y_0)$$

c) 1

$$(x+3)^2 = 4(y-y_0)$$

d) -3

$$(x+3)^2 = -12(y-y_0)$$

e) 0

$$(x+3)^2 = 0$$

f) -2

$$(x+3)^2 = -8(y-y_0)$$

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Year 11 Unit 2 Mathematics

Plane Geometry

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Exercise 1

Angles Formed by Transversals

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1) From the diagram below, give examples of the following pairs of angles

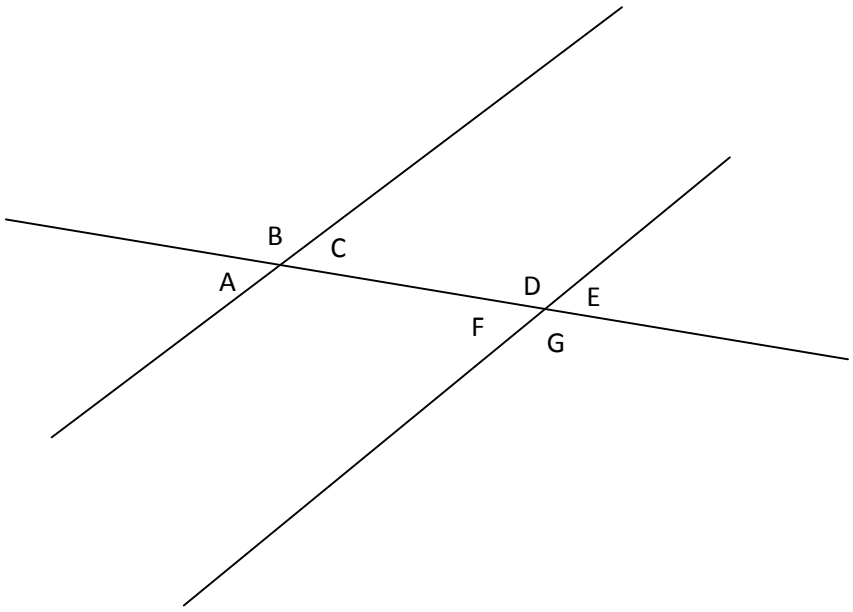
a) Vertically opposite: (A, C) (D, G) (E, F)

b) Alternate interior: (C, F)

c) Corresponding: (B, D) (C, E) (A, F)

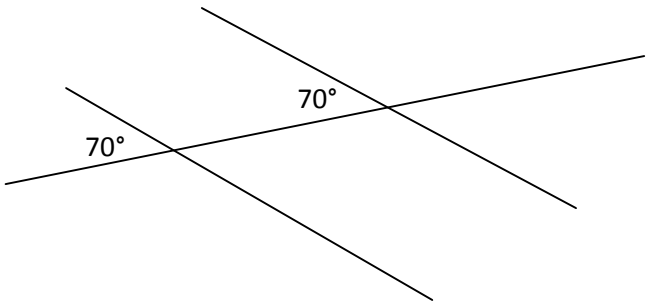
d) Co-interior: (C, D)

e) Alternate exterior: (A, E) (B, G)



2) Identify which diagrams show parallel and which show non parallel lines; give reasons for your answers

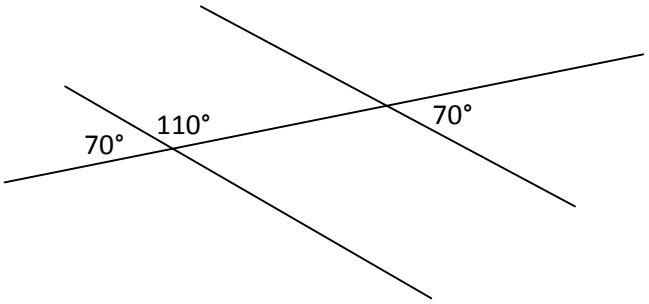
a)



Parallel since corresponding angles are equal

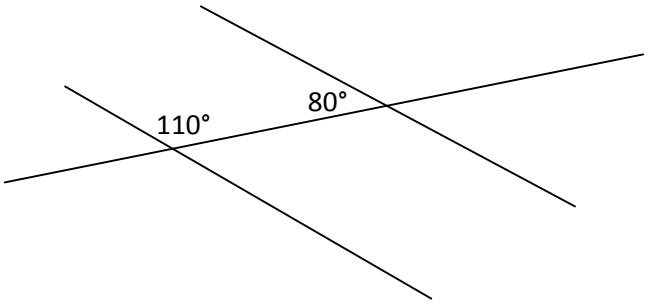
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b)



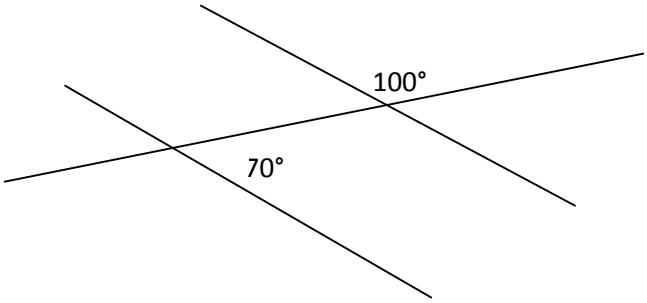
Parallel since alternate exterior angles are equal

c)



Not parallel since co-interior angles do not add to 180°

d)

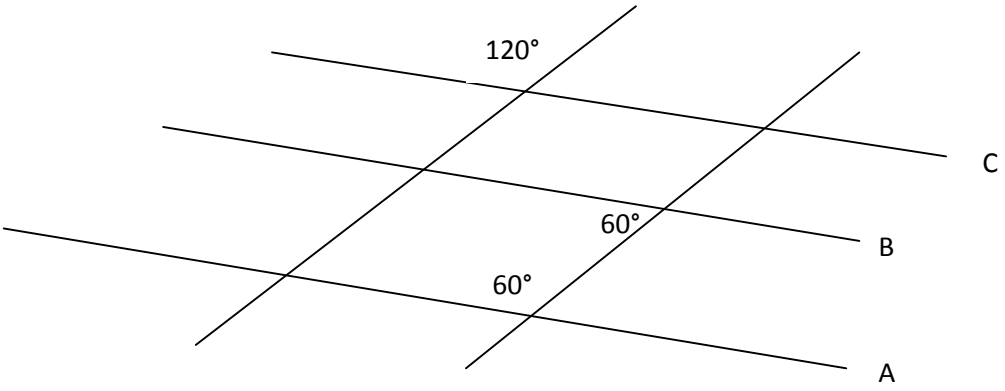


Not parallel since angle to left of 70° is 110°, which is not equal to corresponding angle of 100°

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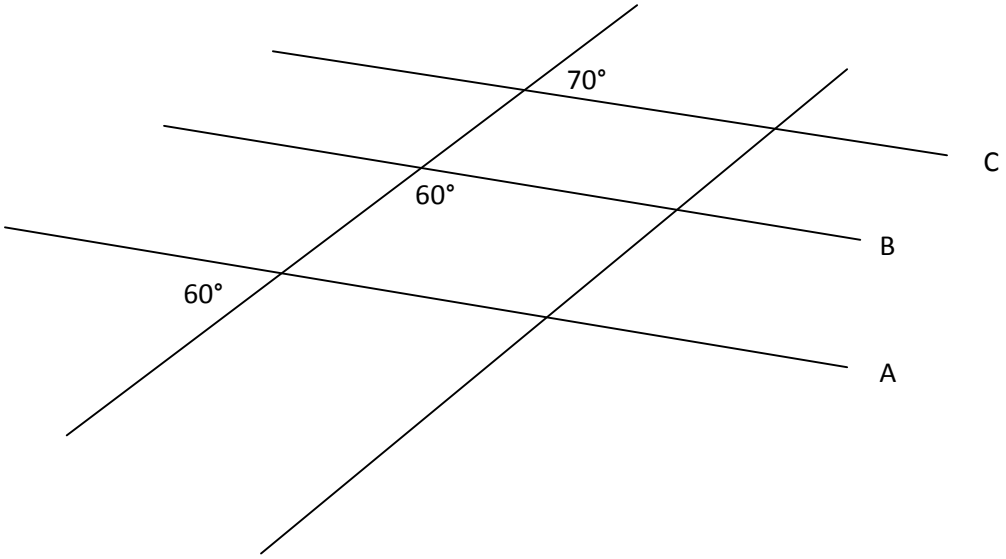
3) For each of the diagrams below, state which of the lines A, B and C are parallel to each other, giving reasons for your answers. Assume that the transversals are parallel to each other

a)



The angle next to 120° is 60°. It is alternate exterior to the angle on line B, which is corresponding to the angle on line A. Therefore all lines are parallel

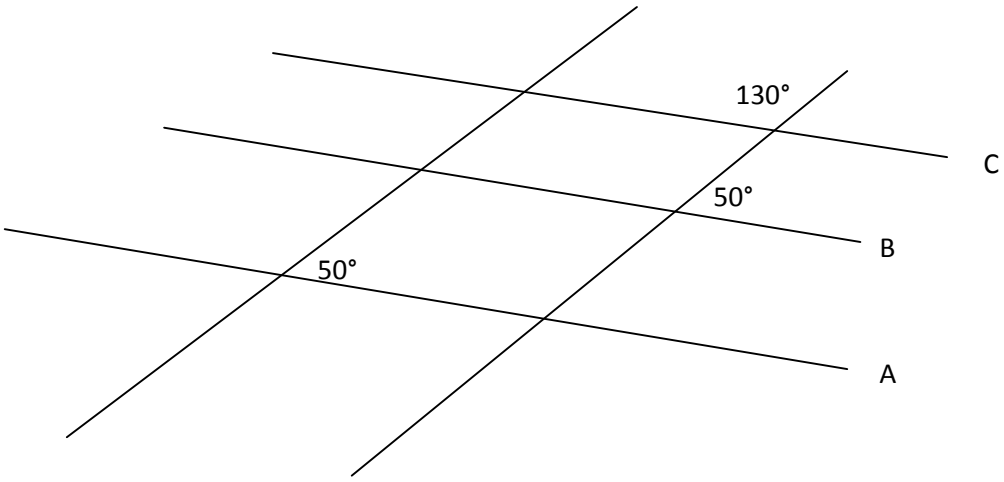
b)



A and B are not parallel due to corresponding angles not being equal, as is the case for B and C

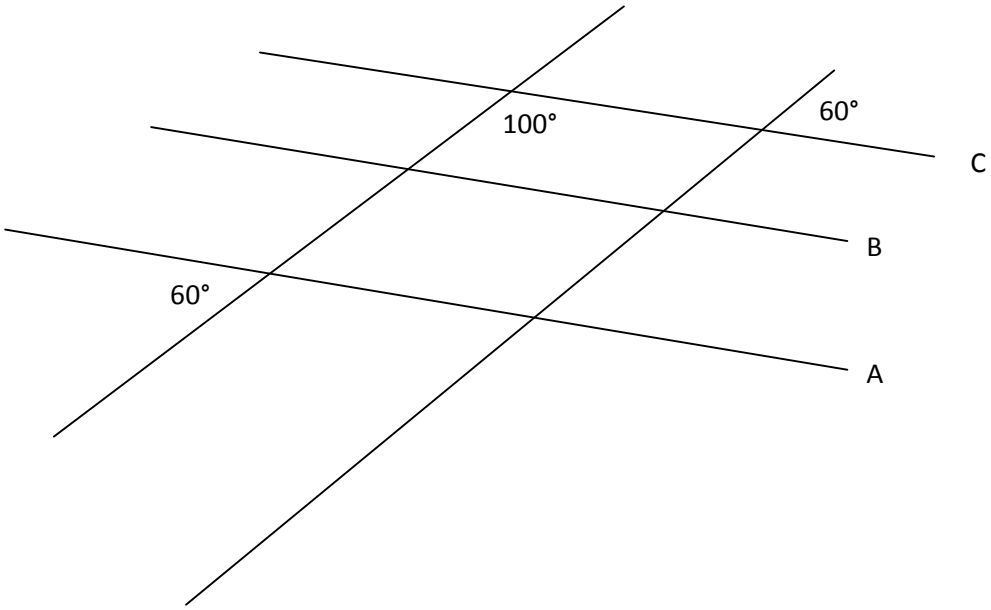
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c)



All lines are parallel; A and B have alternate exterior angles equal, and B and C have co-interior angles equal

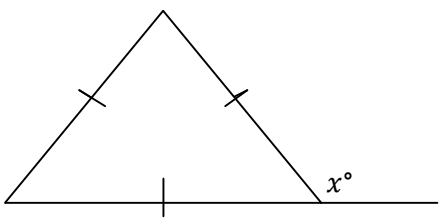
d)



A and C have alternate exterior angles equal and are parallel, B is parallel to neither as vertically opposite angles are not equal between B and C

4) Find the value of x in each of the following

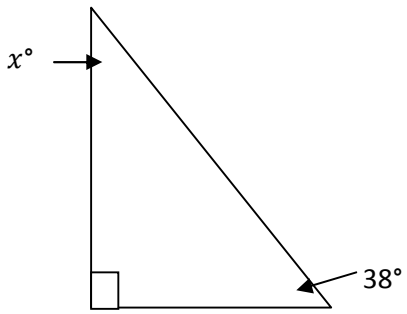
a)



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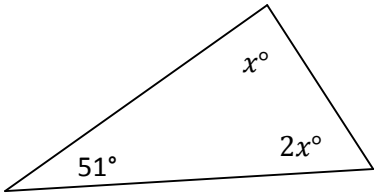
Triangle is equilateral , therefore all angles are 60°, so $x = 120°$

b)



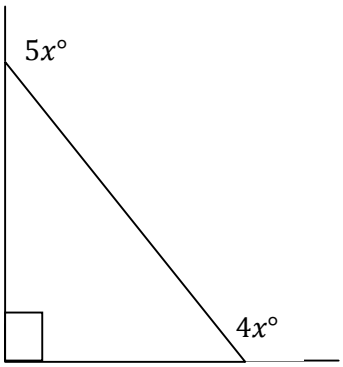
Two known angles are 90° and 38°, therefore $x = 52°$

c)



$3x = 129, x = 43°$

d)



Two internal angles $= 90°$

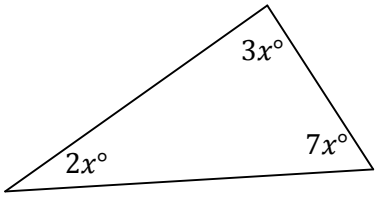
$5x + 4x + 2\ internal\ angles = 360°$

$9x = 270°$

$x = 30°$

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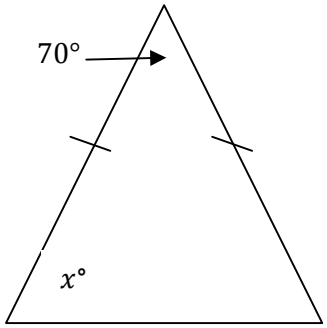
e)



$2x + 7x + 3x = 180°$

$x = 15°$

f)



$2x = 110°$

$x = 55°$

5)

a) Find the size of an interior angle of a regular pentagon

The sum of the interior angles of an n-sided polygon is

Interior angle of a polygon of n sides $= \frac{n-2}{n} \times 180°$

Angle of pentagon (5 sides) is $\frac{3}{5} \times 180° = 108°$

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b) What is the sum of the internal angles of a regular octagon?

One internal angle $= \frac{6}{8} \times 180° = 135°$

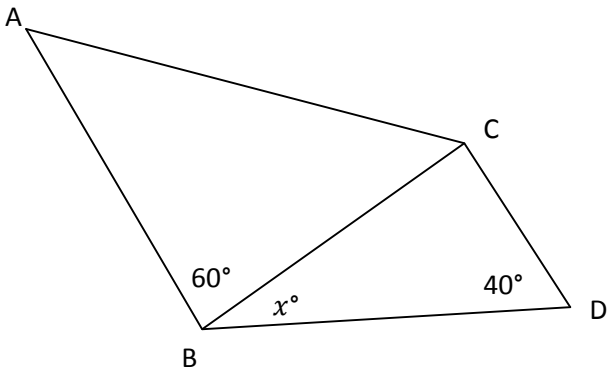
Sum of angles = $8 \times 135° = 1080°$

c) What is the sum of the external angles of a regular nonagon (Taking one angle per vertex)?

The sum of the external angles of any polygon is 360°

6) Find the value of x in the following

a) AB || CD

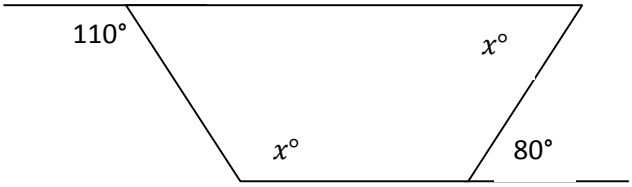


$Angle\ ACB = Angle\ BCD$

Therefore $Angle\ BCD = 60°$

$x = 80°$

b)



Interior angles of a quadrilateral sum to 360°

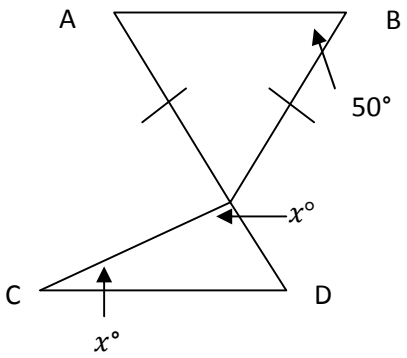
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Two other angles sum to $70° + 100° = 170°$

$2x = 190°$

$x = 95°$

c) AB || CD



$Angle\ BAD = 50°$

Therefore $Angle\ CDA = 50°$

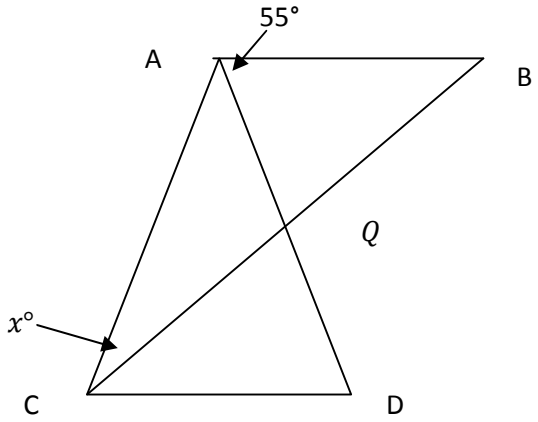
$2x = 130°$

$x = 65°$

d) AB || CD

AD ⊥ BC

AD = AC



Find the size of angle ACB

$Angle\ ADC = 55°, Angle\ ACD = 55°, Angle\ CAD = 70°, Angle\ CQA = 90°$

Therefore $Angle\ ACB = 20°$

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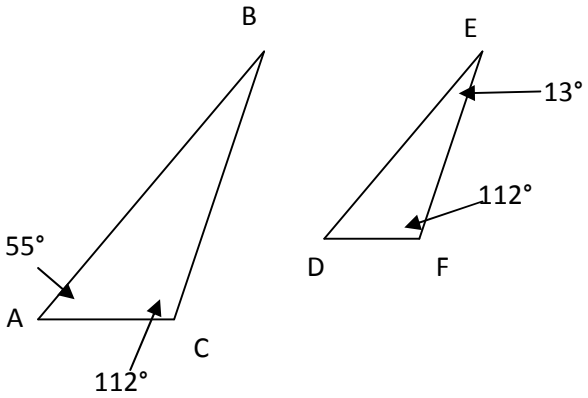
Exercise 2

Similarity & Congruence

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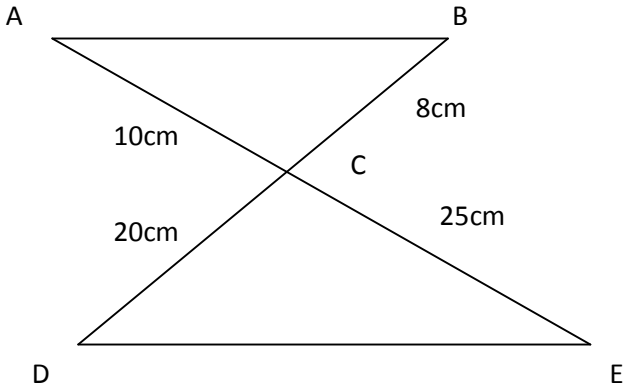
1) Determine if each pair of triangles is similar. If so, state the similarity conditions met

a)



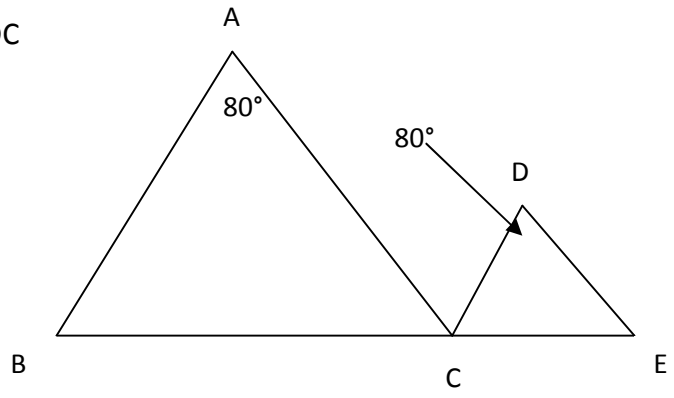
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b)



SAS

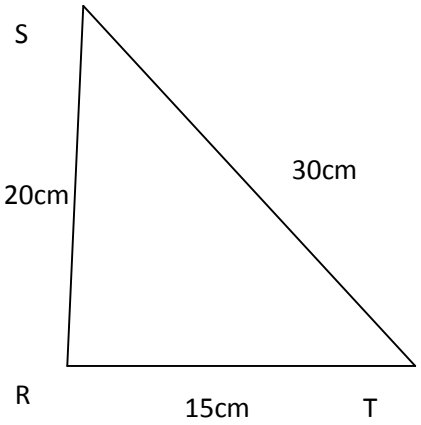
c) AB || DC

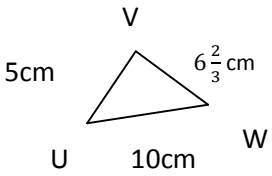


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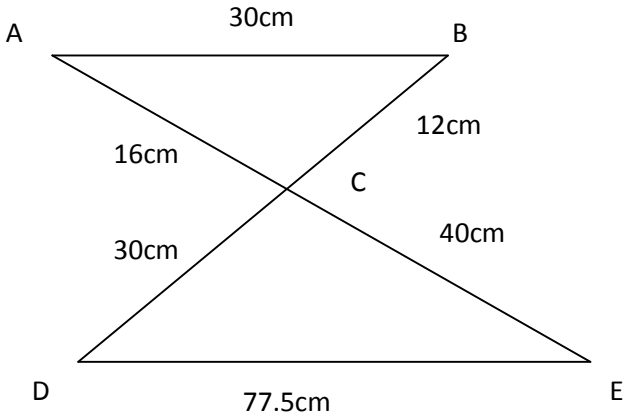
d)





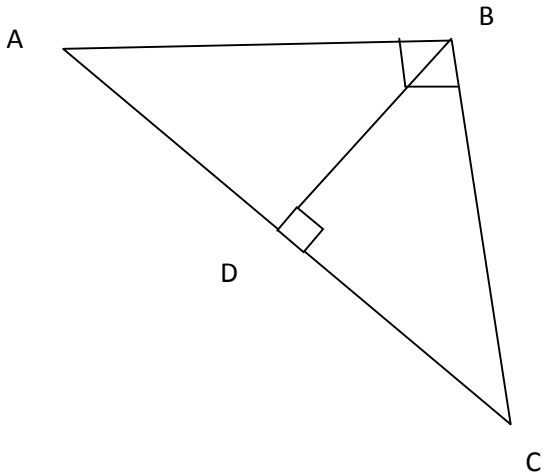
SSS

e)



Not similar, ratios of sides are not equal

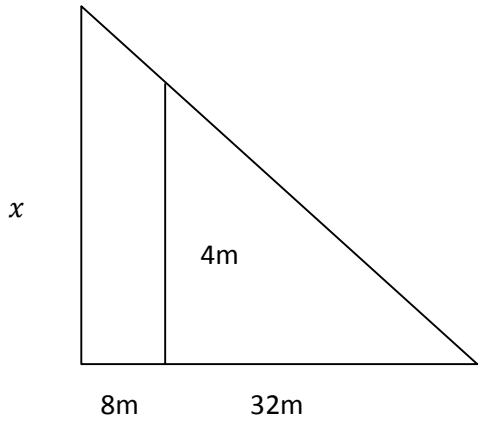
f)



AA

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2) A tower casts a shadow of 40 metres, whilst a 4 metre pole nearby casts a shadow of 32 metres. How tall is the tower?

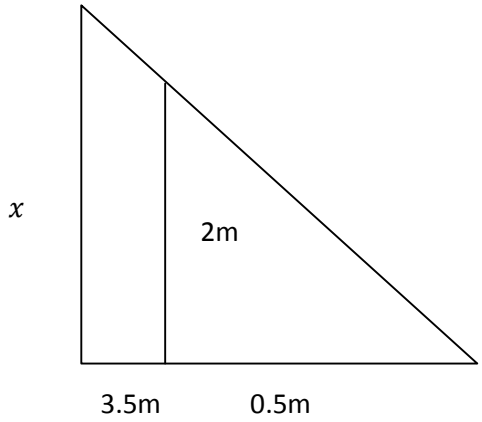


The triangles are similar therefore

$\frac{x}{4} = \frac{40}{32}$

$x = 5m$

3) A pole casts a 4 metre shadow, whilst a man standing near the pole casts a shadow of 0.5 metres. If the man is 2 metres tall, how tall is the pole?



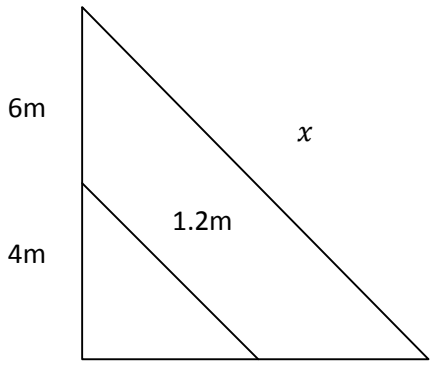
The triangles are similar therefore

$\frac{x}{2} = \frac{4}{0.5}$

$x = 16m$

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4) A ladder of length 1.2 metres reaches 4 metres up a wall when placed on a safe angle on the ground. How long should a ladder be if it needs to reach 10 metres up the wall, and be placed on the same safe angle?

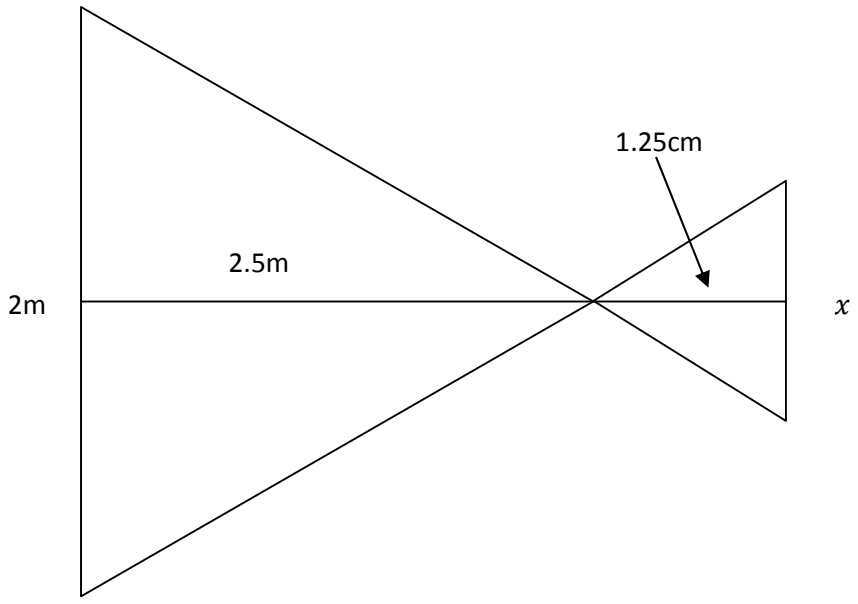


The triangles are similar therefore

$\frac{x}{10} = \frac{1.2}{4}$

$x = 3m$

5) A man stands 2.5 metres away from a camera lens, and the film is 1.25 centimetres from the lens (the film is behind the lens). If the man is 2 metres tall how tall is his image on the film?

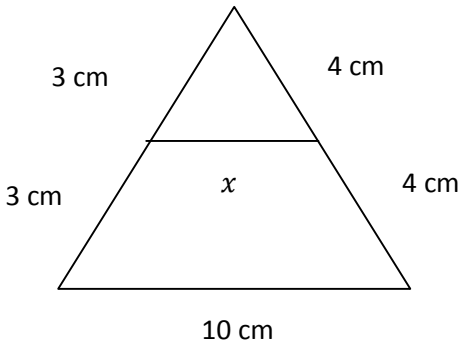


The triangles are similar therefore $\frac{x}{2} = \frac{1.25}{250}$

$x = 1cm$

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6) What is the value of x in the following diagram?

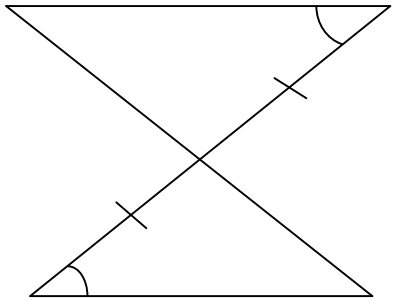


$$\frac{3}{6} = \frac{4}{8} = \frac{x}{10}$$

$$x = 5cm$$

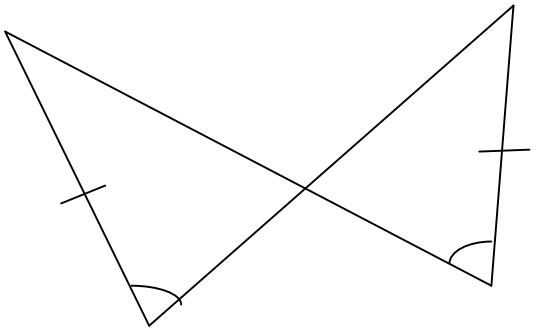
7) State which of the following pairs of triangles are congruent, and the reasons for their congruency

a)



ASA

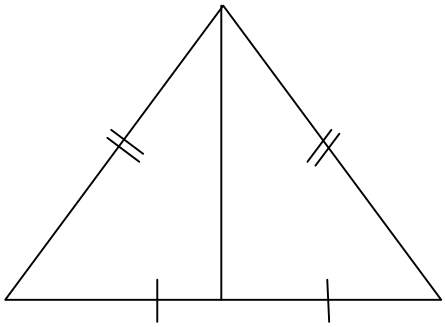
b)



AAS

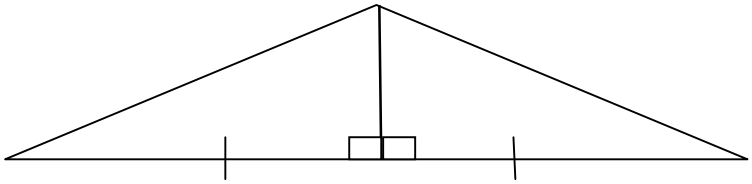
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c)



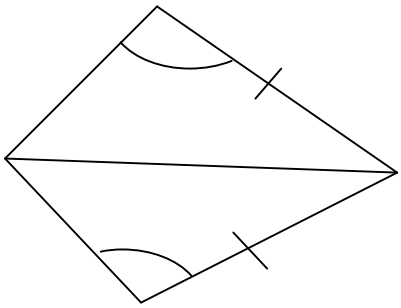
SSS or SAS

d)



SAS

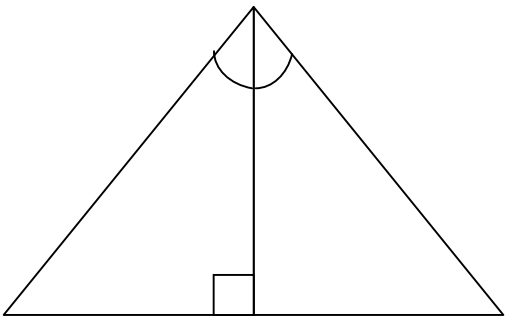
e)



ASS does not prove congruence

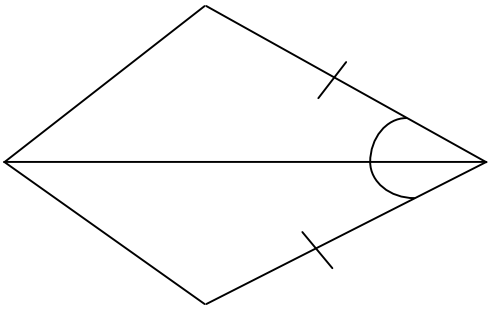
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f)



ASA

g)



SAS

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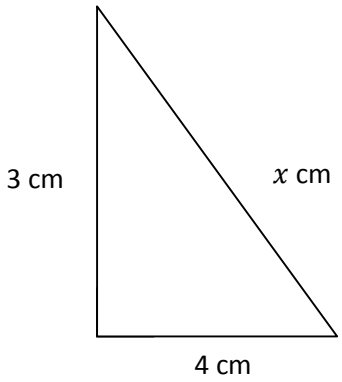
Exercise 3

Pythagoras' Theorem

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1) Find the value of x to 2 decimal places in the following diagrams

a)

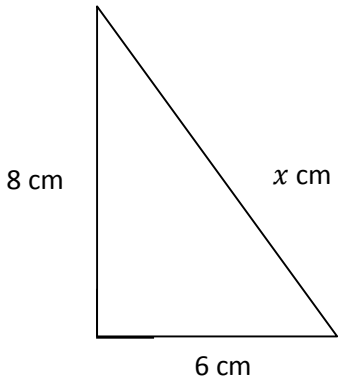


$3^2 + 4^2 = x^2$

$x^2 = 25$

x =5 cm

b)



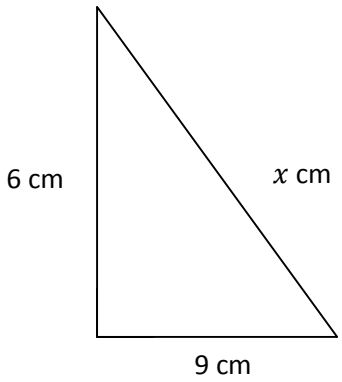
$8^2 + 6^2 = x^2$

$x^2 = 100$

x =10 cm

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c)

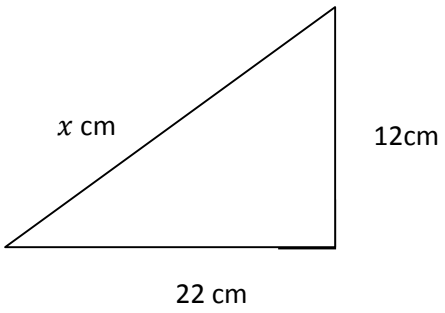


$6^2 + 9^2 = x^2$

$x^2 = 117$

x =10.82 cm

d)

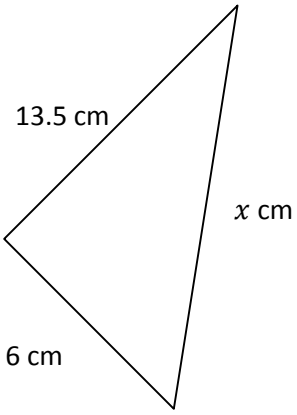


$22^2 + 12^2 = x^2$

$x^2 = 628$

$x = 25.06$ cm

e)



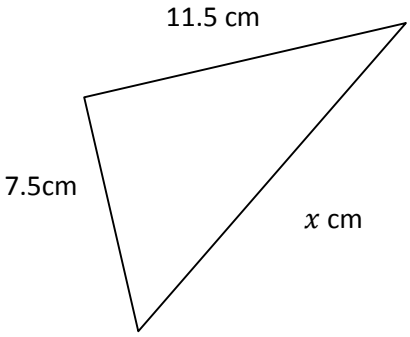
$6^2 + 13.5 = x^2$

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$x^2 = 218.25$

$x = 14.77$ cm

f)



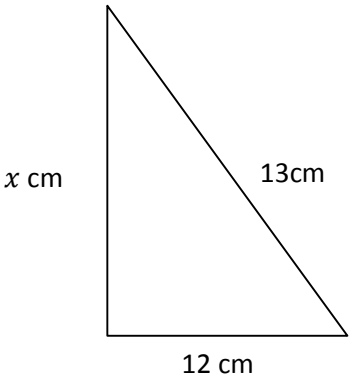
$7.5^2 + 11.5^2 = x^2$

$x^2 = 188.5$

$x = 13.73$ cm

2) Find the value of x to 2 decimal places in the following diagrams

a)



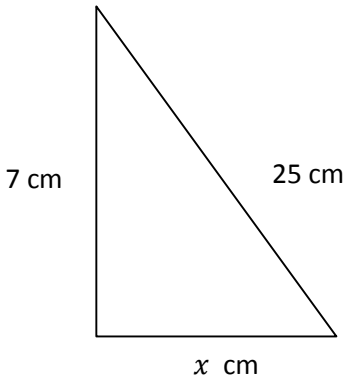
$x^2 + 12^2 = 13^2$

$x^2 = 25$

$x = 5$ cm

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b)

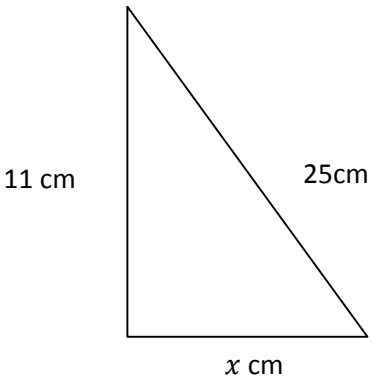


$7^2 + x^2 = 25^2$

$x^2 = 576$

$x = 24$ cm

c)



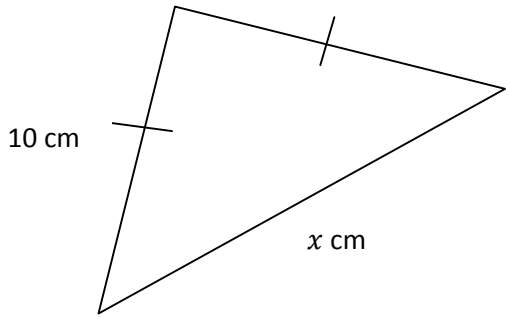
$11^2 + x^2 = 25^2$

$x^2 = 504$

$x = 22.45$ cm

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d)

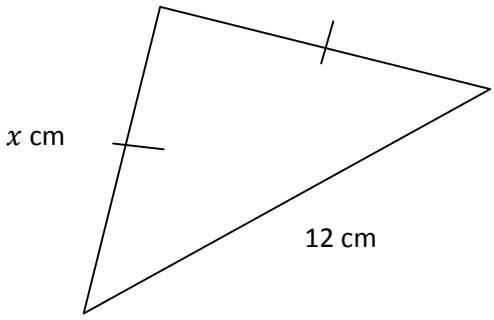


$10 + 10^2 = x^2$

$x^2 = 200$

$x = 14.14$ cm

e)



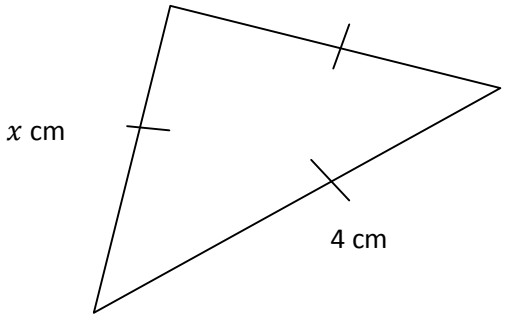
$x^2 + x^2 = 12^2$

$2x^2 = 144$

$x = 8.49$ cm

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f)

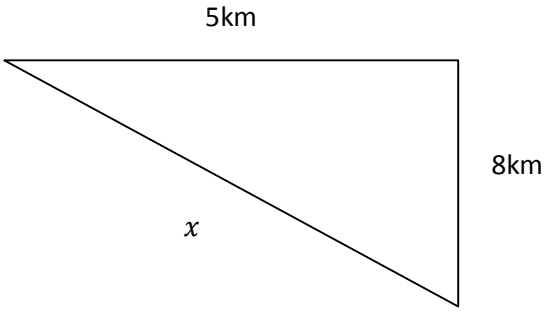


$x^2 + x^2 = 4^2$

$2x^2 = 16$

$x = 2.83$ cm

3) A man walks 5 km east then turns and walks 8 km south. How far is the shortest distance to his starting position?

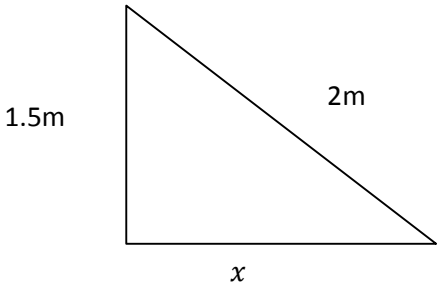


$5^2 + 8^2 = x^2$

$x^2 = 89$

$x = 9.43$ km

4) A ladder 2 meters long is placed against a wall and reaches 1.5 meters up the wall. How far is the foot of the ladder from the base of the wall?



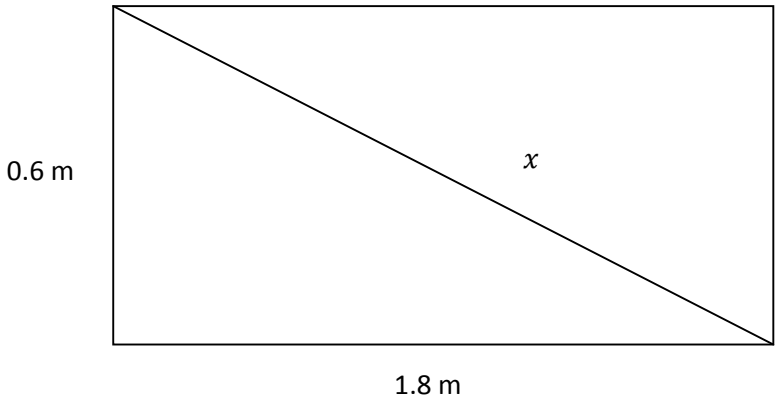
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$1.5^2 + x^2 = 2^2$

$x^2 = 1.75$

$x = 1.32$ m

5) A farmer wishes to place a brace across the diagonal of a rectangular gate that is 1.8 metres long and 0.6 metres wide. How long will the brace be?

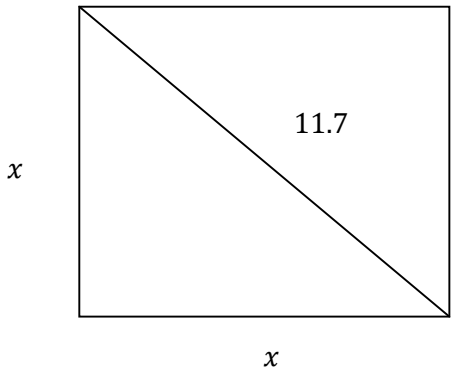


$0.6^2 + 1.8^2 = x^2$

$x^2 = 3.6$

$x = 1.9\ m$

6) A square room measures 11.7 metres from corner to corner. How wide is it?



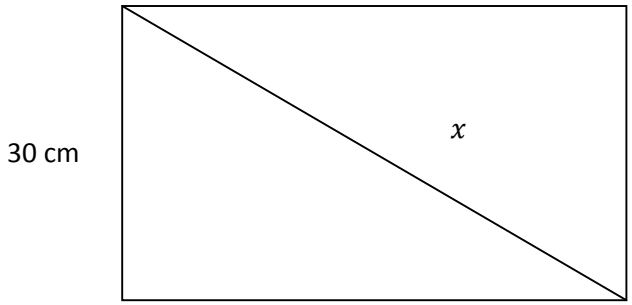
$x^2 + x^2 = 11.7^2$

$2x^2 = 136.89$

$x = 8.27\ m$

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7) The size of television sets are stated in terms of the diagonal distance across the screen. If the screen of a set is 40 cm long and 30 cm wide, how should it be advertised?

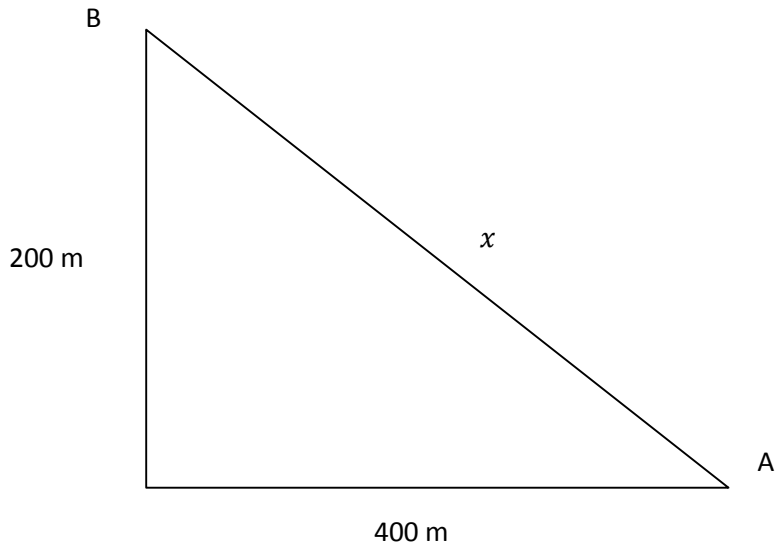


$30^2 + 40^2 = x^2$

$x^2 = 2500$

$x = 50\ cm$

8) A student has two choices when walking to school. From point A, he can walk 400 metres, then turn 90° and walk a further 200 metres to point B (school), or he can walk across the field that runs directly from A to B. How much further does he have to walk if he takes the path instead of the field?



$200^2 + 400^2 = x^2$

$x^2 = 200000$

$x = 447.21\ m$

He will walk an extra 152.79 m by taking the path

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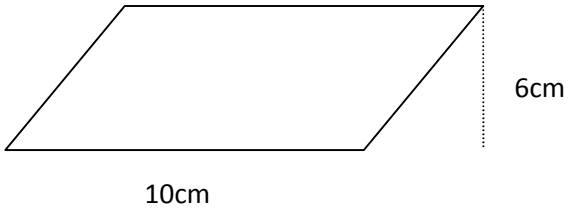
Exercise 4

Area Calculations

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1) Find the area of the following

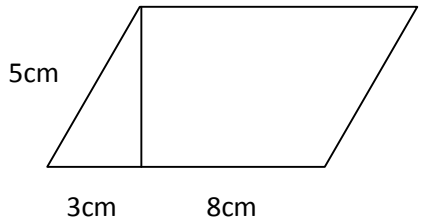
a)



Area $= \frac{1}{2} base \times perpendicular\ height$

$A = \frac{1}{2} \times 10 \times 6 = 30cm^2$

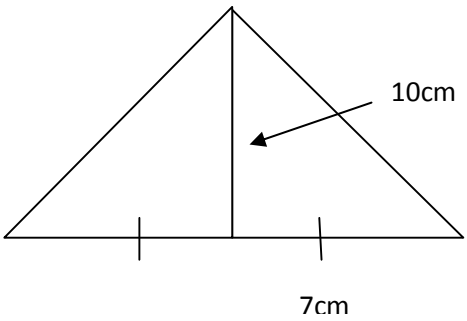
b)



Area $= \frac{1}{2} base \times perpendicular\ height$

$A = \frac{1}{2} \times 11 \times 4 = 22cm^2$

c)

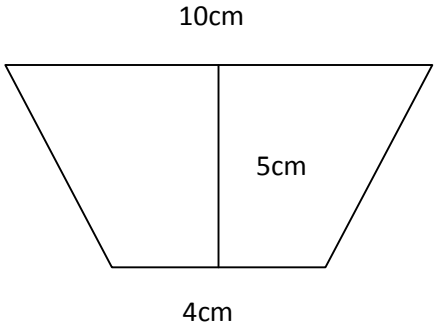


Area $= \frac{1}{2} base \times perpendicular\ height$

$A = \frac{1}{2} \times 14 \times 10 = 70cm^2$

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d)

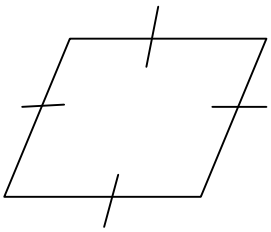


Area of trapezium = $\frac{a+b}{2} \times height$

A = $\frac{10+4}{2} \times 5 = 35\ cm^2$

e) Perimeter = 12 cm

Perpendicular height = 4cm

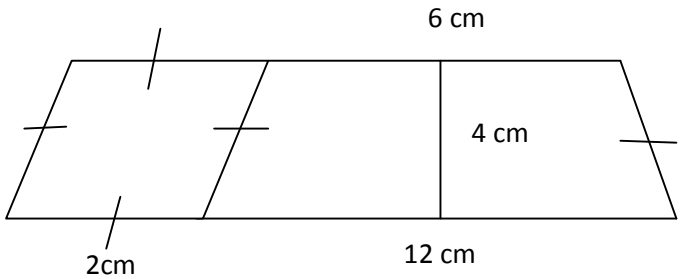


Area of rhombus = $base \times perpendicular\ height$

A= $3 \times 4 = 12cm^2$

2) Calculate the area of the following composite shapes

a)

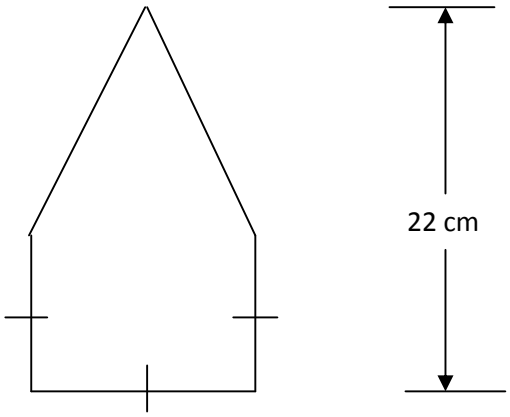


Area = Area of rhombus + area of trapezium

A= $(2 \times 4) + \left(\frac{12+6}{2} \times 4\right) = 8 + 36 = 44cm^2$

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b)

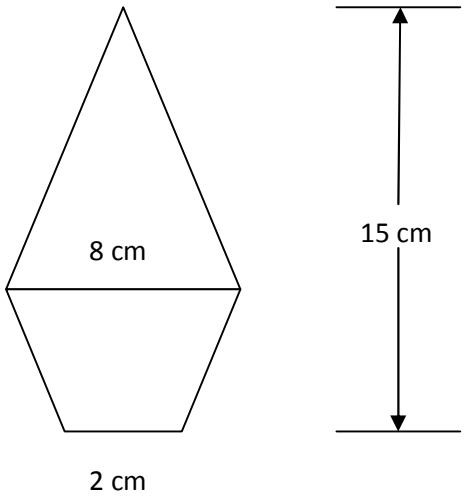


Height of triangle $= 22 - 8 = 14\ cm$

Area of shape = Area of triangle + Area of square

Area $= \left(\frac{1}{2} \times 8 \times 14\right) + (8 \times 8) = 120cm^2$

c) Area of triangle = 40 cm^2



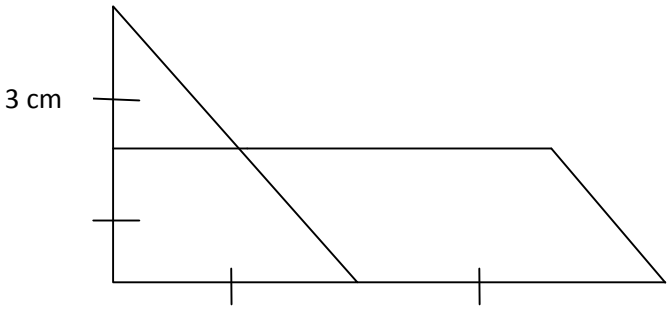
Height of trapezium $= 15$ –height of triangle

Height of triangle $= \frac{2 \times Area}{Base} = \frac{2 \times 40}{8} = 10$

Area$= 40 + \left(\frac{8+2}{2} \times 5\right) = 40 + 25 = 65cm^2$

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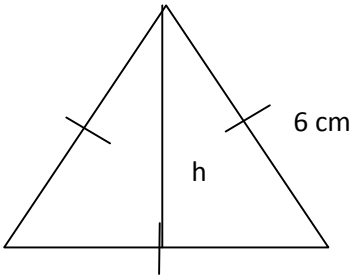
d)



Area = Area of triangle + area of parallelogram

Area $\left(\frac{1}{2}\times 6\times 3\right)+(3\times 3)=18cm^2$

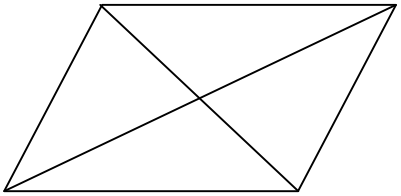
3) A badge is in the shape of an equilateral triangle with a perimeter of 18cm. What is the area of the badge?



$h^2+3^2=6^2$

Area $\frac{1}{2}\times 6\times\left(\sqrt{6^2-3^2}\right)=3\times\sqrt{27}=9\sqrt{3}cm^2$

4) A rhombus has one diagonal measuring 8cm and the other measuring 6cm. What is its area?



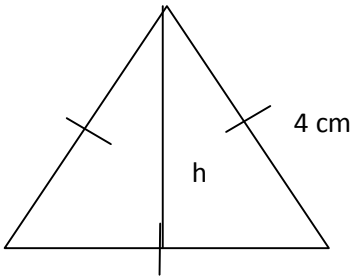
Diagonals of a rhombus bisect at right angles

Rhombus can be divided into 4 right angles triangles of base 4 and height 3

Area $=4\times\left(\frac{1}{2}\times 4\times 3\right)=24cm^2$

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5) What height must an isosceles triangle of base 2cm be in order to have an area the same as an equilateral triangle of side length 4cm?



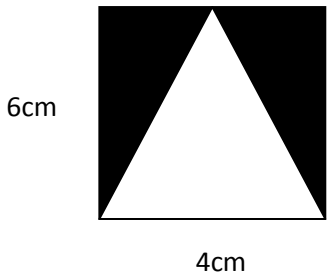
Area $= \frac{1}{2} \times 4 \times \left(\sqrt{4^2 - 2^2}\right) = 2\sqrt{12} = 4\sqrt{3}$

Area of isosceles $= \frac{1}{2} \times 2 \times h = h$

Therefore $h = 4\sqrt{3}$

6) Calculate the area of the shaded regions

a)

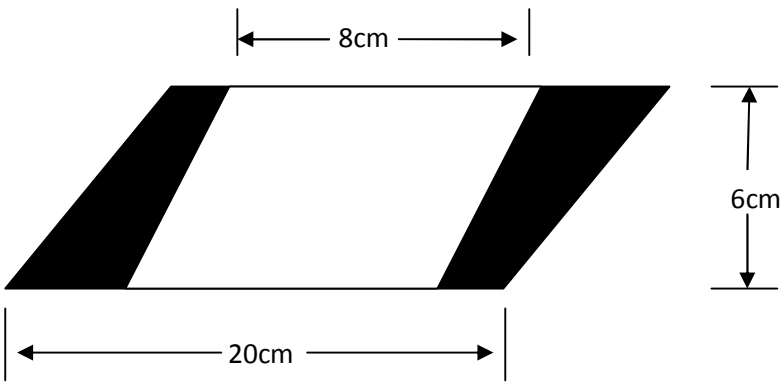


Area of rectangle $= 6 \times 4 = 24$

Area of triangle $= \frac{1}{2} \times 4 \times 6 = 12$

Area of shaded $= 24 - 12 = 12cm^2$

b)



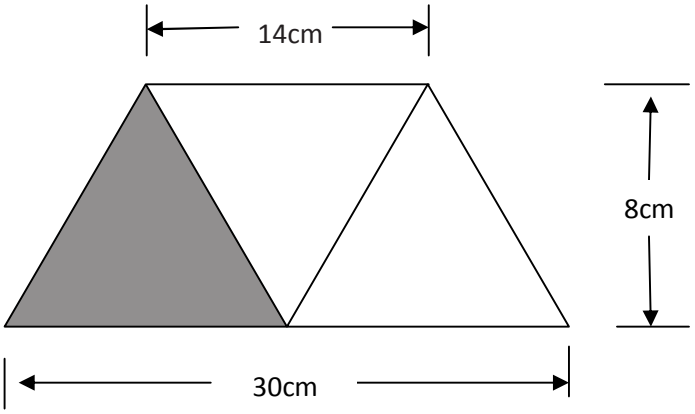
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Area of large parallelogram $= 20 \times 6 = 120$

Area of small parallelogram$= 8 \times 6 = 48$

Area of shaded$= 120 - 48 = 72cm^2$

c)

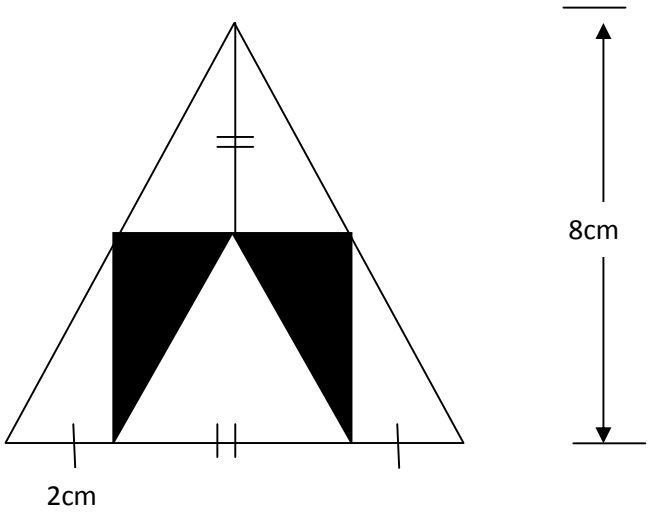


Area of trapezium$= \frac{30+14}{2} \times 8 = 176$

Area of parallelogram$= 14 \times 8 = 112$

Area of shaded$= 176 - 112 = 64cm^2$

d) Area of large triangle = 32 cm^2



Base of large triangle$= \frac{2 \times Area}{Height} = \frac{2 \times 32}{8} = 8\ cm$

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Centre portion of base $= 4\ cm$

Height of rectangle and small triangle $= 8 - 4 = 4$

Area of rectangle $= 4 \times 4 = 16$

Area of small triangle $= \frac{1}{2} \times 4 \times 4 = 8$

Area of shaded $= 16 - 8 = 8\ cm^2$

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Year 11 Unit 2 Mathematics

Derivative of a Function

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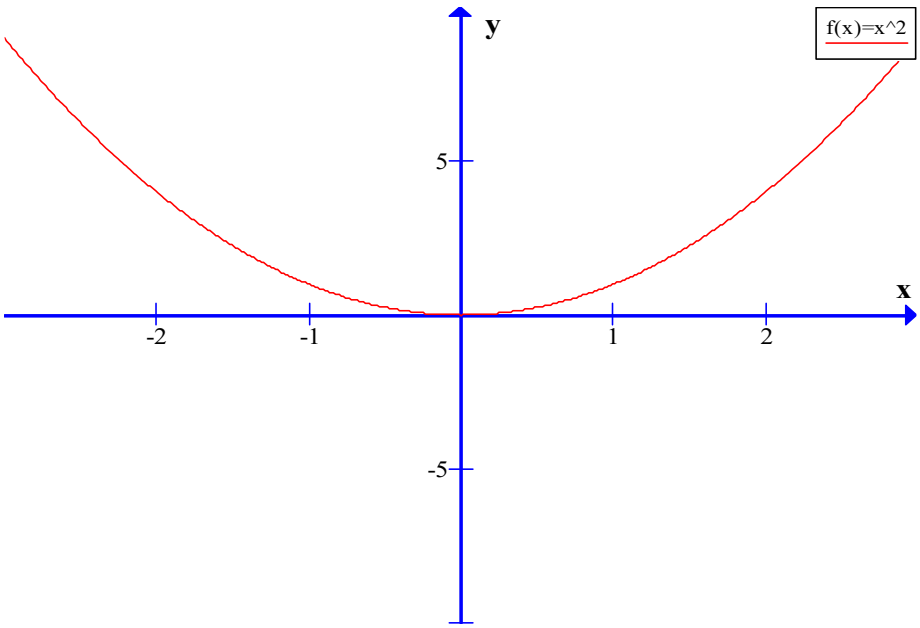
Exercise 1

Continuity

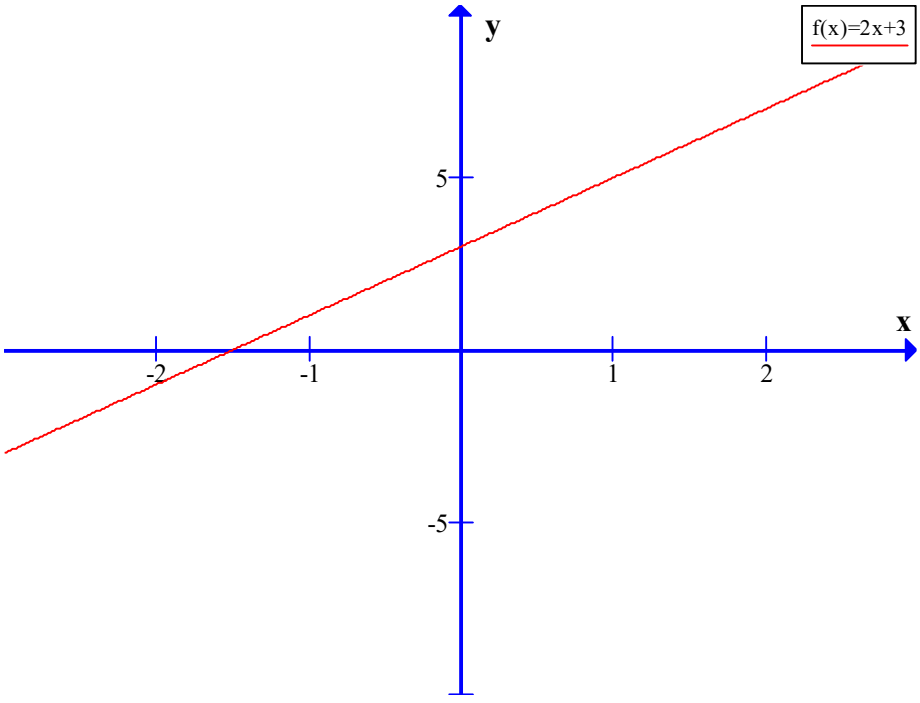
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1) Graph the following functions in the domain $-3 \leq x \leq 3$

a) $f(x) = x^2$

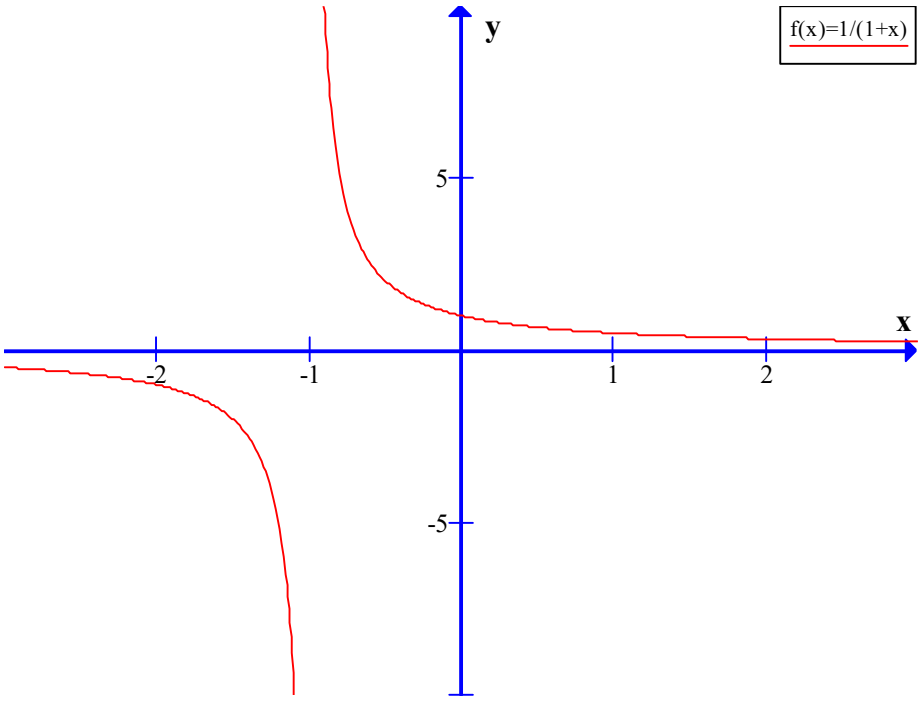


b) $f(x) = 2x + 3$

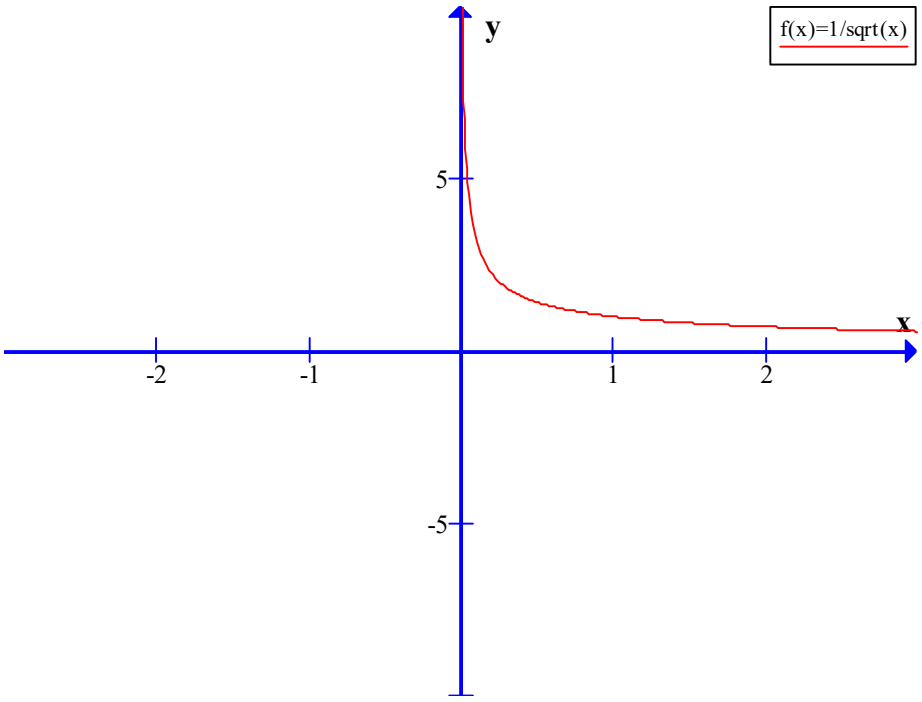


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c) $f(x) = \frac{1}{1+x}$

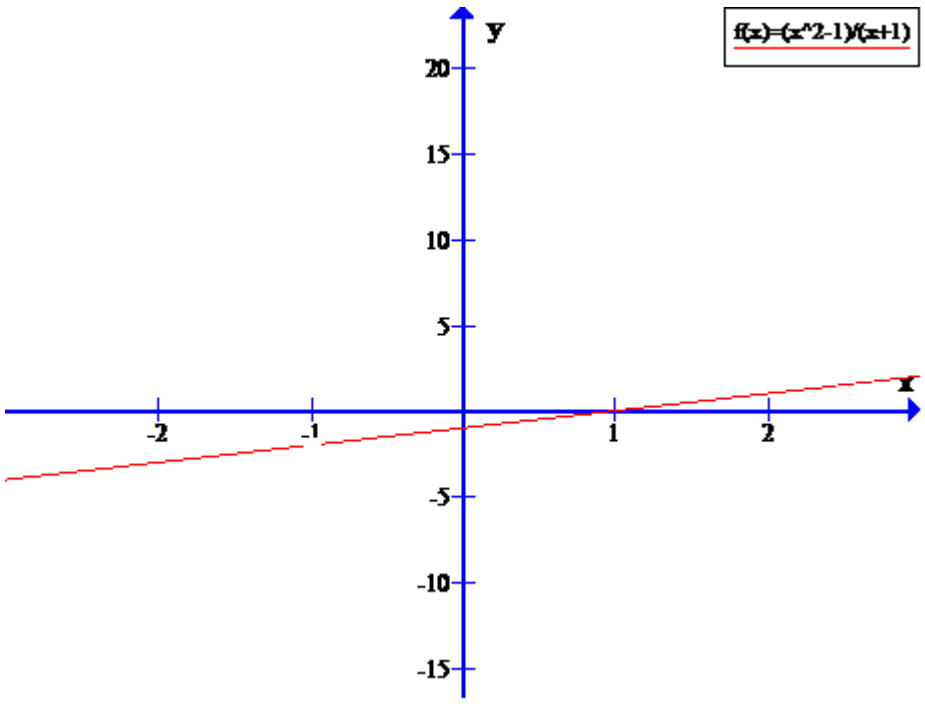


d) $f(x) = \frac{1}{\sqrt{x}}$

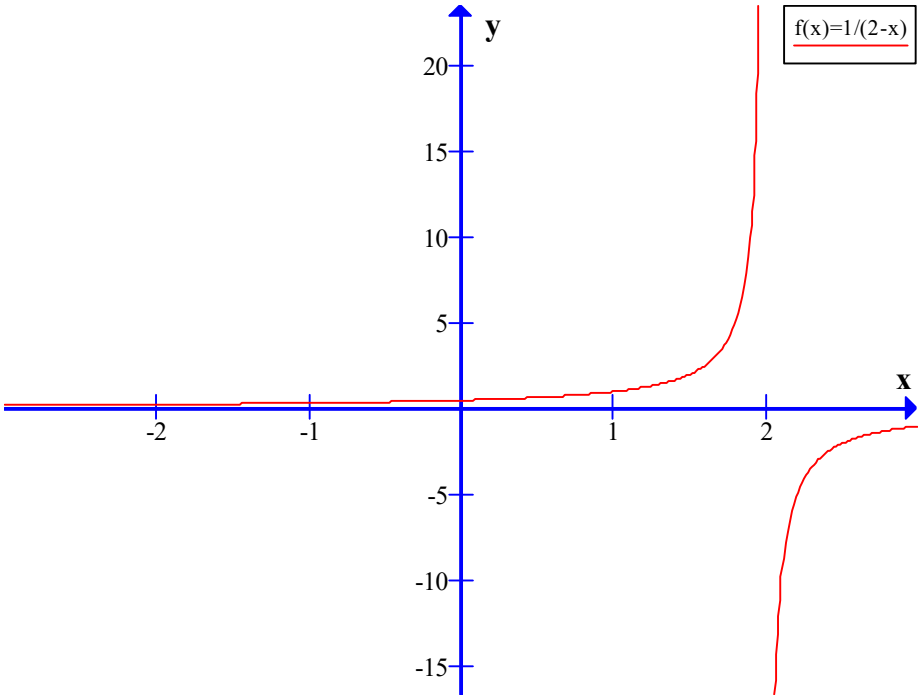


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e) $f(x) = \frac{x^2-1}{x+1}$



f) $f(x) = \frac{1}{2-x}$



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2) Using your graphs in question 1 as a guide, state whether functions are continuous or discontinuous over the domain. Give mathematical proof

c, e and f are discontinuous

None of the functions are defined at the points $x = -1, x = -1, and\ x = 2$ respectively

3) Show at what point(s) the following functions are discontinuous

a) $f(x) = \frac{2}{x^2 - x}$

Function is undefined and hence discontinuous at $x = 1, x = 0, x = -1$

b) $f(x) = \frac{x+1}{x+3}$

Function is undefined and hence discontinuous at $x = -3$

c) $f(x) = \begin{bmatrix} x + 2 & for\ x < 0 \\ \frac{3}{x^2+4} & for\ x \geq 0 \end{bmatrix}$

Function is defined for all values of x

$$f(0) = \frac{3}{4}$$

$$\lim_{x \to 0^+} f(x) = \frac{3}{4}$$

$$\lim_{x \to 0^-} f(x) = 2$$

Therefore function is discontinuous at $x = 0$

d) $f(x) = \frac{1}{x^2 - 1}$

Function is undefined and hence discontinuous at $x = 1, x = -1$

e) $f(x) = \frac{|x|}{x}$

Case 1: $x > 0, f(x) = \frac{x}{x} = 1$

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$\lim_{x\to 0^+} f(x) = 1$

Case 2: $x < 0, f(x) = \frac{-x}{x} = -1$

$\lim_{x\to 0^-} f(x) = -1$

Function is discontinuous at $x = 0$

4) Let $f(x) = x^2, g(x) = \frac{1}{1-x}, h(x) = \frac{\cos x}{x}, r(x) = x^3 - 1$

State whether the following functions are continuous, and give reasons

Let the two functions be $f(x) and\ g(x) respectively$

a) $k(x) = x^2 + \frac{1}{1-x}$

$f(x)$ is continuous

$g(x)$ is discontinuous at $x = 1$

Therefore $k(x)$ is discontinuous

b) $k(x) = x^2 - \frac{\cos x}{x}$

Both functions are continuous, therefore $k(x)$ is continuous

c) $k(x) = x^2(x^3 - 1)$

Both functions are continuous, therefore $k(x)$ is continuous

d) $k(x) = \frac{1}{1-x} + \frac{\cos x}{x}$

$f(x)$ is discontinuous at $x = 1$

$g(x)$ is continuous

Therefore $k(x)$ is discontinuous

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e) $k(x) = \frac{1}{1-x} - x^2$

$f(x)$ is discontinuous at $x = 1$

$g(x)$ is continuous

Therefore $k(x)$ is discontinuous

f) $k(x) = \frac{x^3-1}{1-x}$

$f(x)$ is continuous

$g(x)$ is discontinuous at $x = 1$

Therefore $k(x)$ is discontinuous

g) $k(x) = \frac{\cos x}{x} + (x^3 - 1)$

Both functions are continuous, therefore $k(x)$ is continuous

h) $k(x) = \frac{x^2(\cos x)}{x} = xcos(x)$

Therefore function is continuous

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Exercise 2

Secant to a Curve

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1) Using the curve $y = x^2$, determine the gradient of the line joining the following points on the curve(the secant)

a) (-4,16) and (-2,4)

$$m = \frac{16-4}{-4-2} = \frac{12}{-6} = -2$$

b) (0,0) and (-1,1)

$$m = \frac{1-0}{-1-0} = -1$$

c) (2,4) and (5,25)

$$m = \frac{25-4}{5-2} = \frac{21}{3} = 7$$

d) (2,4) and (-2,4)

$$m = \frac{4-4}{-2-2} = \frac{0}{-4} = 0$$

e) $\left(\frac{1}{2},\frac{1}{4}\right)$ and $\left(-\frac{1}{4},\frac{1}{16}\right)$

$$m = \frac{\frac{1}{16}-\frac{1}{4}}{\frac{-1}{4}-\frac{1}{2}} = \frac{-\frac{3}{16}}{-\frac{3}{4}} = \frac{1}{4}$$

2) For the same curve, determine the gradient of the secant from the point (1,1) to the following points

a) (-4,16)

$$m = \frac{16-1}{-4-1} = -\frac{15}{3} = -5$$

b) (-3,9)

$$m = \frac{9-1}{-3-1} = \frac{8}{-4} = -2$$

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c) (-2,4)

$$m = \frac{4-1}{-2-1} = -\frac{3}{3} = -1$$

d) (-1,1)

$$m = \frac{1-1}{-1-1} = \frac{0}{2} = 0$$

e) (0,0)

$$m = \frac{0-1}{0-1} = \frac{-1}{-1} = 1$$

f) $\left(\frac{1}{2},\frac{1}{4}\right)$

$$m = \frac{\frac{1}{4}-1}{\frac{1}{2}-1} = \frac{-\frac{3}{4}}{-\frac{1}{2}} = \frac{3}{2}$$

g) $\left(\frac{3}{2},\frac{9}{4}\right)$

$$m = \frac{\frac{9}{4}-1}{\frac{3}{2}-1} = \frac{\frac{5}{4}}{\frac{1}{2}} = \frac{5}{2}$$

h) (2,4)

$$m = \frac{4-1}{2-1} = \frac{3}{1} = 3$$

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i) (3,9)

$$m = \frac{9-1}{3-1} = \frac{8}{2} = 4$$

j) (4,16)

$$m = \frac{16-1}{4-1} = \frac{15}{3} = 5$$

k) (5,25)

$$m = \frac{25-1}{5-1} = -\frac{24}{4} = 6$$

3)

a) Does the pattern of numbers in question 2 suggest that there is a limiting value for the gradient of the secant to the point (1, 1) as $x \to 1$? If so what is that value?

2

b) What is the general equation for the limit of the gradient of the secant to the point (1, 1) as $x \to 1$?

$$\lim_{x \to 1} \frac{f(x) - f(c)}{x - c}$$

c) Calculate the limit of the gradient of the secant to the point (1, 1) as $x \to 1$

$$\lim_{x \to 1} \frac{f(x) - f(1)}{x - 1}$$

$$= \lim_{x \to 1} \frac{x^2 - 1}{x - 1}$$

$$= \lim_{x \to 1} \frac{(x+1)(x-1)}{x - 1}$$

$$= 2$$

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4) Calculate and hence construct a table of the limits of the gradient of the secant to the function $f(x) = x^2$ at the following points

a) (-1,1)

$$\lim_{x \to -1} \frac{f(x) - f(-1)}{x - (-1)}$$

$$= \lim_{x \to 1} \frac{x^2 - 1}{x + 1}$$

$$\lim_{x \to 1} \frac{(x - 1)(x + 1)}{x + 1}$$

$$= -2$$

b) (2,4)

$$\lim_{x \to 2} \frac{f(x) - f(c)}{x - c}$$

$$= \lim_{x \to 2} \frac{f(x) - f(2)}{x - 2}$$

$$= \lim_{x \to 2} \frac{x^2 - 4}{x - 2}$$

$$= \lim_{x \to 2} \frac{(x - 2)(x + 2)}{x - 2}$$

$$= 4$$

c) (-4,16)

$$\lim_{x \to -4} \frac{f(x) - f(c)}{x - c}$$

$$= \lim_{x \to -4} \frac{f(x) - f(-4)}{x + 4}$$

$$= \lim_{x \to -4} \frac{x^2 - 16}{x + 4}$$

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$$= \lim_{x \to -4} \frac{(x-4)(x+4)}{x+4}$$

$$= -8$$

d) (3,9)

$$\lim_{x \to 3} \frac{f(x) - f(c)}{x - c}$$

$$= \lim_{x \to 3} \frac{f(x) - f(3)}{x - 3}$$

$$= \lim_{x \to 3} \frac{x^2 - 9}{x - 3}$$

$$= \lim_{x \to 3} \frac{(x-3)(x+3)}{x-3}$$

$$= 6$$

e) (10,100)

$$\lim_{x \to 10} \frac{f(x) - f(c)}{x - c}$$

$$= \lim_{x \to 10} \frac{f(x) - f(10)}{x - 10}$$

$$= \lim_{x \to 10} \frac{x^2 - 100}{x - 10}$$

$$= \lim_{x \to 10} \frac{(x-10)(x+10)}{x-10}$$

$$= 20$$

5) Formulate a rule for the value of the gradient of the secant to the curve $f(x) = x^2$ at any point

Gradient of secant at point x is $2x$

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Exercise 3

Methods of Differentiation

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1) Using the equation: $f'(c) = \lim_{h\to 0} \frac{f(c+h)-f(c)}{h}$, calculate the derivative of the following functions at the nominated values of x.

a) $f(x) = x^2$ at $x = 1, x = -2$

$$f'(1) = \lim_{h\to 0} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h\to 0} \frac{1 + 2h + h^2 - 1}{h}$$

$$= \lim_{h\to 0} \frac{h(2+h)}{h}$$

$$= \lim_{h\to 0} 2 + h$$

$$= 2$$

$$f'(-2) = \lim_{h\to 0} \frac{f(-2+h) - f(-2)}{h}$$

$$= \lim_{h\to 0} \frac{h^2 - 4h + 4 - 4}{h}$$

$$\lim_{h\to 0} \frac{h(h-4)}{h}$$

$$= -4$$

b) $f(x) = x^2 + 3$ at $x = 2, x = 1$

$$f'(2) = \lim_{h\to 0} \frac{f(2+h) - f(2)}{h}$$

$$= \lim_{h\to 0} \frac{h^2 + 4h + 7 - (7)}{h}$$

$$= \lim_{h\to 0} \frac{h(h+4)}{h}$$

$$= 4$$

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$$f'(1) = \lim_{h\to 0} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h\to 0} \frac{h^2 + 2h + 4 - 4}{h}$$

$$= \lim_{h\to 0} \frac{h(h+2)}{h}$$

$$= 2$$

c) $f(x) = x^3 - 3x$ at $x = -1, x = 3$

$$f'(-1) = \lim_{h\to 0} \frac{f(-1+h) - f(-1)}{h}$$

$$= \lim_{h\to 0} \frac{f(-1+h) - f(-1)}{h}$$

$$= \lim_{h\to 0} \frac{(h-1)^3 - 3(h-1) - (-1+3)}{h}$$

$$= \lim_{h\to 0} \frac{h^3 - 3h^2}{h}$$

$$= \lim_{h\to 0} \frac{h(h^2 - 3h)}{h}$$

$$= 0$$

$$f'(3) = \lim_{h\to 0} \frac{f(3+h) - f(3)}{h}$$

$$= \lim_{h\to 0} \frac{h^3 + 9h^2 + 27h + 27 - 3h - 9 - 18}{h}$$

$$= \lim_{h\to 0} \frac{h^3 + 9h^2 + 24h}{h}$$

$$= \lim_{h\to 0} \frac{h(h^2 + 9h + 24)}{h}$$

$$=24$$

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d) $f(x) = x^2 + 2x + 4$ at $x = 1, x = -1$

$$f'(1) = \lim_{h \to 0} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \to 0} \frac{(h+1)^2 + 2(h+1) + 4 - 7}{h}$$

$$= \lim_{h \to 0} \frac{h^2 + 2h + 1 + 2h + 2 - 3}{h}$$

$$= \lim_{h \to 0} \frac{h^2 + 4h}{h}$$

$$= \lim_{h \to 0} \frac{h(h+4)}{h}$$

$$= 4$$

$$f'(-1) = \lim_{h \to 0} \frac{f(-1+h) - f(-1)}{h}$$

$$= \lim_{h \to 0} \frac{(h-1)^2 + 2(h-1) + 4 - 3 - f(-1)}{h}$$

$$= \lim_{h \to 0} \frac{h^2 - 2h + 1 + 2h - 2 + 4 - 3}{h}$$

$$= \lim_{h \to 0} \frac{h^2}{h}$$

$$= \lim_{h \to 0} h$$

$$= 0$$

e) $f(x) = x^3 - 6x$ at $x = 0, x = 2$

$$f'(0) = \lim_{h \to 0} \frac{f(h) - f(0)}{h}$$

$$= \lim_{h \to 0} \frac{h^3 - 6h}{h}$$

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$$= \lim_{h \to 0} \frac{h(h^2 - 6)}{h}$$

$$= -6$$

$$f'(2) = \lim_{h \to 0} \frac{f(h+2) - f(2)}{h}$$

$$= \lim_{h \to 0} \frac{(h+2)^3 - 6(h+2) - (-4)}{h}$$

$$= \lim_{h \to 0} \frac{h^3 + 6h^2 + 6h + 8 - 6h - 12 + 4}{h}$$

$$= \lim_{h \to 0} \frac{h^3 + 6h^2}{h}$$

$$= 6$$

f) $f(x) = x^3 + 2x + 1$ at $x = 1, x = 2$

$$f'(1) = \lim_{h \to 0} \frac{f(h+1) - f(1)}{h}$$

$$= \lim_{h \to 0} \frac{h^3 + 2(h+1) + 1 - 4}{h}$$

$$= \lim_{h \to 0} \frac{h^3 + 3h^2 + 3h + 1 + 1 + 2h + 2 - 4}{h}$$

$$= \lim_{h \to 0} \frac{h^3 + 3h^2 + 5h}{h}$$

$$= \lim_{h \to 0} \frac{h(h^3 + 3h + 5)}{h}$$

$$= 5$$

$$f'(2) = \lim_{h \to 0} \frac{f(h+2) - f(2)}{h}$$
$$= \lim_{h \to 0} \frac{(h+2)^3 + 2(h+2) + 1 - 13}{h}$$

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$$= \lim_{h \to 0} \frac{f(h+2) - f(2)}{h}$$

$$= \lim_{h \to 0} \frac{h^3 + 6h^2 + 12h + 8 + 2h + 4 + 1 - 13}{h}$$

$$= \lim_{h \to 0} \frac{h^3 + 6h^2 + 14h}{h}$$

$$= \lim_{h \to 0} \frac{h(h^2 + 6h + 14)}{h}$$

$$= 14$$

2) From question 1, find the equation of the tangent line to each equation at the specified points

a) When $x = 1, y = 1$

From Q1, $m = 2$

Therefore $y - 1 = 2(x - 1)$

When $x = -2, y = 4$

From Q1, $m = -4$

Therefore $y - 4 = -4(x + 2)$

b) When $x = 2, y = 7$

From Q1, $m = 4$

Therefore $y - 7 = 4(x - 2)$

When $x = 1, y = 4$

From Q1, $m = 2$

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Therefore $y - 4 = 2(x - 1)$

c) When $x = -1, y = 2$

From Q1, $m = 0$

Therefore $y - 2 = 0$

When $x = 3, y = 18$

From Q1, $m = 24$

Therefore $y - 18 = 24(x - 3)$

d) When $x = 1, y = 7$

From Q1, $m = 4$

Therefore $y - 7 = 4(x - 1)$

When $x = -1, y = 3$

From Q1, $m = 0$

Therefore $y - 3 = 0$

e) When $x = 0, y = 0$

From Q1, $m = -6$

Therefore $y = -6x$

When $x = 2, y = -4$

From Q1, $m = 6$

Therefore $y + 4 = 6(x - 2)$

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f) When $x = 1, y = 4$

From Q1, $m = 5$

Therefore $y - 4 = x - 1$

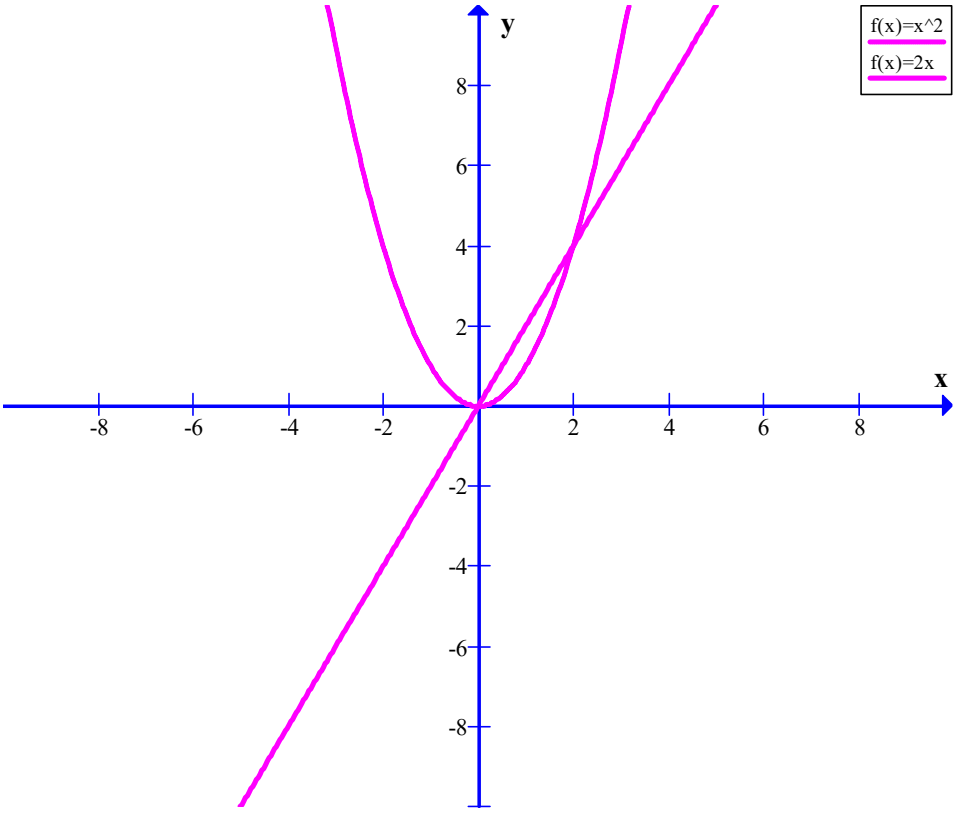
When $x = 2, y = 13$

From Q1, $m = 14$

Therefore $y - 13 = 14(x - 2)$

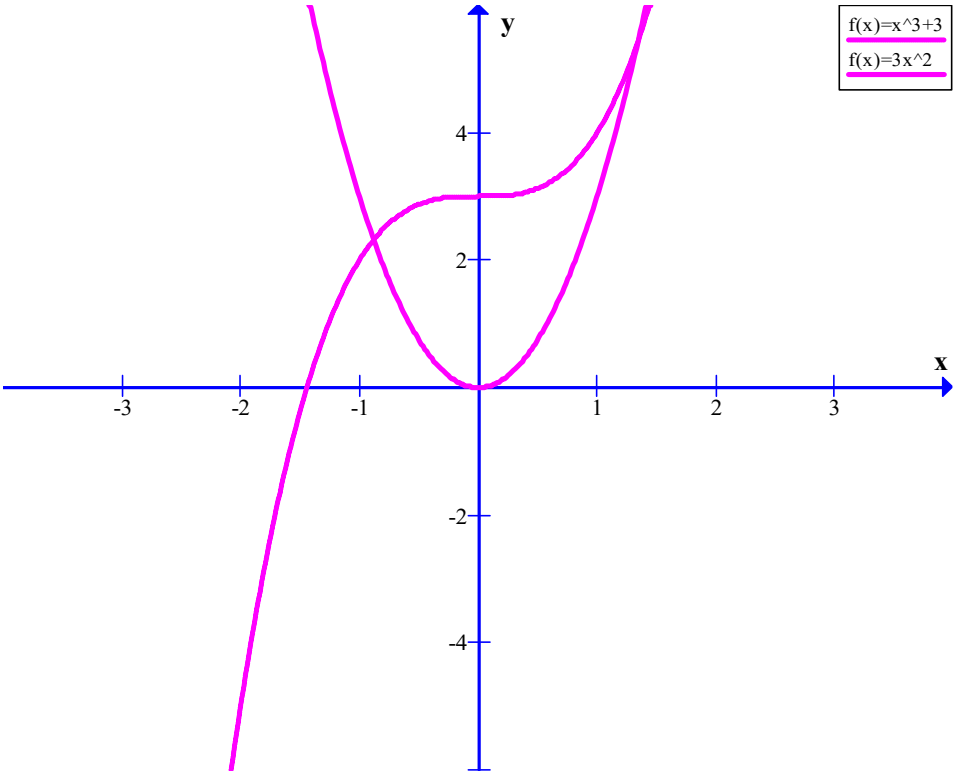
3) Graph each of the functions from question 1 and their derivatives (use the same graph for each function and derivative)

a)

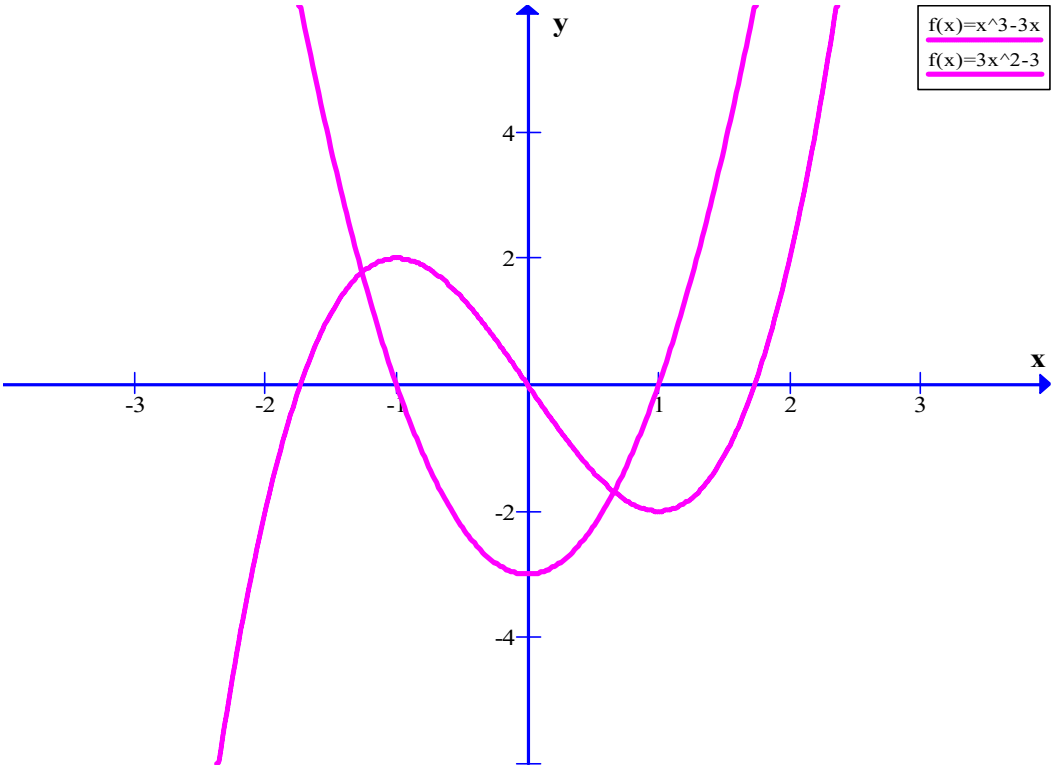


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b)

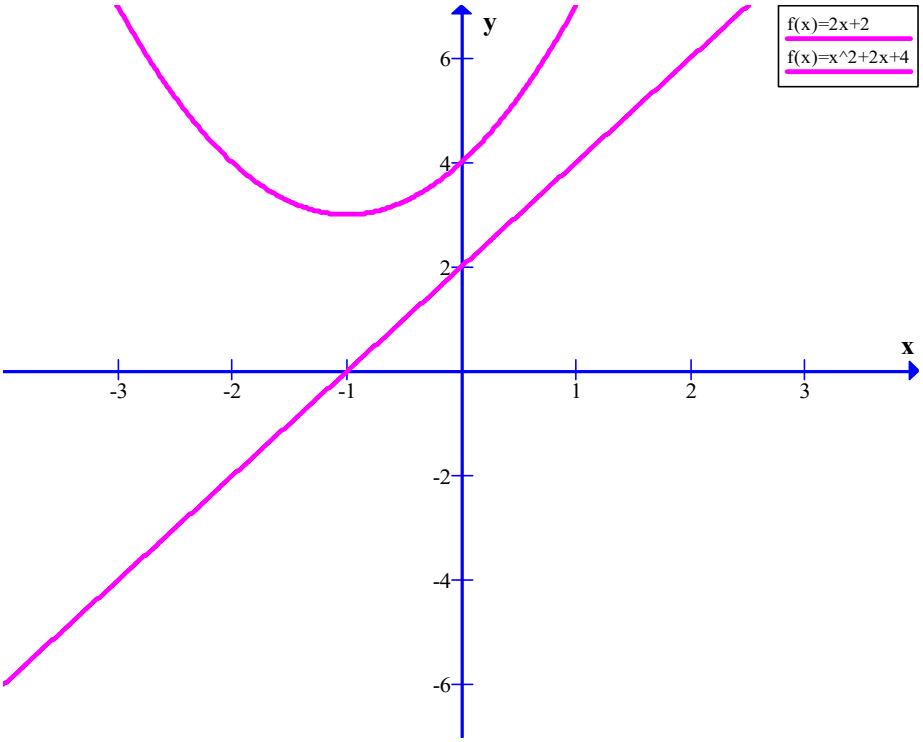


c)

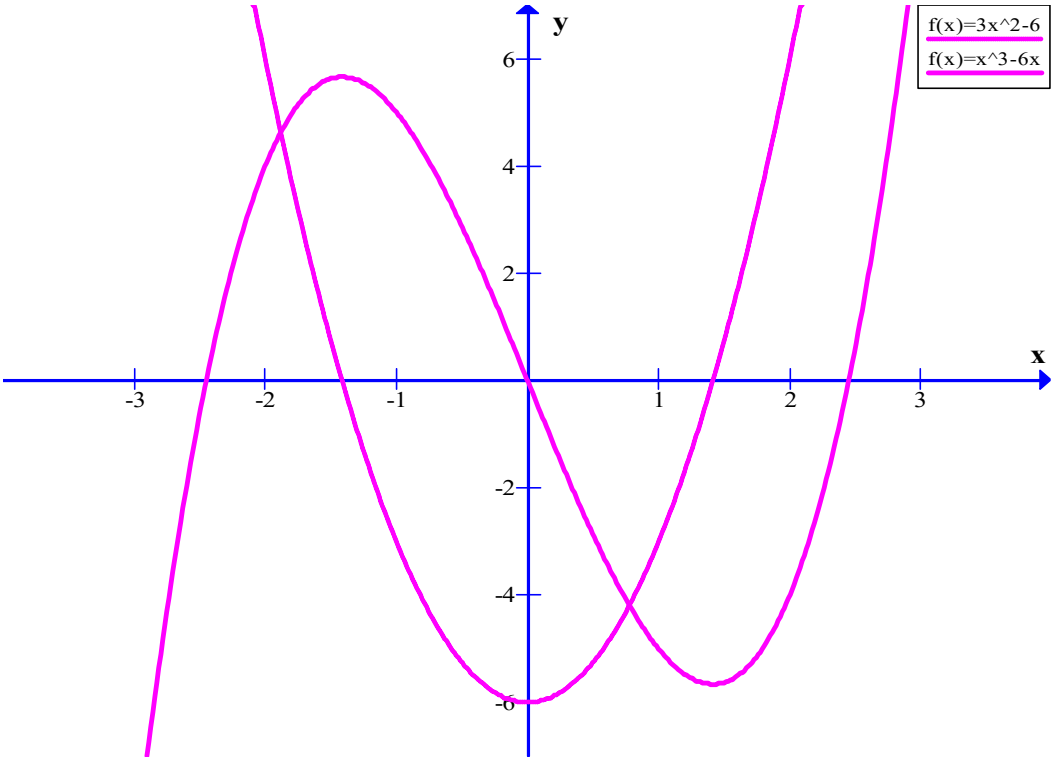


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d)

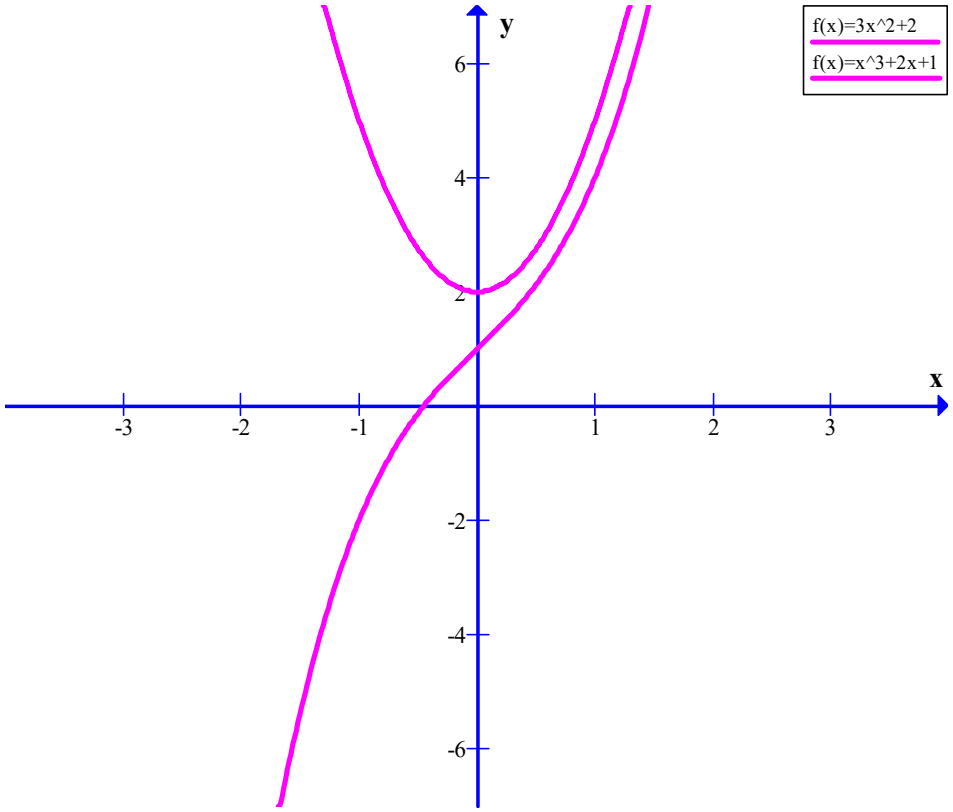


e)



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f)



4) Find $\frac{dy}{dx}$ for each of the following functions

a) $y = x^2$

$$\frac{dy}{dx} = 2x$$

b) $y = x^2 - 2x + 12$

$$\frac{dy}{dx} = 2x - 2$$

c) $y = 2x^2 - 2x$

$$\frac{dy}{dx} = 4x - 2$$

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d) $y = \frac{1}{2}x^2 - x$

$$\frac{dy}{dx} = x - 1$$

e) $y = x^4 + 3x^3 - 4x^2 + \frac{1}{2}x + 2$

$$\frac{dy}{dx} = 4x^3 + 9x^2 - 8x + \frac{1}{2}$$

5) Find $\frac{d}{dx}(f(x))$, where $f(x) =$

a) 4

$$\frac{d}{dx}\big(f(x)\big) = 0$$

b) $2x$

$$\frac{d}{dx}\big(f(x)\big) = 2$$

c) $\frac{1}{3}x^3 - 4x$

$$\frac{d}{dx}\big(f(x)\big) = x^2 - 4$$

d) $2x^5 - 4x^2$

$$\frac{d}{dx}\big(f(x)\big) = 10x^4 - 8x$$

e) $\frac{1}{4}x^4 + \frac{1}{3}x^3 - \frac{1}{4}x^2 + 100$

$$\frac{d}{dx}\big(f(x)\big) = x^3 + x^2 - \frac{x}{2}$$

6) Find the derivative of the following functions

a) $f(x) = \sqrt{x}$

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$$f'(x) = \frac{1}{2}x^{-\frac{1}{2}}$$

$$= \frac{1}{2\sqrt{x}}$$

b) $f(x) = x^{-1}$

$$f'(x) = -x^{-2}$$

$$= -\frac{1}{x^2}$$

c) $f(p) = \frac{1}{p^2}$

$$f'(p) = -2p^{-3}$$

$$\frac{-2}{p^3}$$

d) $f(t) = \frac{1}{\sqrt{t}}$

$$f'(t) = -\frac{1}{2}t^{-\frac{3}{2}}$$

$$= -\frac{1}{2t^{\frac{3}{2}}}$$

e) $f(a) = \sqrt{a} + \frac{1}{a}$

$$f'(a) = \frac{1}{2}a^{-\frac{1}{2}} + (-a^{-2})$$

$$= \frac{1}{2\sqrt{a}} - \frac{1}{a^2}$$

f) $f(x) = \sqrt[3]{x} - x^{-1}$

$$= x^{\frac{3}{2}} - x^{-1}$$

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$$f'(x) = \frac{3}{2}x^{\frac{1}{2}} + x^{-2}$$

$$= \frac{3\sqrt{x}}{2} + \frac{1}{x^2}$$

7) Find $f'(x)$ using the product rule, where $f(x) =$

a) $x^2(\sqrt{x})$

$$f'(x) = 2x(\sqrt{x}) + \frac{x^2}{2\sqrt{x}}$$

$$= \frac{4x^2 + x^2}{2\sqrt{x}}$$

$$= \frac{5x^{\frac{3}{2}}}{2}$$

b) $\frac{1}{x}(2x - 3)$

$$f'(x) = -x^{-2}(2x - 3) + \frac{2}{x}$$

$$= \frac{3 - 2x}{x^2} + \frac{2}{x}$$

$$= \frac{3 - 2x + 2x}{x^2}$$

$$= \frac{3}{x^2}$$

c) $\frac{1}{x^2}(x^2 - 4)$

$$f'(x) = -2x^{-3}(x^2 - 4) + \frac{2x}{x^2}$$

$$= \frac{8 - 2x^2}{x^3} + \frac{2}{x}$$

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$$= \frac{8 - 2x^2 + 2x^2}{x^3}$$

$$= \frac{8}{x^3}$$

d) $2x^{-1}\left(\frac{1}{2}x^3\right)$

$$f'(x) = -2x^{-2}\left(\frac{1}{2}x^3\right) + 2x^{-1}\frac{3x^2}{2}$$

$$= -\frac{x^3}{x^2} + \frac{3x}{4}$$

$$= \frac{7x}{4}$$

e) $\frac{3x}{2}(2x^2 - 1)$

$$f'(x) = \frac{3}{2}(2x^2 - 1) + \frac{3x}{2}(4x)$$

$$= 3x^2 - \frac{3}{2} + 6x^2$$

$$9x^2 - \frac{3}{2}$$

f) $\frac{x^2}{4}\left(x^2 - \frac{1}{4}x\right)$

$$f'(x) = \frac{x}{2}\left(x^2 - \frac{1}{4}x\right) + \frac{x^2}{4}\left(2x - \frac{1}{4}\right)$$

$$= \frac{x^3}{2} - \frac{x^2}{8} + \frac{2x^3}{4} - \frac{x^2}{16}$$

$$\frac{8x^3 - 2x^2 + 8x^3 - x^2}{16}$$

$$= \frac{16x^3 - 3x^2}{16}$$

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8) Find the derivatives of the following functions

a) $f(x) = (x^2 - 2)^2$

$$\frac{dy}{dx} = 2(x^2 - 2).2x = 4x(x^2 - 2)$$

b) $f(t) = (t^2 + 1)^{-1}$

$$\frac{dy}{dt} = -(t^2 + 1)^{-2}.2t$$

$$= \frac{-2t}{(t^2 + 1)^2}$$

c) $f(a) = \frac{a+1}{a-3}$

$$\frac{dy}{da} = \frac{(a - 3) \times 1 - (a + 1) \times 1}{(a - 3)^2}$$

$$= \frac{-4}{(a - 3)^2}$$

d) $g(x) = \frac{x^3-4x+2}{x}$

$$\frac{dg}{dx} = \frac{x(3x^2 - 4) - 1 \times (x^3 - 4x + 2)}{x^2}$$

$$= \frac{3x^3 - 4x - x^3 + 4x - 2}{x^2}$$

$$= \frac{2x^3 - 2}{x^2}$$

e) $h(x) = \frac{1}{(x^2-1)^2}$

$$h(x) = (x^2 - 1)^{-2}$$

$$\frac{dy}{dx} = -2(x^2 - 1)^{-1} \times 2x$$

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$$= \frac{-4x}{(x^2-1)}$$

f) $f(x) = \frac{x^2}{x-2}$

$$\frac{dy}{dx} = \frac{(x-2)2x - x^2}{(x-2)^2}$$

$$= \frac{x^2 - 4x}{(x-2)^2}$$

g) $f(x) = \sqrt{x^2-3}$

$$f(x) = (x^2-3)^{\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{1}{2}(x^2-3)^{-\frac{1}{2}} \times 2x$$

$$= \frac{x}{\sqrt{(x^2-3)}}$$

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